GEK-45167



GE Drive Systems

INSTRUCTIONS

QIII
DC CONTROLLER

"G", "C", And "D" Frames
1 - 800 Horsepower

IMPORTANT
INFORMATION

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GEK-45167
Errata Sheet #1.0
July, 1991

GE Drive Systems

INSTRUCTIONS

QIII DC CONTROLLERS

This Errata Sheet affacts **GEK-45167**. It should be attached inside the front cover and retained as a portion of that publication. The publications text in **Section 12** should be changed in accordance with the following information:

Page 12-4, 12-5, and 12-6 -- QIII DC Controller Spare and Renewal Parts Charts

The referenced **horsepower ratings** and **part numbers** should be as shown in the revised charts on the following pages of this Errata Sheet.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "G" AND "C" FRAMES (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	109AG-	007	007	007	007	011	011	011	021
DC Power Fuse*	109AG-	005	005	007	007	009	015	015	015
Arm SCR Modules	125DC-	054	054	054	054	054	054	054	044
MA Contactor	105CB-	003	003	003	003	005	006	006	013
MA Contactor	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	105CB-	007	007	007	007	007	008	008	011
MA Aux. Interlock	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	118BA-	----	----	----	----	----	----	----	011

Part Description	Cat. # 104X-	Horsepower 50	60	75	100	125	150
AC Power Fuse	109AG-	021	021	021	027	027	027
DC Power Fuse*	109-	AG021	AG021	AG021	AJ019	AJ021	AJ025
Arm SCR Modules	125DC-	044	044	044	045	053	053
MA Contactor	105CB-	013	013	013	----	016	016
MA Contactor	105CA-	----	----	----	069	----	----
MA Aux. Interlock	105CB-	011	011	011	----	----	----
MA Aux. Interlock	105CA-	----	----	----	025	----	----
MA Aux. Interlock	118BA-	011	011	011	----	----	----

**Regenerative drives only.*

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "G" AND "C" FRAMES (HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	109AG-	007	007	007	007	007	007	007	007
DC Power Fuse*	109AJ-	007	007	007	007	007	009	009	013
Arm SCR Modules	125DC-	054	054	054	054	054	054	054	054
MA Contactor	105CB-	003	003	003	003	003	003	003	005
MA Contactor	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	105CB-	007	007	007	007	007	007	007	007
MA Aux. Interlock	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	118BA-	----	----	----	----	----	----	----	----

Part Description	Cat. # 104X-	Horsepower 50	60	75	100	125	150	200	250	300
AC Power Fuse	109AG-	011	011	015	021	021	021	027	027	027
DC Power Fuse*	109AJ-	013	014	015	015	017	017	019	021	025
Arm SCR Modules	125DC-	054	054	054	044	044	044	045	045	053
MA Contactor	105CB-	005	006	012	013	013	013	----	----	016
MA Contactor	105CA-	----	----	----	----	----	----	069	069	----
MA Aux. Interlock	105CB-	007	008	008	----	----	----	----	----	----
MA Aux. Interlock	105CA-	----	----	----	----	----	----	025	025	----
MA Aux. Interlock	118BA-	----	----	----	011	011	011	----	----	----

**Regenerative drives only.*

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "D" FRAMES (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 200	250	300
AC Power Fuse	109AG-	030	030	032
DC Power Fuse	109-	AG030	AG032	AJ025*
Armature SCRs	125DA-	152	152	154
DC Contactor	106AE-	008	009	----
DC Contactor	106MA-	----	----	G01
Armature SCRs With Leads	36A353476AX-	G02	G02	G03
Current Transformer Asm.	36A358218YA-	G02	G02	G02

** Two per drive required.*

NOTE: *On Regenerative Controllers with paralleled DC fuses, replace both fuses even if only one is blown.*

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "D" FRAME (HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 400	500	600	700	800
AC Power Fuse	109AG-	030	032	032	032	034
DC Power Fuse	109AJ-	026	025*	025*	025*	026*
Armature SCRs	125DA-	152	152	154	154	154
DC Contactor	106AE-	008	009	009	----	----
DC Contactor	106MA-	----	----	----	G01	G01
Armature SCRs With Leads	36A353476AX-	G02	G02	G03	G03	G03
Current Transformer Assembly	36A358218YA-	G02	G02	G02	G02	G02

** Two per drive required.*

NOTE: *On Regenerative Controllers with paralleled DC fuses, replace both fuses even if only one is blown.*

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1. SAFETY PRECAUTIONS

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.
WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.
CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.
NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - STRAIN HAZARD: Improper lifting practices can cause serious or fatal injury. Lift only with adequate equipment and trained personnel.

WARNING - ELECTRICAL SHOCK HAZARD: Circuit breakers, if supplied as part of the total system, may not disconnect all power to the equipment (see system elementary diagrams). Whether the AC voltage is grounded or not, high voltage to ground will be present at many points.

WARNING - ELECTRICAL SHOCK AND BURN HAZARD: When using instruments such as oscilloscopes to work on live equipment, the oscilloscope's chassis should be grounded and a differential amplifier input should be used. Care should be used in the selection of probes and leads and in the adjustment of the oscilloscope so that accurate readings may be made. See instrument manufacturers instruction book for proper operation and adjustments to the instrument.

WARNING - ELECTRICAL SHOCK HAZARD: Some controllers are furnished with partial enclosures open at the top and bottom. These are intended only for mounting in another enclosure or in a control room having access by qualified personnel only.

WARNING - FIRE AND EXPLOSION HAZARD: Fires or explosions might result from mounting drive controllers in hazardous areas such as locations where flammable or combustible vapors or dusts are present. Drive controllers should be installed away from hazardous areas, even if used with DC motors suitable for use in these locations.

WARNING - ELECTRICAL SHOCK HAZARD: All motor bases and equipment enclosure housings should be connected to the factory or facility earth grounding systems.

WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT HAZARD: Ensure safety of personnel in vicinity of controller and controlled equipment before operation. It is the responsibility of the user to insure that operation does not result in an unsafe condition. Factory provided interlocks and operating limits should not be bypassed or modified.

WARNING - DRIVE MISOPERATION HAZARD: Care should be exercised in the use of SLD contacts in external circuitry due to "contact bounce" during power-up and initialization that may have an adverse impact on critical or sensitive circuitry.

CAUTION: Do not remove connections or printed circuit cards from the controller while power is applied. This can damage the equipment.

CAUTION: Do not remove input power from the drive until it has fully executed a stop sequence as this can damage the drive system.

CAUTION: Installation wiring must be in accordance with the National Electrical Code and be consistent with all local codes. Secondaries of the 115 volt control transformers typically have one side fused and the other grounded or available for grounding by the user.

CAUTION: The elementary diagrams supplied with the drive controller identify some signal connections as "low level analog and/or digital signals" and indicate special wiring practices. These signals should be isolated from all other wiring (particularly any power wires) and run in a separate conduit or wire run to prevent possible misoperation of the controller.

CAUTION: Meggering can damage electronic components. Do not megger or Hi-Pot the drive system without consulting the supplier of the QIII DC Controller.

CAUTION: Do not connect any external circuits other than those shown on the elementary diagram. Connection of devices such as ammeters on the shunt or voltmeters on the tachometer may degrade the performance of the drive system.

CAUTION: Do not use power factor correction capacitors with this equipment without consulting the supplier of the QIII DC Controller. Damage may result from high voltages generated when capacitors are switched.

CAUTION: If drive is **speed regulated** (tachometer feedback), overspeed protection will only be properly set if the optional Encoder Process Interface card is properly configured. If drive is **CEMF regulated** (no tachometer), the minimum field current must be properly identified to correctly set field loss and avoid overspeed conditions.

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow the start-up procedures step by step.*

NOTE: *Read and heed all* ***WARNING, CAUTION*** *and* ***NOTE*** *labels posted on the equipment.*

2. RECEIVING AND STORAGE

RECEIVING

The controller should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from the supplier of the QIII DC Controller. When seeking assistance please use **serial number, requisition number, model number** and **drive code** to identify the equipment.

STORAGE

Hardware:

If this controller is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from -20° C (-4° F) to 55° C (131° F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage enviroment exceeds a 15° C (27° F) drop in temperature at 50% humidity over a four hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes can also cause condensation.

Software:

Software provided on magnetic media, such as floppy disks and magnetic tape, should be stored in a clean dry location at ambient temperatures of 10° C to 55° C (50° F to 125° F). Care should be taken to protect the media from magnetic fields, high heat or humidity, and physical damage.

NOTES:

3. INSTALLATION GUIDELINES

LOCATION

QIII DC Controllers are suitable for most areas where industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0° C (32° F) to 40° C (104° F) and relative humidities up to 90%. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20° C (68° F) and 30° C (87° F). Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

WIRING LEVELS

1. Wiring should be run in separate conduits or wireways for signal, control and power wiring levels.
   a. Signal
      -- Low level analog and digital signals
      -- Speed and/or position signals
      -- Power supplies and logic signals
      (See system schematics and notes to identify signal level wires)
   b. Control
      -- AC or DC control circuits
   c. Power
      -- Field leads or armature leads
      -- Brakes, 115 VAC circuits

2. Signal wiring and power wiring may cross at right angles with a **minimum** one-inch separation. Avoid parallel runs between signal level wires and power or control wires. If signal wires must be run in parallel to control or power wires, a **minimum** of a four-inch separation must be maintained between the wires.

GROUNDING

1. Drive common (PCOMX) should be grounded at **only one** point. If the reference is supplied by numerical control or process instrument with a grounded common, the drive common should not be grounded separately.

2. If an isolation transformer is used and its neutral must be grounded, it is recommended that a high resistance ground be used.

3. Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

4. Proper case grounding should utilize wire sizes and crimp-type terminals per the following chart:

Circuit Rating (Amps, Max.)	Copper Ground Wire AWG	AMP Terminal Number	AMP Tool Number
20	14	34120	49900
60	10	33457	49900
100	8	33458	69062
200	6	33465	69062
300	4	31811	69062
500	2	320383	69062
800	1/0	36917	69120
1000	2/0	36923	69120
1200	3/0	36925	69120
1600	4/0	321878	69120

NOTE: *"G" frames would use 4 AWG, "C" frames would use 2 AWG, and "D" frames would use 2/0 AWG.*

COMMONS

Connection of reference, meter, or any other external commons can only be made at 3TB-PCOMX, unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, solenoids, or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

-- Incoming AC line connections
-- Power connections to motor armature and motor field
-- 1TB, 2TB, 3TB, and 4TB (if furnished) terminal board connections

ADDITIONAL CONSIDERATIONS

Attention should be given to the National Electrical Code and any applicable local codes when installing any drives. Wire size and insulation type, conduit sizing, enclosures, etc., should be determined per these codes.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dust, dirt, or foreign matter.
2. Vibration or shock.
3. Moisture or vapors.
4. Rapid temperature excursions or high ambient temperatures.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference or "**noise**" introduced by:
   a. Radio frequency signals, typically from portable transmitters used in the vicinity of the equipment or its wiring.
   b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay, contactor, or brake coils as a part of, or in the vicinity of, drive control circuits.

MOUNTING

If mounting the QIII DC Controller in an enclosure, attention should be given to the following:

-- Totally enclosed controllers should be positioned to permit heat radiation from all surfaces.
-- A wall-mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the door may be fully opened for easy access.
-- Wall-mounted controllers may be mounted on any firm, reasonably flat, vertical surface.

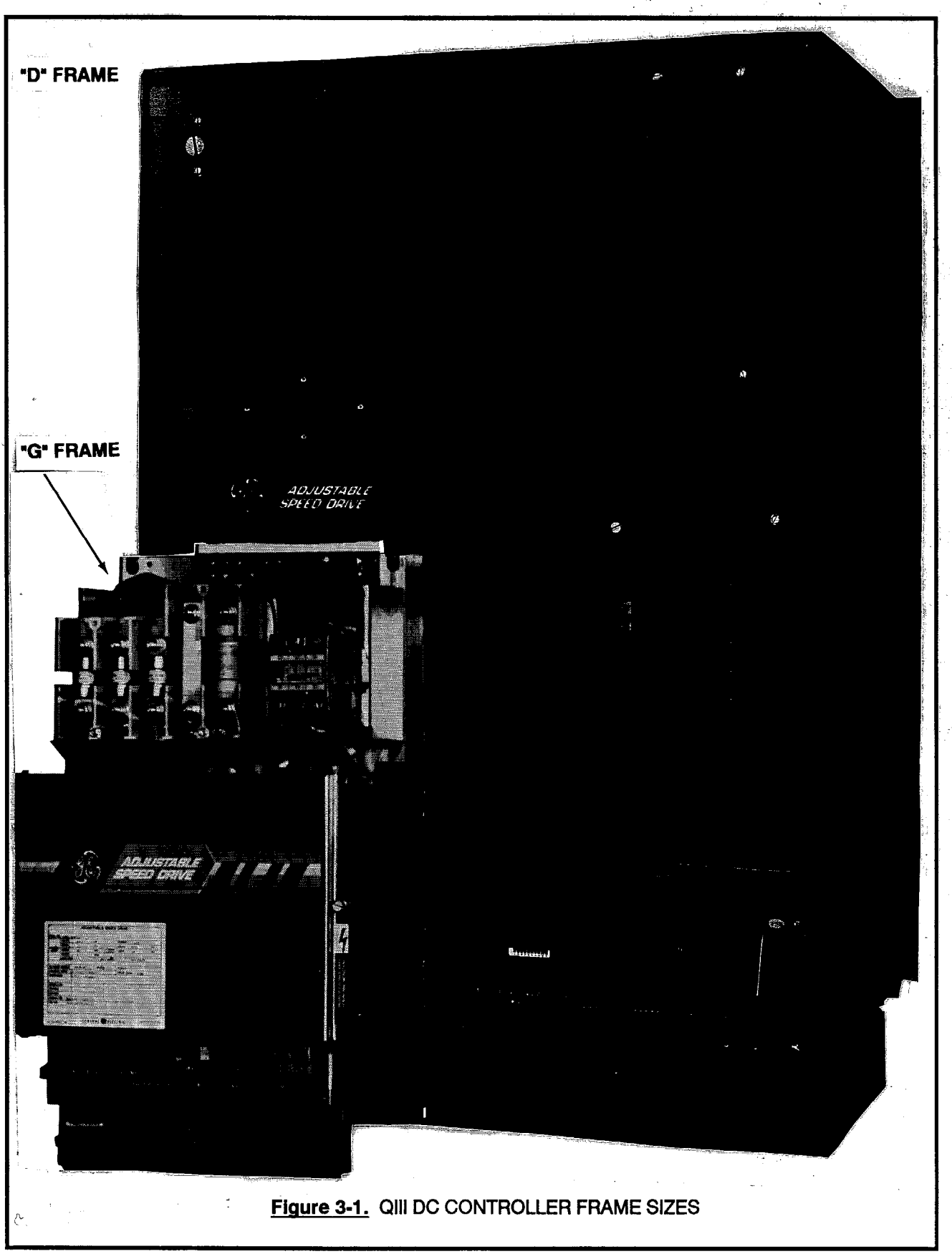


Figure 3-1. QIII DC CONTROLLER FRAME SIZES

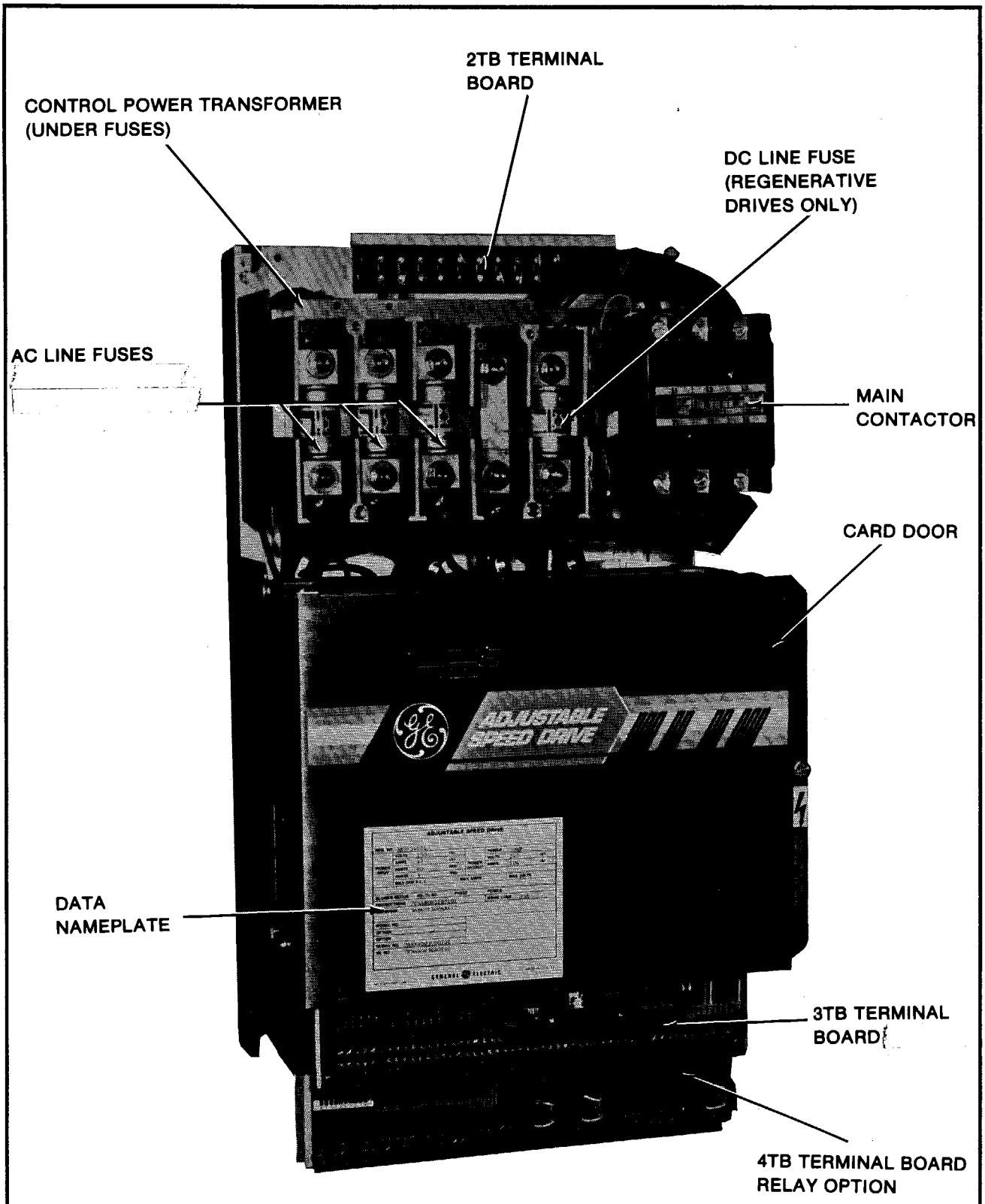


Figure 3-2. QIII DC CONTROLLER, WALL MOUNTED OPEN CHASSIS CONSTRUCTION

4. MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS

GENERAL DESCRIPTION

The QIII DC Controller is an adjustable speed, digitally controlled, programmable drive utilizing microprocessors to provide digital regulation. Drive parameters are stored in EEROM memory, and therefore, are accurate and drift free.

Figure 3-1 shows the three versions or frame sizes that make up the 1-800 HP range of the QIII DC Controllers covered in this instruction book. The **G frame** is rated up to 75 HP, the **C frame** to 300 HP, and the **D frame** to 800 HP (ratings given are for 460 VAC).

This instruction book covers the G, C, and D frame QIII DC controllers, as these have hardware and software commonality. Attention is drawn to any variations between these frame sizes where necessary.

The QIII DC controller provides over 80 diagnostic messages which can be decoded via a LED diagnostic readout. Diagnostic messages can also be read via an optional coded 8-wire output or over the optional RS422/232C serial communication link. Drive operating parameters stored in the EEROM may also be examined and changed via the RS422/232C serial link. In addition, the QIII DC controller provides diagnostics which provides a power-up self check down to the printed circuit board level and can check SCRs by independently initiated operation.

The QIII DC controller contains four printed circuit cards (Control card, Power Supply card, Power Connection card, and 3TB Terminal Board card) for standard operation and offers optional cards and terminal boards for additional functions. Optional card kits include the Application card, Encoder Process Interface card, 4TB Terminal Board card, and Serial Link Interface card. Other optional hardware includes a BCD Option Kit and Balun Option Kit.

The **Control card** provides an outer speed regulator loop with an inner current regulator. In situations where speed accuracy is not a requirement, the speed regulator can be converted to a CEMF regulator. Individually timed linear acceleration and deceleration is adjustable over rates of 0 (current limit acceleration) to 999.9 seconds. Optional "S-Curve" timing is also available.

Output metering signals for armature current, armature voltage, field current, speed, torque, and horsepower are also available from the QIII DC controller. In addition, other signals can be brought out as analog signals via one of three optional digital-to-analog converters or the optional RS422/232C serial link. (These signals and others can be programmed to activate optional programmable signal level detectors.)

The block or "flow" diagram (Figure 4-1) shows the operation of a basic QIII DC controller. Power is fed through the AC fuses, current transformers (regenerative controllers only), and MA ("G" and "C" frames) contactor, and enters the power conversion SCR modules where it is converted to adjustable DC voltage. DC current is fed through a shunt and a DC link fuse (regenerative controllers only) to the DC motor armature.

Analog voltage (or optional current) or digital inputs can be used as the references. Likewise, a wide variety of tach generators can be interfaced to the drive, including both AC or DC analog tachs or pulse tachs with frequency rates up to 40 KHz (410 KHz if optional Encoder Process Interface card is furnished).

POWER CIRCUITS

(See Figures 4-2 thru 4-11)

The motor field control (MFC) may be provided by an internal 10 amp maximum field supply (for "G" and "C" frames, Figure 4-2) or an external field supply rated to 24 amps ("D" frames, Figure 4-3).

For the 10 Amp MFC, the Power Supply card contains the field supply fuses (CFU1, CFU2 and CFU3), metal oxide varistors (MOVs), field current feedback transformer (T3), and line reactor (L4). The field power module is mounted on the main controller heatsink.

The 24 Amp external MFC provides a separate heatsink with field supply fuses (CFU1-1, CFU2-1 and CFU3-1), line reactor (L4), field power module and MFC Suppression card containing the MOVs and field current feedback transformer (T1).

NOTE: *There is a variation in fuse and transformer designations between the two motor field control versions.*

The operation of the MFC may be best understood by reading the following while referring to the appropriate figure (Figure 4-2 or Figure 4-3). Field power is supplied through fuses CFU1, 2 and 3 (CFU1-1, 2-1 and 3-1 on the external MFC). MOVs provide protection from line spikes and transients.

The power to the motor field control is "two-thirds" wave rectified via the field power module. The SCR in the field power module controls field current as follows:

-- When AC line L1 is high with respect to L2 and L3 the SCR is fired and current passes through the diode between DX and DW to the motor field. The circuit is completed through the SCR and returns through the line reactor to AC line L3.
-- When L2 becomes high, current flows through the diode between DY and DW to the field and back through the SCR, line reactor and L3.
-- When L3 becomes high, the SCR is reversed biased and the energy stored in the field winding circulates current through the diode between DZ and DW (acting as a free wheeling diode) until L1 becomes high again and the SCR is fired.

Field current is sensed by the current feedback transformer (T1 or T3 depending on the MFC rating) and fed into the field gain select circuit on the Power Supply card (through FCPL if an external MFC) and to the Control card.

The Power Supply card provides input voltage to the control power transformer (CPT) through CFU4 and CFU5 via terminals CPT H1 and CPT H2. RT1, RT2, and RT3 are the AC line signals to the Control card that synchronize SCR firing with line voltage.

Fuses CFU6, CFU7 and CFU8 are located on the Power Supply card. The internal DC power supplies are protected by CFU6 and CFU7. The 115V AC internal supply is protected by CFU8. These circuits are fed from the control power transformer through plug CPTPL.

The SCR Power Conversion Bridges for regenerative and non-regenerative drives, G, C, and D frames, are shown in Figures 4-4 thru 4-7. Three-phase power is delivered to the conversion bridge through AC line fuses (FU1-3). Two line current transformers (CT1 and CT3) provide a current feedback signal for commutation failure indication (regenerative drives only) and the MA ("G" and "C" frames) or MD ("D" frames) contactor and reactors (regenerative drives only) provide protection against AC line transients.

RC snubber networks, with capacitors located on the Power Connection card and resistors mounted on the heat sink, provide protection for the SCRs and are connected across a forward/reverse SCR pair on regenerative drives or forward SCR on non-regenerative drives. The SCR firing pulses are generated through pulse transformers located on the Power Connection card in the firing order 1-6-2-4-3-5 for both regenerative and non-regenerative drives, forward or reverse.

DC power is supplied to the armature through the shunt and DA1 and returns through DA2 (which is fused on regenerative drives only). An RC network across P1 and DA2 aids latching of the SCRs. The voltage feedback signal is in parallel with the snubber and current feedback is derived from the shunt. A thermal switch is also mounted on the heatsink.

AC POWER REQUIREMENTS

The QIII DC controller is available in either 230 VAC +10%, -5% (1 - 300 HP) or 460 VAC +10%, -5% (2 - 800 HP), three-phase power configurations. Nominal line frequency is 60 Hz with a tolerance of $\pm$2 Hz.

AC LINE FUSES (FU1-3)

AC line fuses provide short circuit protection for the SCRs and the wiring inside the controller. Additional overload protection is provided by a timed overcurrent function and motor thermostat.

FERRITE CORE ASSEMBLY (REACTORS)

The ferrite core assembly is provided on the AC power lines between the AC line fuses and the SCRs to protect the SCRs from possible misoperation due to transient currents and voltages during operation.

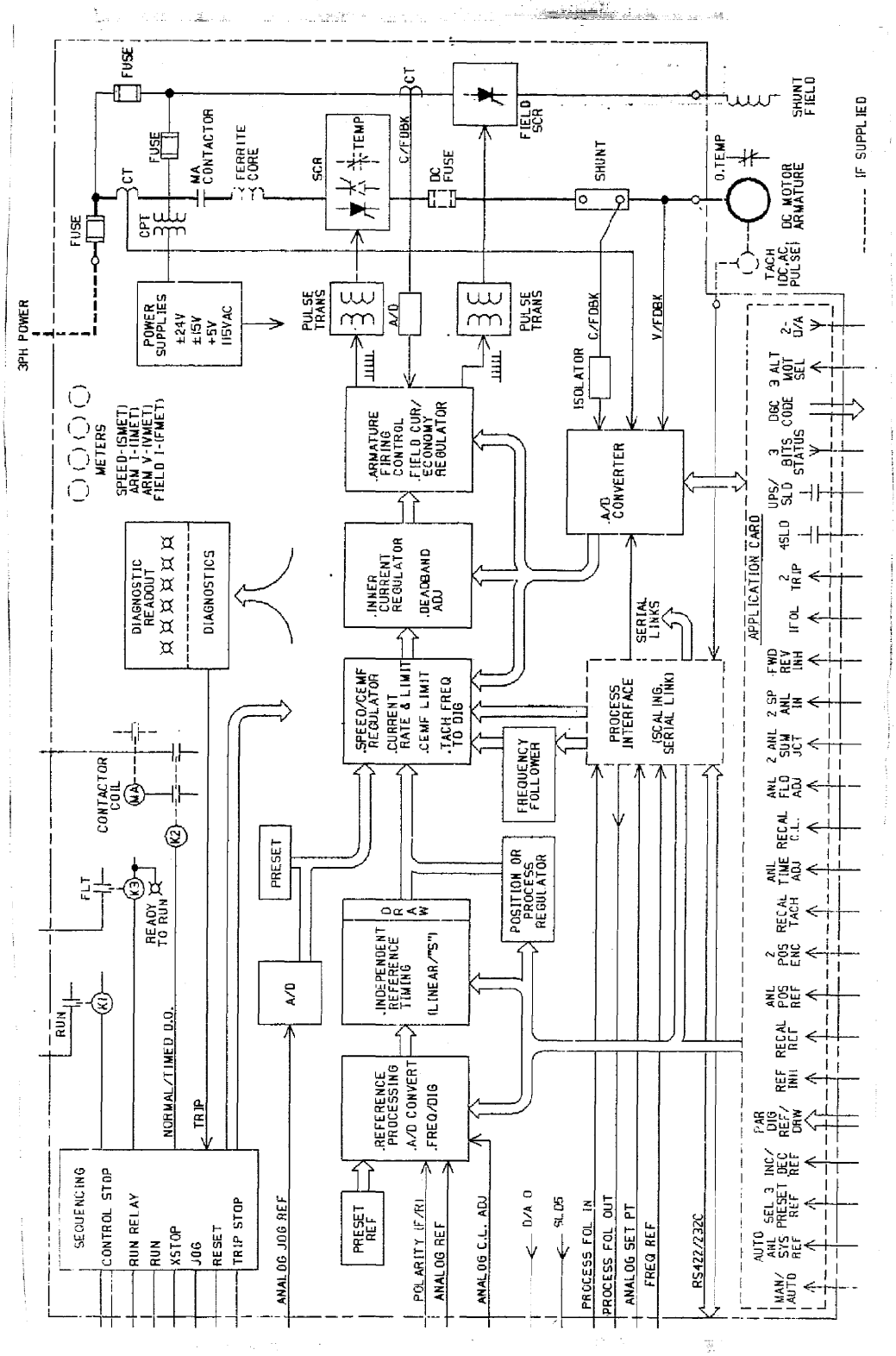


Figure 4-1. QIII DC CONTROLLER BLOCK DIAGRAM

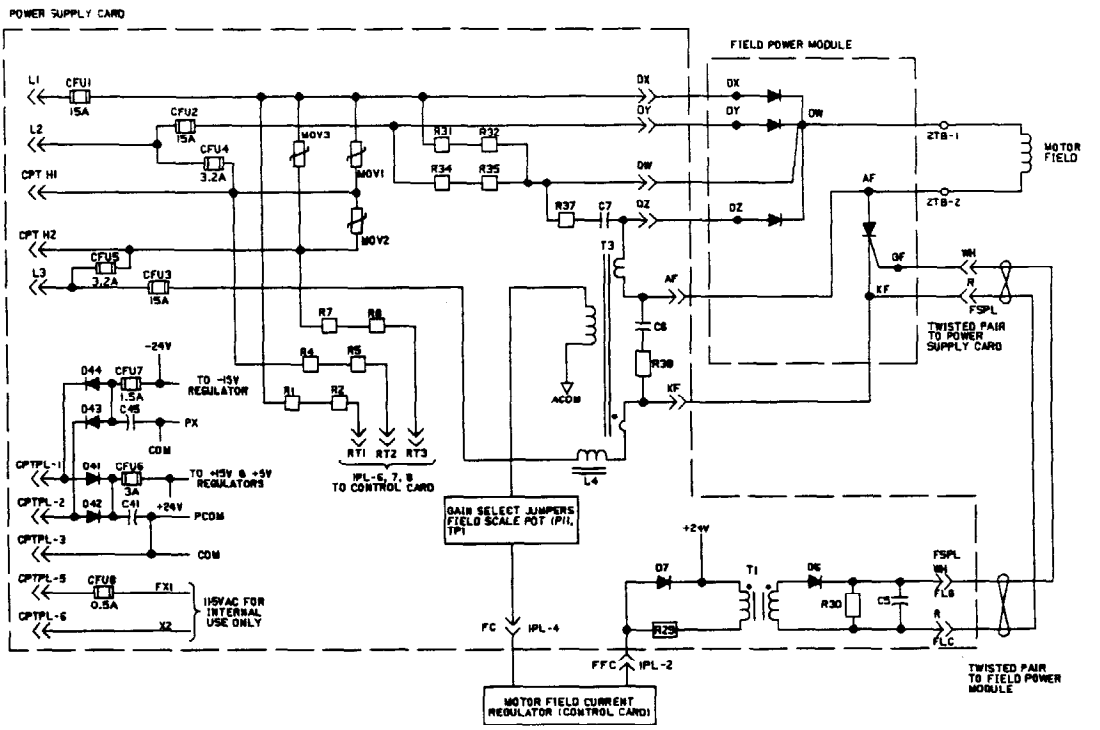


Figure 4-2. MOTOR FIELD CONTROL DIAGRAM (10 AMP)

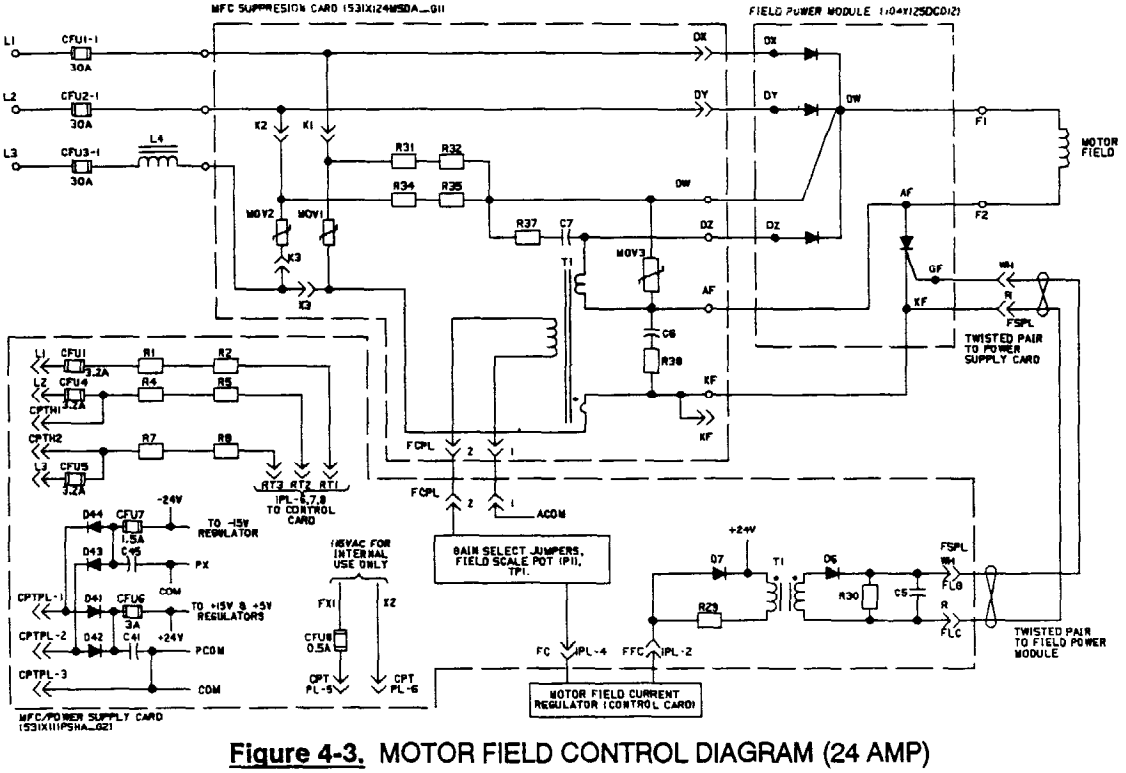


Figure 4-3. MOTOR FIELD CONTROL DIAGRAM (24 AMP)

MAIN CONTACTOR

(MA, "G" and "C" Frames)
The MA contactor provides a controllable disconnect between the AC power line and the power conversion bridge. The contactor picks up when running mode is selected and no faults exist in the drive. The MA contactor drops out under the following conditions:

1. When "STOP" is commanded on **non-regenerative drives** (coast stop via "Control On", 1 or 4 quadrant).
2. When "STOP" is commanded and the motor slows to near zero speed on **regenerative drives.**
3. When a fault condition occurs.

(MD, "D" Frames)
The MD contactor provides a controllable disconnect between one side of the DC motor and the power conversion bridge. The contactor picks up when running mode is selected and no faults exist in the drive. The MD contactor drops out under the following conditions:

1. When "STOP" is commanded on **non-regenerative drives.**
2. When "STOP" is commanded and the motor slows to near zero speed on **regenerative drives.**
3. When a fault condition occurs.

DC LINE FUSE (FU4)

(Regenerative Controllers Only)
A DC line fuse is provided in the DC output line to the motor armature and protects the SCRs and motor from overcurrent sourced from the motor.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuses (CFU4 and CFU5) on the Power Supply card. The secondary winding provides two isolated voltages:

1. 115 VAC to operate the coil of the main contactor (MA or MD), RUN (K1) Relay, and controller cooling fan (if required).
2. 38 VAC (center tapped) for the Power Supply card to generate the DC control voltages necessary for controller operation.

SHUNT (SH)

The shunt provides the armature current feedback signal to the control circuit. A nominal 100mV output signal is generated at the current rating stamped on the shunt.

CURRENT TRANSFORMER ASM (CT)

The current transformer assembly is used to provide AC line current feedback signals (regenerative controllers only) that are used for circulating current fault detection.

CONNECTION DIAGRAMS

(See Figures 4-8 thru 4-11)
The connection diagrams for regenerative and non-regenerative controllers are shown in Figures 4-8 thru 4-11. These graphically describe the internal connections for all plugs and cables. These diagrams aid control hook up as well as troubleshooting when connection integrity of one of the cables is in question. The functions of the connectors are described in the "Electrical Connection" section.

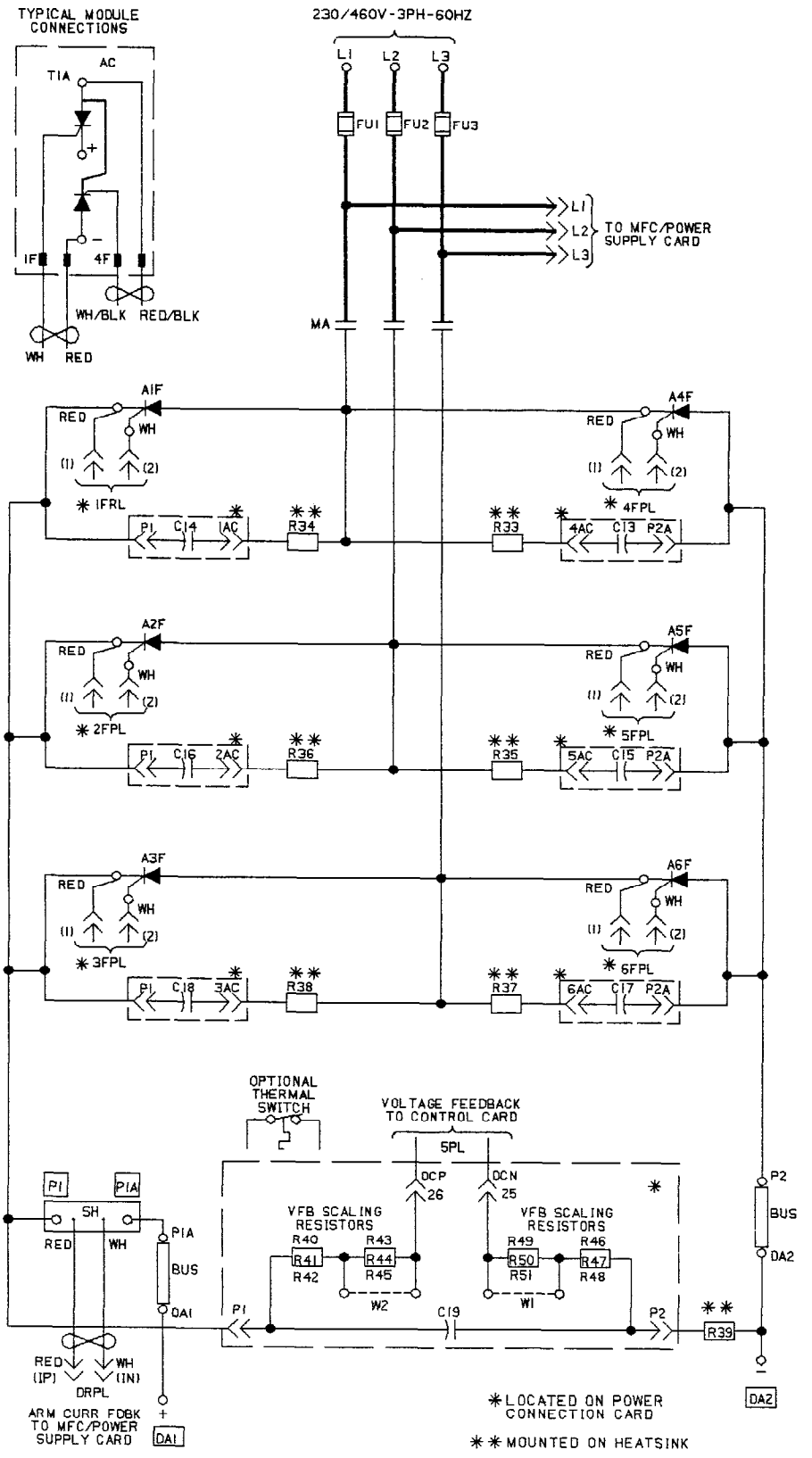


Figure 4-4. NON-REGENERATIVE SCR POWER CONVERSION BRIDGE (G, C, Frames)

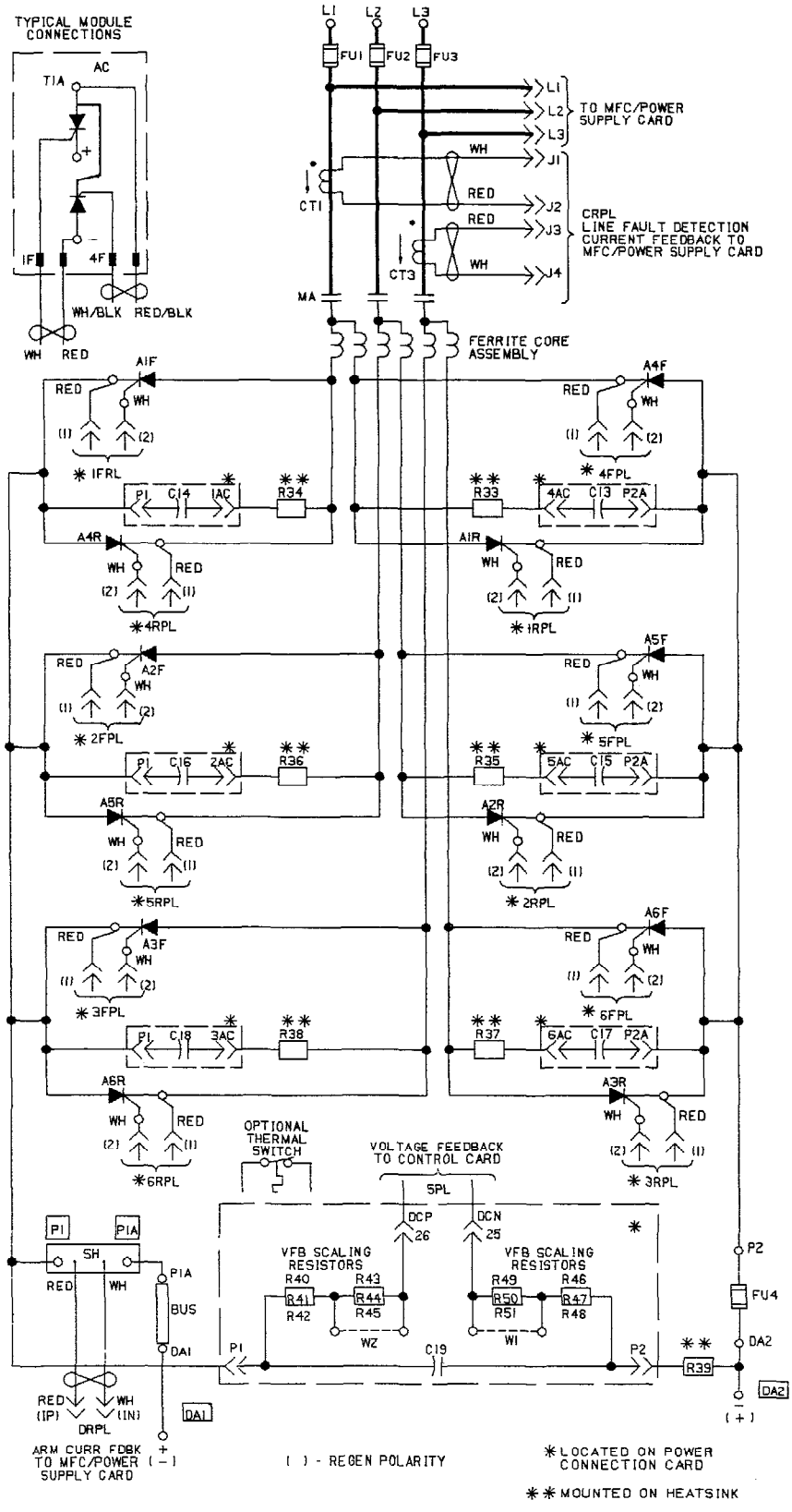


Figure 4-5. REGENERATIVE SCR POWER CONVERSION BRIDGE (G, C Frames)

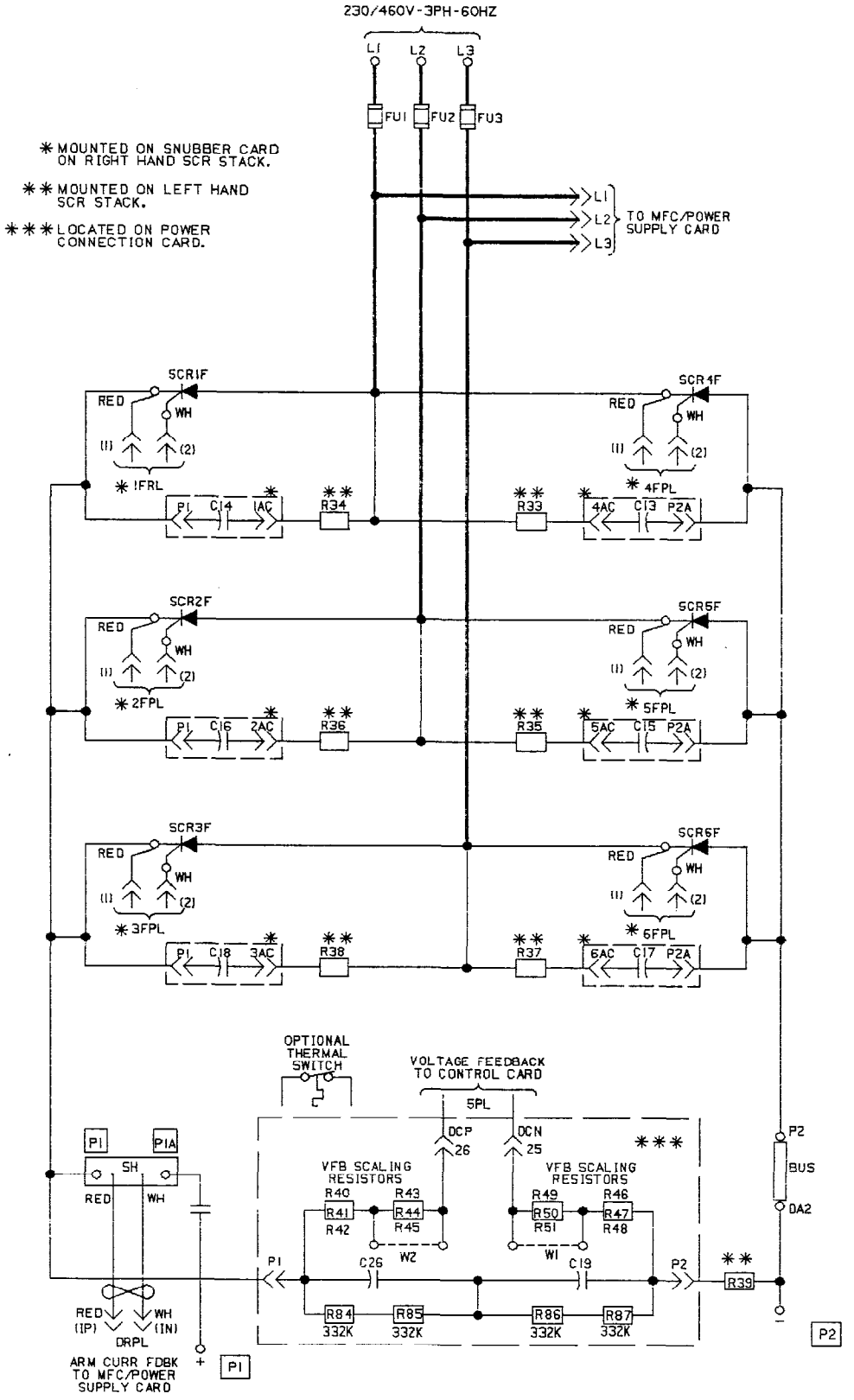


Figure 4-6. NON-REGENERATIVE SCR POWER CONVERSION BRIDGE (D Frames)

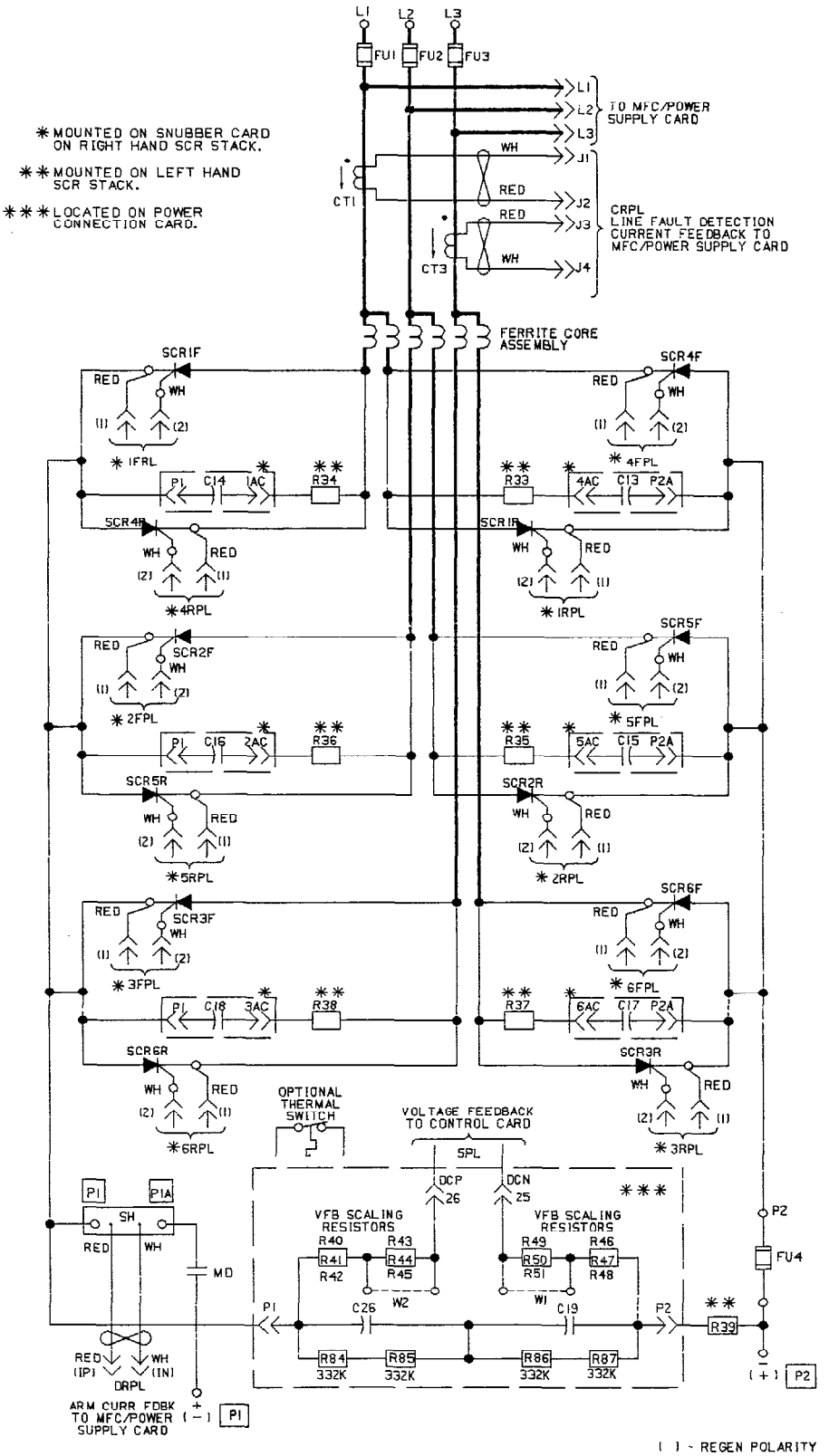


Figure 4-7. REGENERATIVE SCR POWER CONVERSION BRIDGE (D Frames)

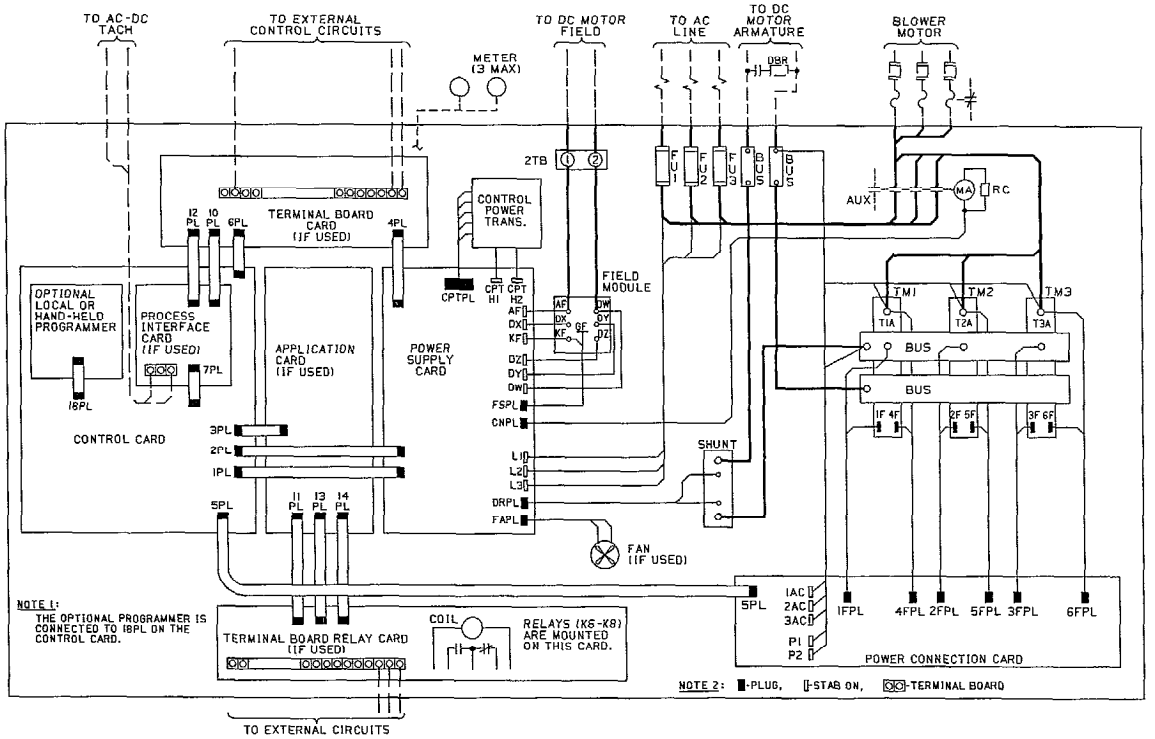


Figure 4-8. CONNECTION DIAGRAM, NON-REGENERATIVE (G, C Frames)

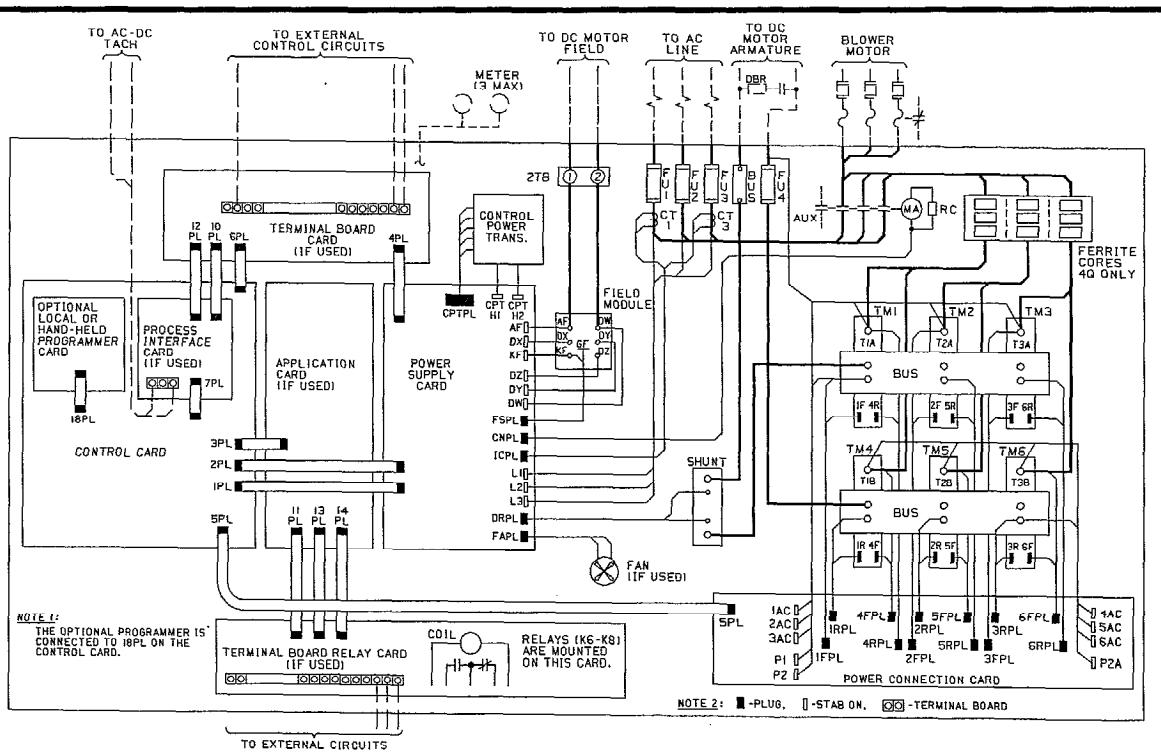


Figure 4-9. CONNECTION DIAGRAM, REGENERATIVE (G, C Frames)

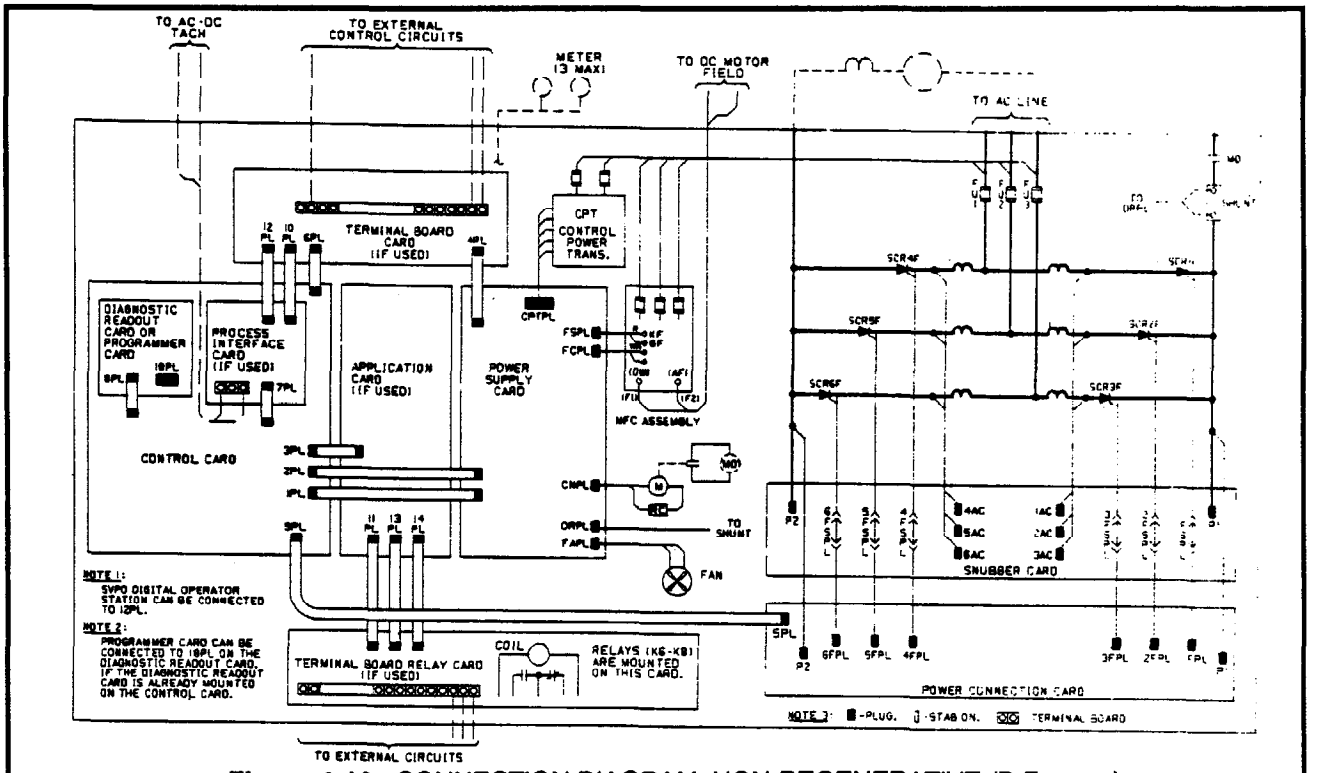


Figure 4-10. CONNECTION DIAGRAM, NON-REGENERATIVE (D Frames)

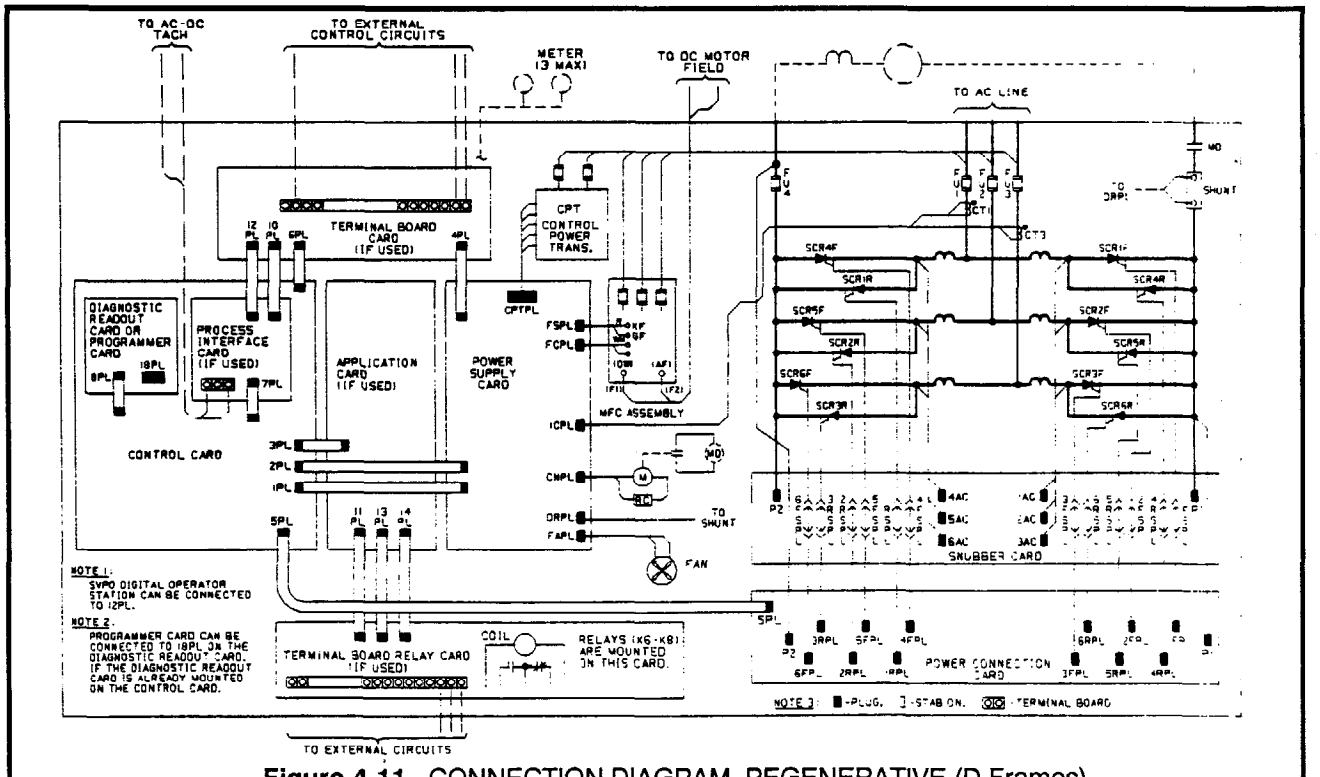


Figure 4-11. CONNECTION DIAGRAM, REGENERATIVE (D Frames)

CONTROL CARD

(See Figure 4-12)

This card provides the intelligence for the QIII DC controller. It contains three microprocessors and performs most of the regulating functions, SCR burst firing, diagnostic, and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented in hardware and/or software.

The control card (Figure 4-12) contains six Diagnostic LEDs to display the status of the QIII DC controller. The LEDs that are lit are interpreted as a binary number indicative of the fault. See the Troubleshooting section for listing of Fault Codes. Note that during proper controller operation, the LEDs will shift sequentially right to left when the drive is stopped and sequentially right to left in pairs when the drive is running.

MFC/POWER SUPPLY CARD

(See Figures 4-13 and 4-14)

This power supply card provides the necessary +5 VDC, ±15 VDC, and ±24 VDC power used throughout the controller. The armature current feedback isolation circuit as well as three relays, "RUN" (K1), "MAX" (K2), and "FAULT" (K3) are also on this card. Additionally, the motor field control is provided by this card (G3 version, "G" and "C" frames). This version card is used with motors requiring field current up to 10 amps and contains the pulse transformer, current transformer, choke, and fuses for the motor field circuitry as previously described. The G2 version card is used in conjunction with an external motor field control assembly (Figure 4-15) for field current up to 24 amps (for "D" frames).

The external motor field control assembly (Figure 4-15) is used with the version G2 MFC/Power Supply card for controllers requiring up to 24 amps for the motor field (for "D" frames). The assembly contains the fuse (CFU1-3), power module, current feedback transformer, and choke.

NOTE: *The version G2 MFC/Power Supply card and 24 Amp motor field control assembly are used only for "D" frame controllers.*

POWER CONNECTION CARDS

(See Figure 4-16 and Figure 4-17)

NOTE: *Four different version Power Connection Cards are used in the QIII DC Controller depending on the frame size and application. They are as follows:*

4 quadrant, w/snubbers, "G" and "C" Frames ..... 531X308PCSA_G1

1 quadrant, w/snubbers, "G" and "C" Frames ..... 531X308PCSA_G2

4 quadrant, w/o snubbers, "D" Frames ..... 531X308PCSA_G3

1 quadrant, w/o snubbers, "D" Frames ..... 531X308PCSA_G4

("G" and "C" Frames)

These cards contain SCR and DC output snubber capacitors and the gate pulse transformers for the SCRs. The DC voltage feedback isolation resistors are also mounted on this card. SCR and DC output snubber resistors are located on the heat sink.

("D" Frames)

These cards contain the gate pulse transformers for the SCRs and DC voltage feedback isolation resistors. They are used in conjunction with an additional Snubber card (see Snubber Card paragraph below).

SNUBBER CARD ("D" FRAMES)

(See Figure 4-18)

This card contains the SCR and DC output capacitor snubbers and is located on the right-hand SCR stack assembly. The snubber resistors are located on the heat sinks on the left-hand SCR stack.

3TB TERMINAL BOARD CARD

(See Figure 4-19)

This card is used to convert the input and output signals of the QIII DC Controller from ribbon cable connections to discrete terminals via terminal board points. Additional functions provided are a programmable signal detector relay and serial communications connector (D-type) with set-up switches.

METAL OXIDE VARISTORS (MOVs)

On **"G" and "C" frames,** the MOVs are mounted in a potted assembly located under the 3TB Terminal Board bracket. On **"D" frames,** the MOV assembly is located on the back of the door.

OPTIONAL APPLICATION CARD

(See Figure 4-20)

This optional card provides a combination of analog interfaces between the Control card and external devices. These interfaces allow the core drive to be expanded to handle optional functions required in certain applications. The card provides the following optional functions:

- Analog Inputs
- Digital Inputs
- Sixteen-Bit Parallel Word Input
- Five Programmable Signal-Level-Detector Relays (SLDs)
- Seven-Bit Digital Fault Status Output
- Three Programmable Status Bit Outputs
- Two Analog Outputs

WARNING - DRIVE MISOPERATION HAZARD: Care should be exercised in the use of SLD contacts in external circuitry due to "contact bounce" during power-up and initialization that may have an adverse impact on critical or sensitive circuitry.

OPTIONAL ENCODER PROCESS INTERFACE CARD

(See Figure 4-21)

This optional card provides a mixture of analog and digital interfaces between the Control card and the equipment. The optional functions on this card are:

- Analog (AC or DC) tachometer interface.
- Analog speed meter driver (analog tachs only).
- Digital (square wave) tachometer interface.
- Process Follower (current or voltage reference) interface.
- RS232C and RS422 serial link interface.

CAUTION: If drive is **speed regulated** (tachometer feedback), overspeed protection will only be properly set if the optional Encoder Process Interface card is properly configured. If drive is **CEMF regulated** (no tachometer), the minimum field current must be properly identified to correctly set field loss and avoid overspeed conditions.

OPTIONAL 4TB TERMINAL BOARD CARD

(See Figure 4-22)

This optional card provides convenient termination points for external devices to interface to the controller's. optional Application card. In addition, three 115 VAC relays are available on this card.

SERVICE AND PARTS INSTRUCTION LABELS

(See Figure 4-23)

Each drive contains, inside the door, a multi-color label similar in appearance to that shown in Figure 14. This label provides information on connection point and component locations essential in connecting or troubleshooting the drive. The label contains the following information in color-coded format:

- Card location
- Thyristor module locations
- Fuse information and nomenclature
- Plug locations
- Incoming control connections at 3TB and 4TB
- Power connections

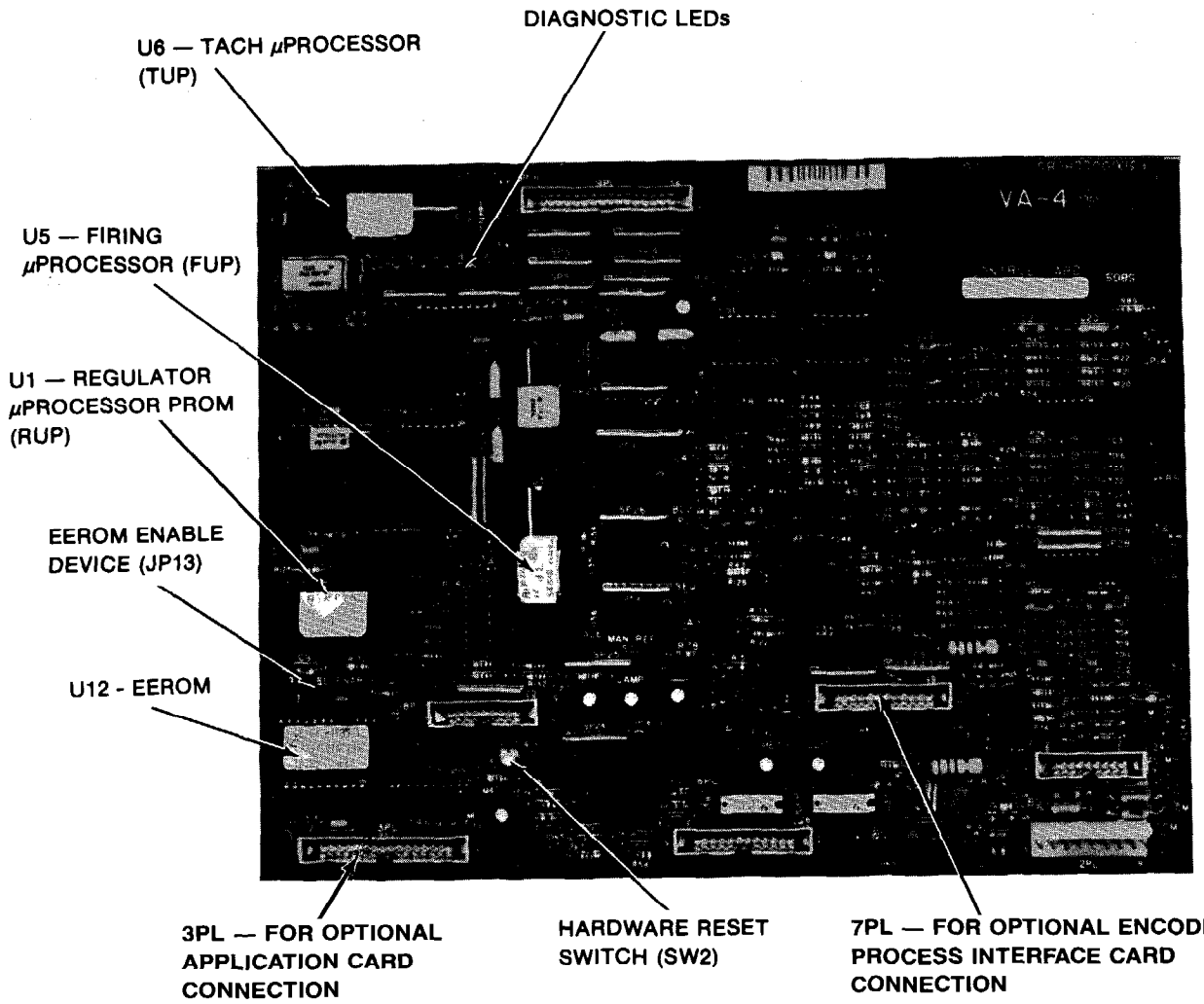


Figure 4-12. CONTROL CARD

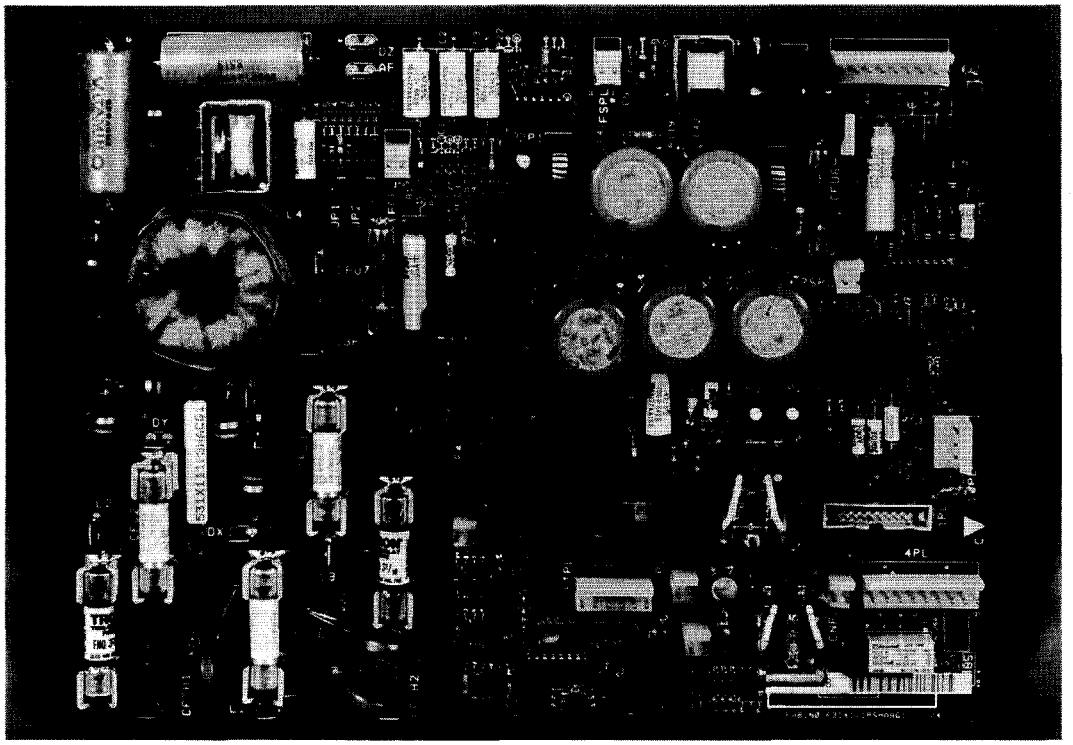


Figure 4-13. MFC/POWER SUPPLY CARD (Version G3)

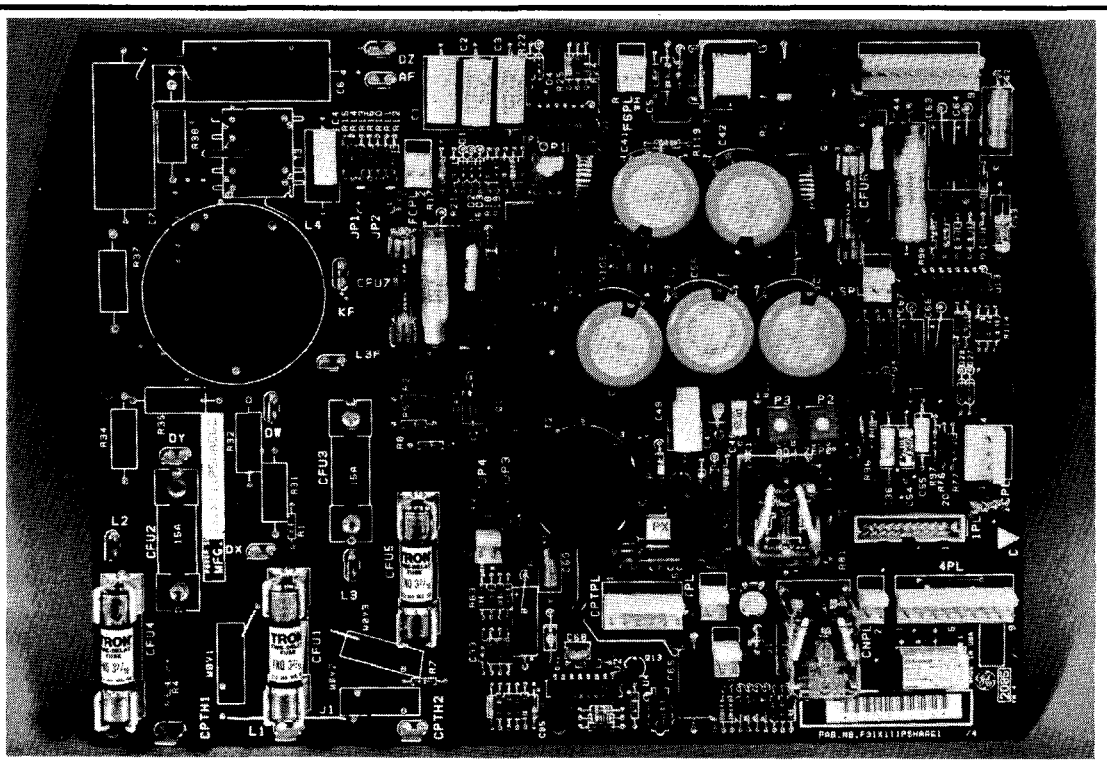


Figure 4-14. MFC/POWER SUPPLY CARD (Version G2 for "D" Frame, Requires External MFC Asm.)

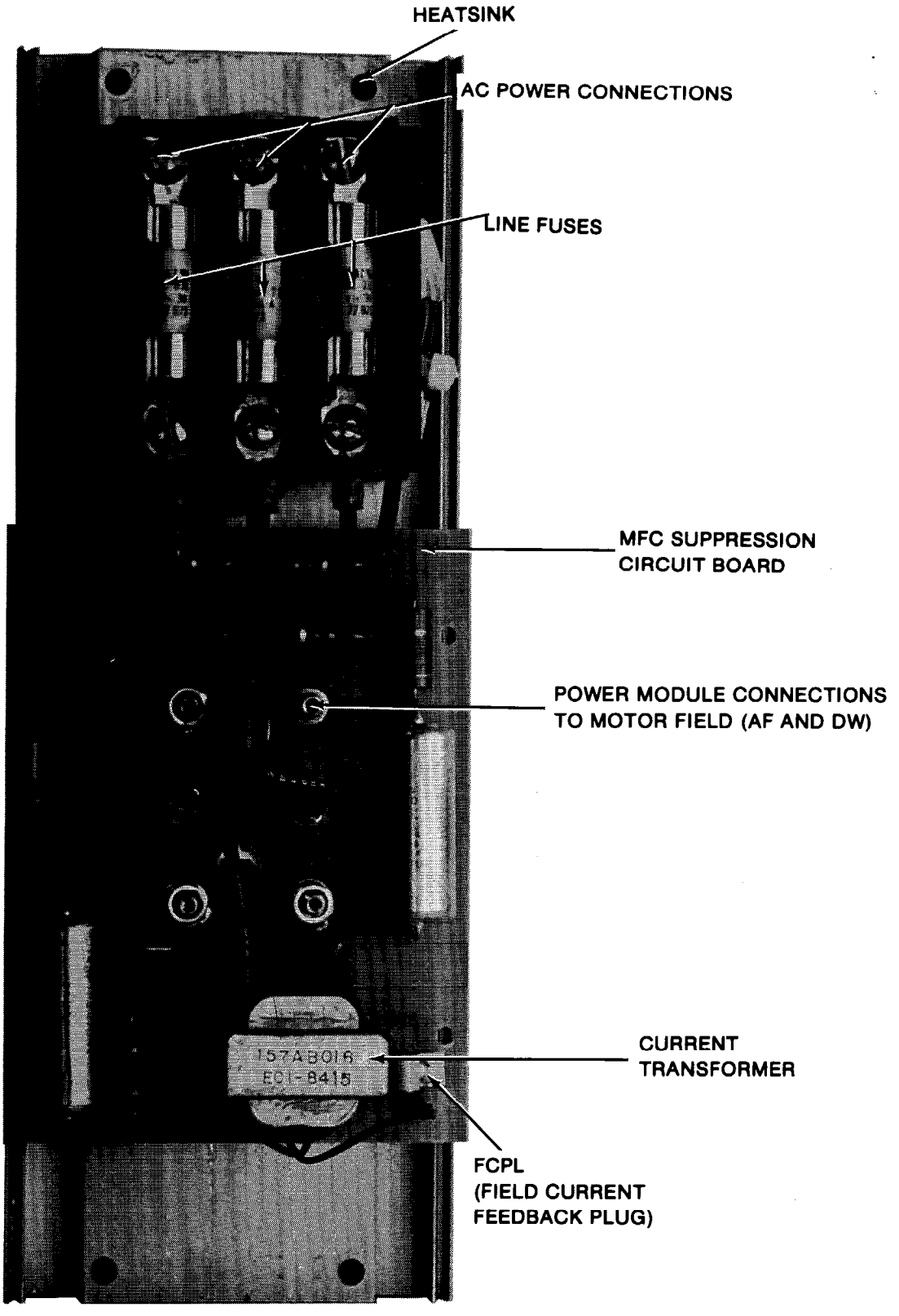


Figure 4-15. EXTERNAL MOTOR FIELD CONTROL ASSEMBLY

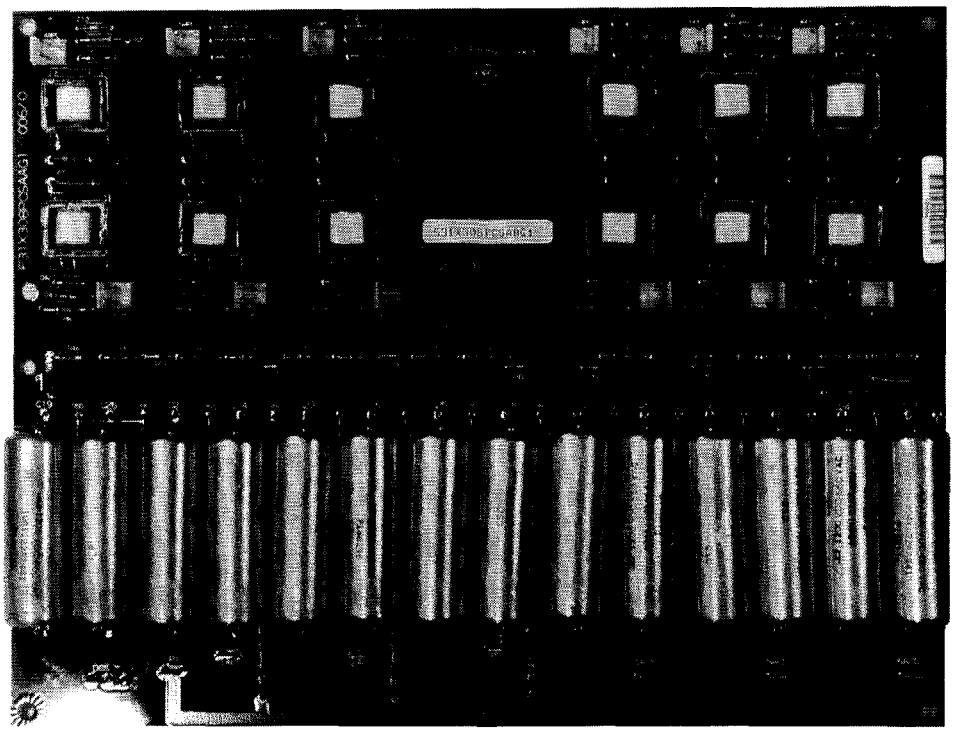


Figure 4-16. POWER CONNECTION CARD (Version G1, G2 - With Snubbers)

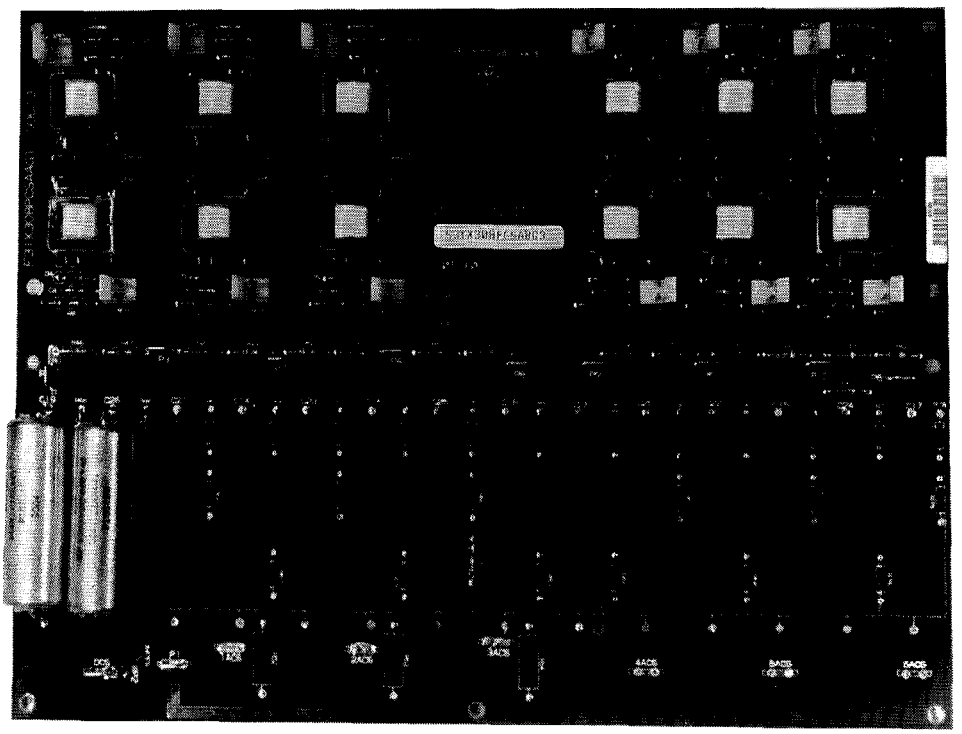


Figure 4-17. POWER CONNECTION CARD (Version G3, G4 - Without Snubbers)

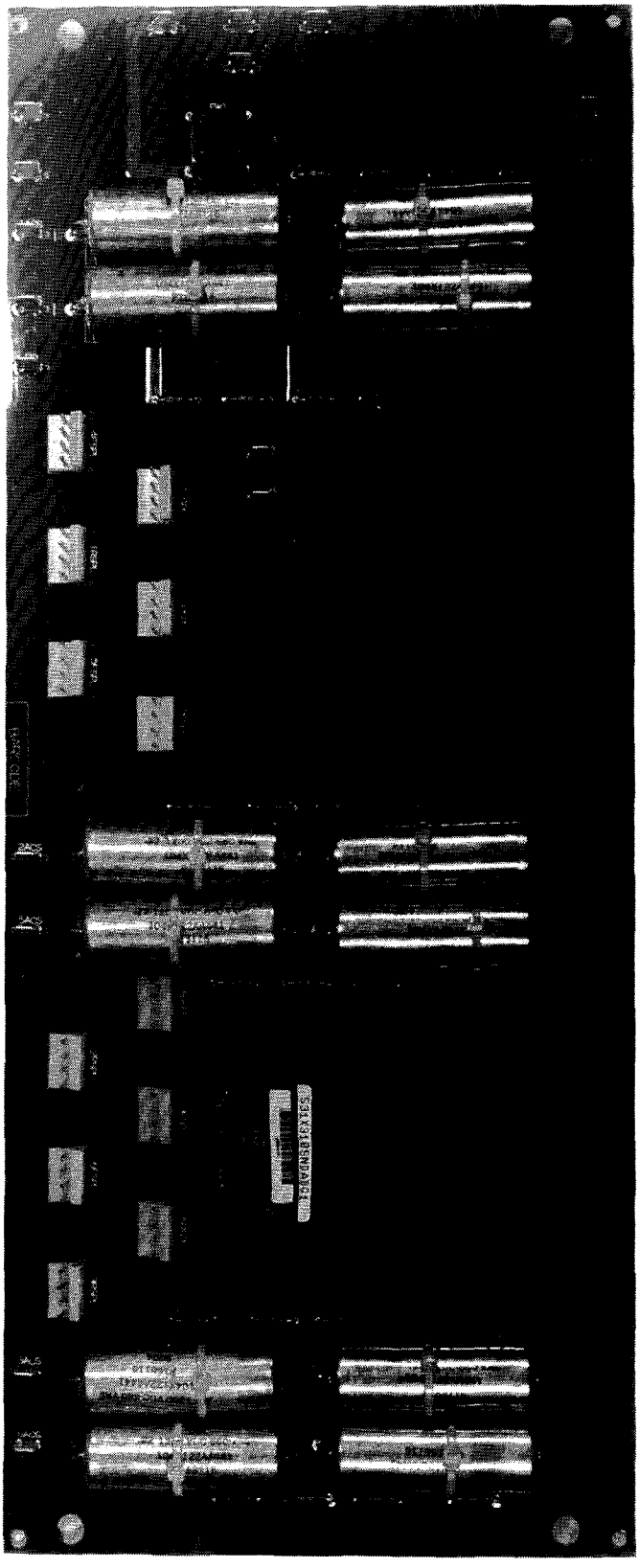


Figure 4-18. SNUBBER CARD (D Frame)

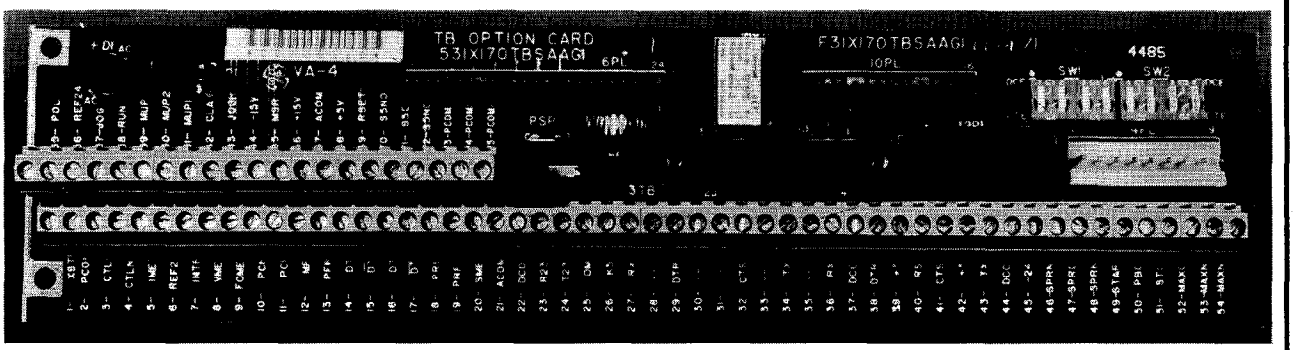


Figure 4-19. 3TB TERMINAL BOARD CARD

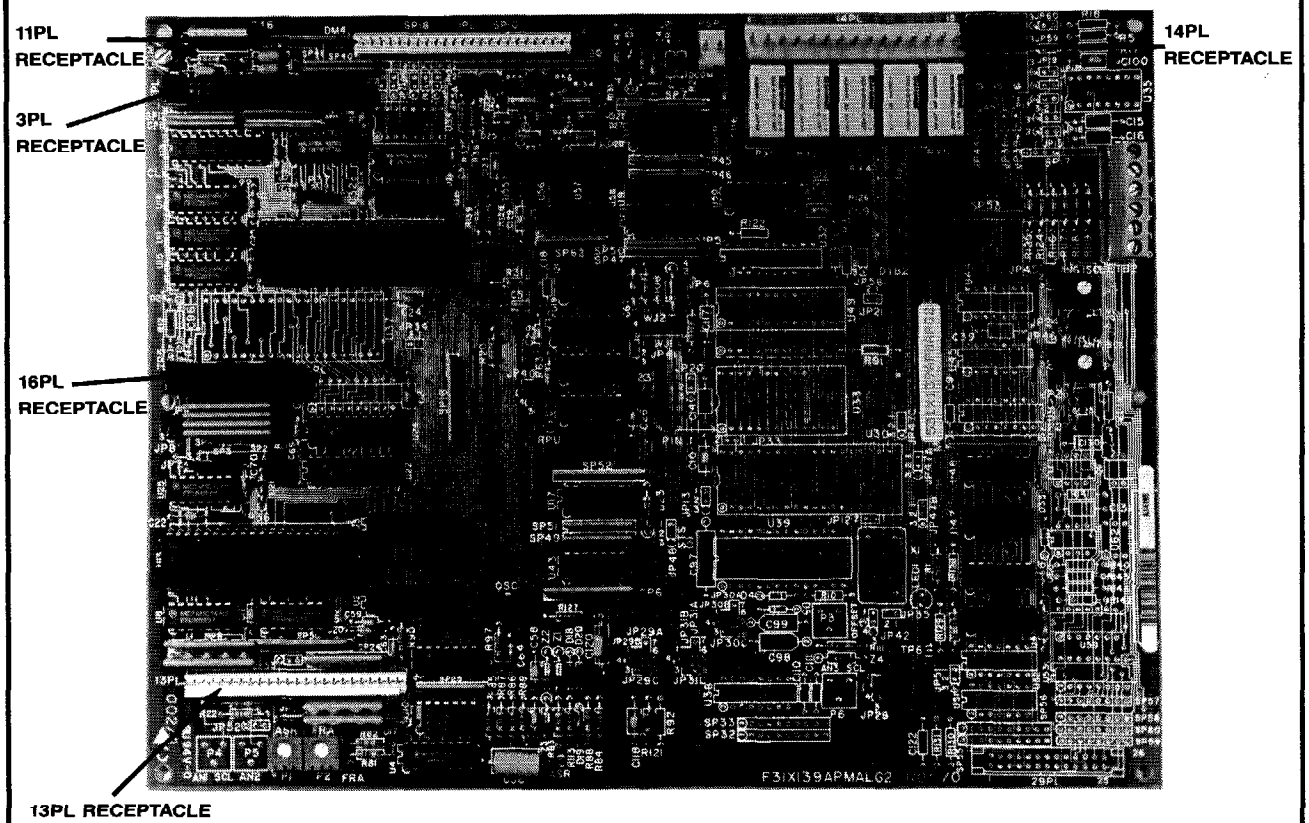


Figure 4-20. OPTIONAL APPLICATION CARD

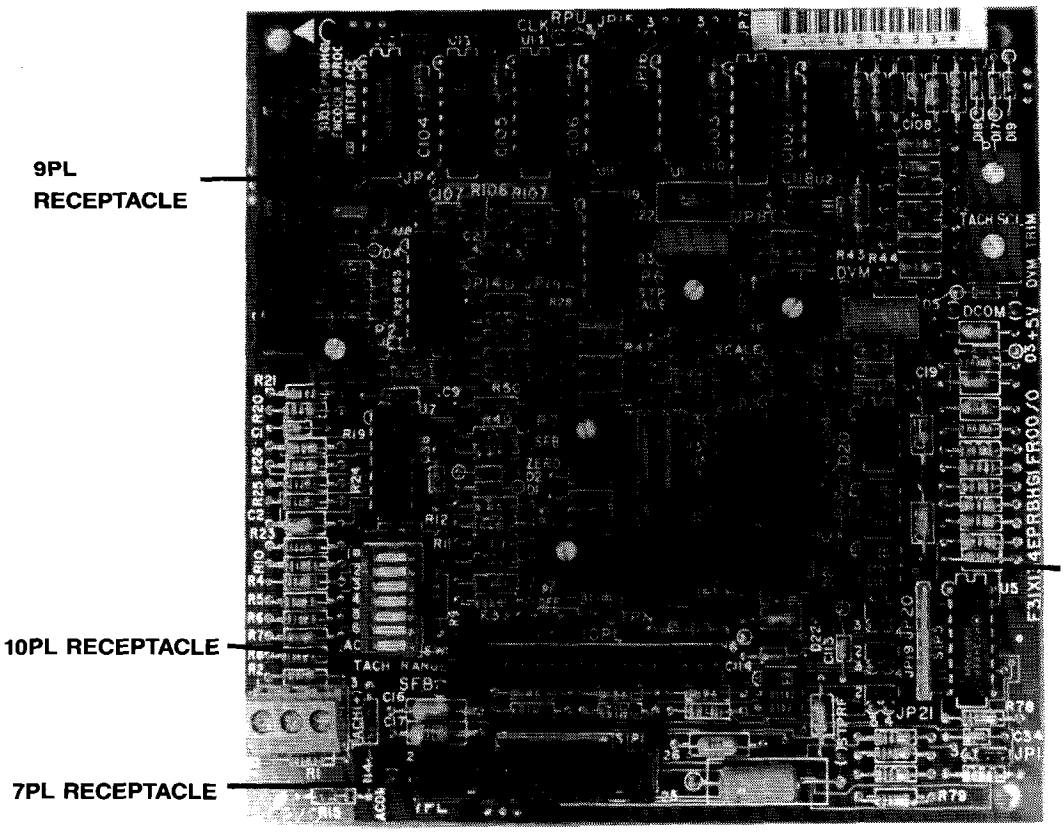


Figure 4-21. OPTIONAL ENCODER PROCESS INTERFACE CARD

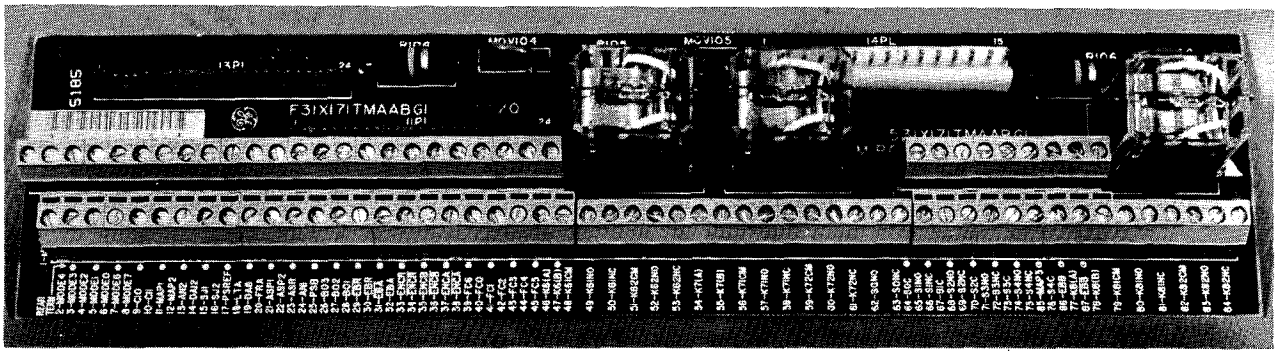


Figure 4-22. OPTIONAL 4TB TERMINAL BOARD CARD

Instructions

For Service

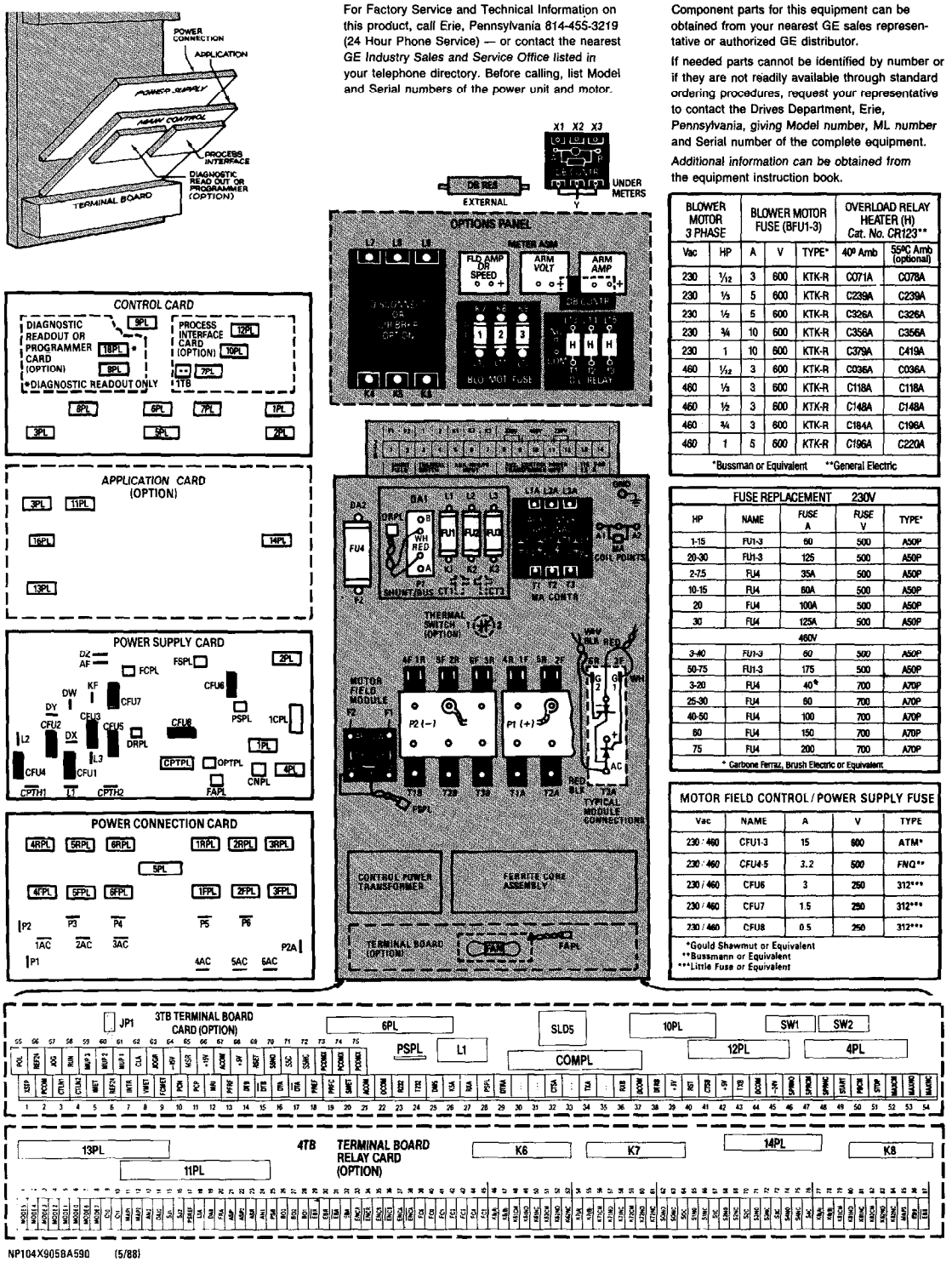
For Factory Service and Technical Information on this product, call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service) — or contact the nearest *GE Industry Sales and Service Office listed in* your telephone directory. Before calling, list Model and Serial numbers of the power unit and motor.

For Parts

Component parts for this equipment can be obtained from your nearest GE sales representative or authorized GE distributor.

If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures, request your representative to contact the Drives Department, Erie, *Pennsylvania, giving Model number, ML number* and Serial number of the complete equipment.

Additional information can be obtained from the equipment instruction book.



BLOWER MOTOR 3 PHASE		BLOWER MOTOR FUSE (BFU1-3)			OVERLOAD RELAY HEATER (H) Cat. No. CR123**	
Vac	HP	A	V	TYPE*	40° Amb	55°C Amb (optional)
230	1/12	3	600	KTK-R	C071A	C078A
230	1/3	5	600	KTK-R	C239A	C239A
230	1/2	5	600	KTK-R	C326A	C326A
230	3/4	10	600	KTK-R	C356A	C356A
230	1	10	600	KTK-R	C379A	C419A
460	1/12	3	600	KTK-R	C036A	C036A
460	1/3	3	600	KTK-R	C118A	C118A
460	1/2	3	600	KTK-R	C148A	C148A
460	3/4	3	600	KTK-R	C184A	C196A
460	1	5	600	KTK-R	C196A	C220A

*Bussman or Equivalent **General Electric

FUSE REPLACEMENT 230V				
HP	NAME	FUSE A	FUSE V	TYPE*
1-15	FU1-3	60	500	A50P
20-30	FU1-3	125	500	A50P
2-7.5	FU4	35A	500	A50P
10-15	FU4	60A	500	A50P
20	FU4	100A	500	A50P
30	FU4	125A	500	A50P
460V				
3-40	FU1-3	60	500	A50P
50-75	FU1-3	175	500	A50P
3-20	FU4	40*	700	A70P
25-30	FU4	60	700	A70P
40-50	FU4	100	700	A70P
60	FU4	150	700	A70P
75	FU4	200	700	A70P

* Carbone Ferraz, Brush Electric or Equivalent

MOTOR FIELD CONTROL / POWER SUPPLY FUSE				
Vac	NAME	A	V	TYPE
230 / 460	CFU1-3	15	600	ATM*
230 / 460	CFU4-5	3.2	500	FNQ**
230 / 460	CFU6	3	250	312***
230 / 460	CFU7	1.5	250	312***
230 / 460	CFU8	0.5	250	312***

*Gould Shawmut or Equivalent
**Bussmann or Equivalent
***Little Fuse or Equivalent

NP104X905BA590 (5/88)

Figure 4-23. TYPICAL INFORMATION LABEL

NOTES:

5. ELECTRICAL CONNECTOR PIN CHARTS

This section contains electrical connector pin charts for all connectors (standard and optional) in the controller. The major connectors are listed in numerical order and any additional smaller connectors are listed at the end of the section under the heading **"ADDITIONAL CONNECTORS"**.

CONNECTOR 1PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

Pin	Nomen-clature	Description
1	DCOM	Digital Common.
2	FFC	Field firing control-- drives field gate pulse transformer.
3	DCOM	
4	FC	Field current feedback from PSC.
5	DCOM	
6	RT1	AC phase 1 input from PSC.
7	RT2	AC phase 2 input from PSC.
8	RT3	AC phase 3 input from PSC (RT1, RT2, & RT3 are resistance isolated).
9	FRC	Fault relay control - Drives the 24 volt fault relay on PSC.
10	-24V	Unregulated -24V return path for FRC and RP.
11	RP	MA contactor pilot relay driver (MAX - K2 on PSC).
12	RUN	Drive run signal from RUN (K1) relay contact on PSC.
13	REF24	Supplies +/-24V to RUN (K1) relay on PSC depending on setting of jumper JP20.
14	W5V	Watchdog Reference - Provides 5V reference from PSC to undervoltage detector on Control card.
15	J1	Input from line phase 1 CT.
16	J2	Input from line phase 1 CT.
17	J3	Input from line phase 3 CT.
18	J4	Input from line phase 3 CT.
19	DCOM	
20	CFB	Armature current feedback input from PSC.

CONNECTOR 2PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

Pin	Nomen-clature	Description
1	ACOM	Analog signal return for +/-15V.
2	-15V	-15V regulated analog power supply input.
3	+15V	+15V regulated power supply input.
4	+24V	+24V unregulated raw power supply input.
5	-24V	-24V unregulated raw power supply input.
6	+5V	+5V regulated digital power supply input.
7	DCOM	Digital signal return for +5V.
8	+5V	
9	PCOM	Raw power supply return for +/-24V.

NOTE: *+5 VDC is **not** to be used for any purpose external to the controller.*

CONNECTOR 3PL -- BETWEEN CONTROL CARD AND OPTIONAL APPLICATION CARD

Pin	Nomen-clature	Description
1	D0	U4 data bus bit 0 (least significant bit).
2	D1	U4 data bus bit 1.
3	DCOM	Digital common.
4	D2	U4 data bus bit 2.
5	D3	U4 data bus bit 3.
6	D4	U4 data bus bit 4.
7	DCOM	
8	D5	U4 data bus bit 5.
9	D6	U4 data bus bit 6.
10	D7	U4 data bus bit 7.
11	DCOM	
12	A8	U4 address bus bit 8.
13	A9	U4 address bus bit 9.
14	A10	U4 address bus bit 10.
15	DCOM	
16	A11	U4 address bus bit 11.
17	SEL	Signal from U3. Used for Micro-application card synchronization.
18	ALE	Address latch from U4.
19	DCOM	
20	/RD	Read control line from U4.
21	/WR	Write control line from U4.
22	A15	U4 address bus bit 15, used for Application card I/O select.
23	RST	Card reset signal for Application card.
24	+5V	+5V regulated digital supply to Application card.
25	+5V	
26	+5V	
27	PCOM	+/-24V power supply return.
28	+24V	+24V non-regulated raw power supply to Application card.
29	-24V	-24V non-regulated power supply to Application card.
30	DM4	Provides "dummy" connection to Application card.
31	+15V	+15V regulated analog supply to Application card.
32	-15V	-15V regulated analog supply to Application card.
33	ACOM	Analog signal return for +/-15V.
34	OPTA	Input to A-D converter from Application card multiplexer.

CONNECTOR 4PL -- BETWEEN POWER SUPPLY CARD (PSC) AND 3TB TERMINAL BOARD CARD

Pin	Term. #	3TB clature	Nomen- Description
1	46	SPRNO	Fault Relay Normally Open Contact
2	47	SPRCM	Fault Relay Common Connection
3	48	SPRNC	Fault Relay Normally Closed Contact
4	49	START	Connection Point for Start PB to "RUN" relay (K1)
5	50	PBCM	Common Point for Start and Stop Pushbuttons
6	51	STOP	Connection for Stop Pushbutton
7	52	MAXCM	MA Auxiliary Relay Common Connection
8	53	MAXNO	MA Auxiliary Relay Normally Open Contact
9	54	MAXNC	MA Auxiliary Relay Normally Closed Contact

NOTE: *Contact rating K2 relay is 1.2 amp at 115 volts AC or 28 volts DC. Contact rating K3 relay is 1 amp at 115 volts AC or 28 volts DC.*

CONNECTOR 5PL -- BETWEEN CONTROL CARD AND POWER CONNECTION CARD

Pin	Nomen-clature	Description
1	A6F	Drives cell 6F gate pulse transformer.
2	DFP	Delayed firing power for A6F and A5F.
3	A5F	Drives cell 5F gate pulse transformer.
4	PCOM	Shield wire.
5	A4F	Drives cell 4F gate pulse transformer.
6	DFP	Delayed firing power for A4F and A3F.
7	A3F	Drives cell 3F gate pulse transformer.
8	PCOM	Shield wire.
9	A2F	Drives cell 2F gate pulse transformer.
10	DFP	Delayed firing power for A2F and A1F.
11	A1F	Drives cell 1F gate pulse transformer.
12	PCOM	Shield wire.
13	A1R	Drives cell 1R gate pulse transformer.
14	DFP	Delayed firing power for A1R and A2R.
15	A2R	Drives cell 2R gate pulse transformer.
16	PCOM	Shield wire.
17	A3R	Drives cell 3R gate pulse transformer.
18	DFP	Delayed firing power for A3R and A4R.
19	A4R	Drives cell 4R gate pulse transformer.
20	PCOM	Shield wire.
21	A5R	Drives cell 5R gate pulse transformer.
22	DFP	Delayed firing power for A5R and A6R.
23	A6R	Drives cell 6R gate pulse transformer.
24	PCOM	Shield wire.
25	DCN	Resistance isolated armature voltage.
26	DCP	Resistance isolated armature voltage.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND 3TB TERMINAL BOARD CARD

Pin	3TB Term. #	Nomen-clature	Description
1	69	RSET	A hard reset of drive is initiated when RSET is connected to +15V. Must be open to run. RSET stops firing of all SCRs, removes delayed firing power, drops out the MA contactor and fault relay, and holds all microprocessors in a reset state until RSET is opened. The drive will not automatically restart if reset while running. Do not apply RSET while running.
2	68	+5V	Not for use external to drive.
3	67	ACOM	Not for use external to drive.
4	66	+15V	Signal voltage for System Reference pot, RSET circuit, and others.
5	65	MSR	Manual system reference input to drive, ±15 VDC max. Positive for forward motoring with POL at zero or open.
6	64	-15V	Signal voltage for System Reference pot and others.
7	63	JOGR	Jog reference input (can be selected to bypass linear time), ±14V = Top speed. Programmable to 7 VDC = top speed for fast update rate. Refer to Programming Software Package.
8	62	CLA	Current limit adjust - Adds to software setting of current limit. +10V increases current limit by 100% of rated, -10 VDC decreases it by 100%.
9	61	MUP1	Multiuse point 1 - connects to connection test point MUP1, which may be jumpered to another connection point. **Example:** DA0 may be jumpered to MUP1 for output on 3TB.
10	60	MUP2	Multiuse point 2 - connects to connection test point MUP2, which may be jumpered to another connection point.
11	59	MUP3	Multiuse point 3 - connects to connection test point MUP3, which may be jumpered to another connection point.
12	58	RUN	Alternate run command input point in place of "RUN" relay. Do not pull high if run relay is also being used. Drive runs when connected to 3TB56 (6PL-14), REF24. Input is active (controller runs) when RUN is pulled down to -24 to 0.4 VDC. Input is inactive (controller stops) when RUN is open or tied to +3.5 to +24 VDC.
13	57	JOG	Jog command input point. Drive jogs when connected to 3TB56 (6PL-14), REF24. Polarity controlled same as RUN above.
14	56	REF24	24 VDC supply for RUN, JOG, POL or XTSP input. Inputs are active when pulled down between -24 to 0.4 VDC. Inputs are inactive when open or tied to +3.5 to +24 VDC.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND 3TB TERMINAL BOARD CARD (contd.)

Pin	3TB Term. #	Nomen-clature	Description
15	55	POL	Control input which reverses the direction of the drive for any applied reference; e.g. activating POL low is equivalent to reversing the polarity of the input at MSR. Activated/deactivated same as RUN above. Programmable for multiple functions. Refer to Programming Software Package and elementaries if used.
16	1	XSTP	Must be activated to enable drive. Activated/deactivated same as RUN described previously. Causes a normal, coast, quick or trip stop when open (see ADR004-STPJP in the Programming Software Package).
17	2	PCOM	Not for use external drive.
18	3	CTLN1	Control On - Must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally drops MA contactor after a 50 millisecond attempt by microprocessor to precondition the drive.
19	4	CTLN2	See CTLN1.
20	5	IMET	Analog load meter output proportional to armature current, adjustable for 1 to 8V DC output at rated load current. See P2 and JP15 on the Control card (Table 1) for details. Load not to exceed 1000 ohms/volt.
21	6	REF24	24V supply for RUN, JOG, POL or XSTP input. Set by JP19 and JP20 on the Control card.
22	7	SLD5	Programmable 24 VDC drive from Control card to the optional signal level detector SLD5 on 3TB terminal board. Controls contacts at 3TB70, 71 and 72. See JP1 on the 3TB Terminal Board card (Table 1) for details.
23	8	VMET	Analog armature voltmeter output. This output is bipolar, normally 4 volts at rated armature voltage output. (See P8 on the Control card, Table 1, for details.) Maximum load is 10 milliamps.
24	9	FCMET	Analog field current meter output, 1 to 5 VDC output. See P5 on the Control card (Table 1) for details. Load not to exceed 1000.
--	26	K5A	Optional SLD5 control input from REF24 for external control of SLD5. Controls contacts at 3TB70, 71, and 72. See JP1 on the 3TB Terminal Board card (Table 1) for details.
--	70	S5NO	Optional SLD5 normally open contact.
--	71	S5C	Optional SLD5 common contact.
--	72	S5NC	Optional SLD5 normally closed contact.

CONNECTOR 7PL -- BETWEEN CONTROL CARD AND OPTIONAL ENCODER PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	ACOM	Analog signal return.
2	DVM	Output from digital voltmeter.
3	SFB	Output from analog tach circuit.
4	ASFB	Amplified signal from analog tach circuit, used for improved resolution at low speed operation.
5	ACOM	
6	-15V	-15V supply to analog circuits on Encoder Process Interface card.
7	+15V	+15V supply to analog circuits on Encoder Process Interface card.
8	+5V	+5V supply to logic circuits on Encoder Process Interface card.
9	+5V	
10	TRS	RS422 driver tri-state command to the Encoder Process Interface card.
11	DCOM	Digital signal return.
12	DIR	Signal from digital tach interface signalling direction of motor rotation when a quadrature track tach is used.
13	TPN	Net pulse train signal from digital tach or AC analog tach interface used to determine tach speed.
14	PRF	Pulse train from digital reference interface used to determine desired reference when used.
15	DCOM	
16	DCOM	
17	RXD	Input from either the RS422 or RS232C serial link receiver from the Encoder Process Interface card.
18	TXD	Output to the RS422 and RS232C serial link drivers from the Encoder Process Interface card.
19	/DTR	Data terminal ready - Output indicating the Encoder Process Interface card is ready to receive a byte on the serial link. Low when ready, high when not.
20	/CTS	Clear to send - Input from serial link signalling when the receiver on the other end of the link is ready to receive a byte. When low, drive may transmit. When high, drive must not send anything out on TXD.
21	RST	Reset line from Control card to Encoder Process Interface card, active high.
22	+5V	
23	+5V	
24	DCOM	
25	DCOM	
26	-24V	-24V raw power to P.I. card.

CONNECTOR 9PL -- BETWEEN CONTROL CARD AND OPTIONAL ENCODER PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	P10	Least significant bit of position (position bit 0)
2	P11	Position bit 1
3	P12	Position bit 2
4	P13	Position bit 3
5	P14	Position bit 4
6	P15	Position bit 5
7	P16	Position bit 6
8	P17	Position bit 7
9	P20	Position bit 8
10	P21	Position bit 9
11	P22	Position bit 10
12	P23	Position bit 11 (most significant bit)
13	P24	May be jumpered to DM5 using JP18
14	P25	May be jumpered to MKRPLS using JP17
15	P26	Connects to 9PL-16 for test purposes
16	P27	Connects to 9PL-15 for test purposes
17 thru 22	----	No Connection
23	DCOM	Digital common
24	DCOM	Digital common
25 thru 32	----	No Connection
33	+5V	Power supply from Control card
34	----	No Connection

CONNECTOR 10PL -- BETWEEN ENCODER PROCESS INTERFACE CARD AND 3TB TERMINAL BOARD CARD

Pin	3TB Term. #	Nomen-clature	Description
1	10	PCN	Optional negative input for process follower current (1-5, 4-20, 5-50, or 5-65mA.) or voltage (0-10 VDC) reference.
2	11	PCP	Optional positive input for process follower reference.
3	12	MR1	Optional reference for manual system reference potentiometer to allow coordinated max speed scaling when PFRF is connected to Control card reference input via auto/manual selection.
4	13	PFRF	Optional process follower interface output (0-10V) that connects to a speed reference or other analog process input.
5	14	DTB	Optional square wave tach quadrature input (track B high). Operating Voltage: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
6	15	/DTB	Optional inverted square wave differential tach quadrature input (track B low).
7	16	DTA	Optional square wave tach quadrature input (track A high). Operating Voltage: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
8	17	/DTA	Optional inverted square wave differential tach quadrature input (track A low).
9	18	PREF	Optional pulse reference frequency positive input, 0 to 6KHz Max.
10	19	PRFC	Optional high impedance common for pulse reference input.
11	20	SMET	Analog speed meter output, 10 VDC. See P2 on the Encoder Process Interface card (Table 1).
12	21	ACOM	Common for optional RS232C communications.
13	22	DCOM	Not for use external drive.
14	23	R232	Optional RS232C receive line.
15	24	T232	Optional RS232C transmit line.
16	25	DM5	Spare connection tied to undedicated point DM5.

NOTE: / = NOT; **Example:** /C = Not C

CONNECTOR 11PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB TERMINAL BOARD CARD

Pin	4TB Term. #	Nomen-clature	Description
1	30	EBR	No connection.
2	29	/EBR	No connection.
3	85	MAP3	Not Used
4	25	PSB	Strobe for optional 16 bit parallel input on 16PL. See Application card JP7 (Table 1). Inputs must be set 50 microseconds before strobe, and strobe held for 100 microseconds.
5	26	B03	Optional programmable 24V bit output #3.
6	27	B02	Optional programmable 24V bit output #2.
7	28	B01	Optional programmable 24V bit output #1.
8	32	EBA	No connection.
9	31	/EBA	No connection.
10	86	EBB	No connection.
11	87	/EBB	No connection.
12	33	ENCR	Optional non-inverting differential or single-ended encoder marker pulse input.
13	34	/ENCR	Optional inverting differential encoder marker pulse input.
14	35	ENCB	Optional non-inverting differential or single-ended encoder Track B input.
15	36	/ENCB	Optional inverting differential encoder Track B input.
16	37	ENCA	Optional non-inverting differential or single-ended encoder Track A input.
17	38	/ENCA	Optional inverting differential encoder Track A input.
18	39	FC6	Bit 6 of optional fault code 24V output.
19	40	FC0	Bit 0 of optional fault code 24V output.
20	41	FC1	Bit 1 of optional fault code 24V output.
21	42	FC2	Bit 2 of optional fault code 24V output.
22	43	FC3	Bit 3 of optional fault code 24V output.
23	44	FC4	Bit 4 of optional fault code 24V output.
24	45	FC5	Bit 5 of optional fault code 24V output.

CONNECTOR 12PL -- BETWEEN OPTIONAL ENCODER PROCESS INTERFACE CARD AND 3TB TERMINAL BOARD CARD

Pin	3TB Term. #	Nomen-clature	Description
1	--	--	No connection.
2	--	--	No connection.
3	--	--	No connection.
4	--	--	No connection.
5	--	--	No connection.
6	--	--	No connection.
7	32	CTSA	Optional inverted RS422 clear-to-send input from serial link signaling when the Control card may transmit a byte, active high. (EIA standard nomenclature is CTSA.)
8	29	DTRA	Optional inverted RS422 data-terminal-ready output from Control card signaling when it is ready to receive a byte on the serial link, active high. (EIA standard nomenclature is DTRA.)
9	34	TXA	Optional inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXA.)
10	27	RXA	Optional inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXA.)
11	36	RXB	Optional non-inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXB.)
12	43	TXB	Optional non-inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXB.)
13	38	DTRB	Optional non-inverted RS422 data-terminal-ready output from Control card. (EIA standard nomenclature is DTRB.)
14	41	CTSB	Optional non-inverted RS422 clear-to-send input from serial link. (EIA standard nomenclature is CTSB.)
15	40	RST	Reset signal from Control card, active high, not for use external to drive.
16	39	+5V	Not for use external to drive.
17	42	+5V	Not for use external to drive.
18	37	DCOM	Not for use external to drive.
19	44	DCOM	Not for use external to drive.
20	45	-24V	Not for use external to drive.

CONNECTOR 13PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB TERMINAL BOARD CARD

NOTE: *Refer to the Programming Software Package for definitions of I/O through 13PL for particular application.*

Pin	4TB Term. #	Nomen-clature	Description
1	6	MODE 0	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
2	5	MODE 1	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
3	4	MODE 2	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
4	3	MODE 3	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
5	2	MODE 4	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
6	1	MODE 5	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
7	7	MODE 6	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
8	8	MODE 7	Programmable mode select input. Refer to elementaries and Programming Software Package* for information.
9	9	CI0	No connection
10	10	CI1	No connection
11	11	MAP1	Not Used
12	12	MAP2	Not Used
13	13	AN2	No connection.
14	14	DA12	Optional analog output from 12-bit D-A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Update rate is 30 to 40 times/second.
15	15	SJ1	Optional analog input to summing junction #1. Operating range = $\pm$10V; maximum input = $\pm$15V.
16	16	SJ2	Optional analog input to summing junction #2. Operating range = $\pm$10V; maximum input = $\pm$15V.
17	17	PSREF	Optional analog position or process reference input ($\pm$10V).
18	18	LTA	Optional analog linear time adjust. Positive voltages increase timing, negative voltages decrease timing. $\pm$10 volts maximum. 10 volts is equal to 66 second adjust.
19	19	DA8	Optional analog output from 8-bit D-A converter. See DA12 for specifications. Normal update rate of 180 times/sec. Reduced to 30 to 40 times/sec if fast update selected for DA0.

**Functions of MODE 0 - MODE 7 are determined by Addresses 000-015 (ADR000 - ADR015) in the Programming Software Package.*

CONNECTOR 13PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB TERMINAL BOARD CARD (continued)

Pin	4TB Term. #	Nomen-clature	Description
20	20	FRA	Optional analog field reference adjust. Positive voltages up to +10V increase field current. Negative voltages down to -10V decrease field current. See optional Application card JP32 and P2 (Table 1).
21	21	ASP1	Optional spare analog input #1 (programmable).
22	22	ASP2	Optional spare analog input #2 (programmable).
23	23	ASR	Optional auto speed reference input to drive, ±15V maximum, positive for forward cell firing while motoring. ASR is the active speed reference in the auto mode when analog auto reference is selected.
24	24	AN1	No connection.

CONNECTOR 14PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB TERMINAL BOARD CARD

Pin	4TB Term. #	Nomen-clature	Description
1	62	S0NO	Optional SLD #0 normally open contact.
2	63	S0NC	Optional SLD #0 normally closed contact.
3	64	S0C	Optional SLD #0 common contact.
4	65	S1NO	Optional SLD #1 normally open contact.
5	66	S1NC	Optional SLD #1 normally closed contact.
6	67	S1C	Optional SLD #1 common contact.
7	68	S2NO	Optional SLD #2 normally open contact.
8	69	S2NC	Optional SLD #2 normally closed contact.
9	70	S2C	Optional SLD #2 common contact.
10	71	S3NO	Optional SLD #3 normally open contact.
11	72	S3NC	Optional SLD #3 normally closed contact.
12	73	S3C	Optional SLD #3 common contact.
13	74	S4NO	Optional SLD #4 normally open contact.
14	75	S4NC	Optional SLD #4 normally closed contact.
15	76	S4C	Optional SLD #4 common contact.

CONNECTOR 16PL -- LOCATED ON OPTIONAL APPLICATION CARD

Pin	Nomen-clature	Description
1	DCOM	Digital common.
2	+5V	Not used.
3	PI0	Optional parallel word input, least significant binary bit input. For all inputs refer to Table 1, JP8 description for voltage level select. Input must be set for 50 microseconds prior to strobe.
4	PI1	Optional parallel binary bit input 1.
5	PI2	Optional parallel binary bit input 2.
6	PI3	Optional parallel binary bit input 3 (or PI0 - PI3 = BCD units digit).
7	PI4	Optional parallel binary bit input 4.
8	PI5	Optional parallel binary bit input 5.
9	PI6	Optional parallel binary bit input 6.
10	PI7	Optional parallel binary bit input 7 (or PI4 - PI7 = BCD tens digit).
11	PI8	Optional parallel binary bit input 8.
12	PI9	Optional parallel binary bit input 9.
13	PI10	Optional parallel binary bit input 10.
14	PI11	Optional parallel binary bit input 11 (or PI8 - PI11 = BCD hundreds digit).
15	PI12	Optional parallel binary bit input 12.
16	PI13	Optional parallel binary bit input 13.
17	PI14	Optional parallel binary bit input 14 (or PI12 - PI14 = BCD thousands digit).
18	PI15	Optional parallel sign bit input.
19	DCOM	Digital common.
20	DCOM	Digital common.

TERMINAL BOARD 1TB LOCATED ON OPTIONAL ENCODER PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	TKN	Negative analog tachometer feedback signal. Maximum input is 310 VDC or 344 VAC.
2	----	No connection (used for clearance)
3	TKP	Positive analog tachometer feedback signal.

ADDITIONAL CONNECTORS

CONNECTOR 1CPL -- BETWEEN POWER SUPPLY CARD AND AC LINE CTs (CT1 AND CT3)
Regenerative Controllers Only

Pin	Nomen-clature	Description
1	J1	Input from line phase 1 Current Transformer.
2	J2	Input from line phase 1 Current Transformer.
3	J3	Input from line phase 3 Current Transformer.
4	J4	Input from line phase 3 Current Transformer.

CONNECTOR CNPL -- BETWEEN POWER SUPPLY CARD AND MA CONTACTOR

Pin	Nomen-clature	Description
1	MAP	MA contactor control output.
2	X2	MA contactor control output.

CONNECTOR CPTPL -- BETWEEN POWER SUPPLY CARD AND CONTROL POWER TRANSFORMER

Pin	Nomen-clature	Description
1	--	38 VAC, center tapped for controller power supplies.
2	--	38 VAC, center tapped for controller power supplies.
3	--	Center tap point for 38 VAC.
4	--	Not used.
5	--	115 VAC for controller cooling fan.
6	--	115 VAC for controller cooling fan.

CONNECTOR DRPL -- BETWEEN POWER SUPPLY CARD AND SHUNT

Pin	Nomen-clature	Description
RED	IP	Armature current feedback from shunt, 0-100mV. Positive when motoring with the forward power circuit, Red wire.
WH	IN	Armature current feedback from shunt. Negative when motoring with the forward power circuit, White wire.

CONNECTOR FAPL -- BETWEEN POWER SUPPLY CARD AND CONTROLLER COOLING FAN

Pin	Nomen-clature	Description
1	X1	115 VAC for controller cooling fan.
2	X2	115 VAC for controller cooling fan.

CONNECTOR FCPL -- BETWEEN POWER SUPPLY CARD AND EXTERNAL MOTOR FIELD CONTROL ASSEMBLY

Pin	Nomen-clature	Description
1	ACOM	Analog common.
2	--	External motor field control feedback. Used only with external motor field control assembly.

CONNECTOR FSPL -- BETWEEN POWER SUPPLY CARD AND FIELD POWER MODULE

Pin	Nomen-clature	Description
1	FLC	Gate pulse transformer output to field power module SCR, Red wire. Not used with external motor field control assembly.
2	FLG	Gate pulse transformer output to field power module SCR, White wire.

CONNECTOR OPTPL -- LOCATED ON POWER SUPPLY CARD

Pin	Nomen-clature	Description
1	FX1	Fused 115 VAC from Control Power Transformer.
2	X2	115 VAC return from Control Power Transformer.

CONNECTOR PSPL -- LOCATED ON POWER SUPPLY CARD

Pin	Nomen-clature	Description
1	+5V	+5V regulated digital supply (not for use external to drive).
2	DCOM	Digital signal return for +5V (not for use external to drive).

CONNECTOR COMPL -- OPTIONAL R232C SERIAL COMMUNICATIONS CONNECTOR ON 3TB

Pin	Nomen-clature	Description
1	GND	Ground.
2	RXA	For Data Communications Equipment (DCE), dumb terminal mode, pin 2 is the or RS232C receive line (RXA).
	TXA	For Data Terminal Equipment (DTE), modem or PDS system, pin 2 is the RS232C transmit line (TXA).
3	TXA or RXA	For DCE, pin 3 is the TXA line. or For DTE, pin 3 is the RXA line.
4	----	Not used.
5	DTRA or CTSA	For DCE, pin 5 is the data-terminal-ready (DTRA) line. or For DTE, pin 5 is the clear-to-send (CTSA) line.
6	DSR	Data set ready, no connection in DCE or DTE mode.
7	DCOM	Pin 7 is the communication system common.
8 thru 19	----	Not used.
20	CTSA or DTRA	For DCE, pin 20 is the clear-to-send (CTSA) line. or For DTE, pin 20 is the data-terminal-ready (DTRA) line.
21 thru 25	----	Not used.

NOTE: *Refer to 3TB SW1 and SW2 description in Table 1, for setting DCE and DTE modes.*

6. STARTUP PROCEDURE

This startup procedure is meant as a general guide for the initial checkout and startup of a QIII DC Controller. It references some general hardware and software checks relative to most basic applications. Refer to the sections on "Hardware Jumpers and Pot Adjustments" and "Key Parameter Adjustments" and the accompanying "Parameter Map" (Figure 8-1) for more complete information on adjustments available on the QIII DC Controller. Refer to the Programming Software Package and Elementaries for information on wiring and adjustments specific to your drive application.

NOTE: *Prior to beginning actual drive startup, read this entire manual and be aware of all pertinent information.*

WARNING - ELECTRICAL SHOCK HAZARD: Circuit breakers, if supplied as part of the total system, may not disconnect all power to the equipment (see system elementary diagrams). Whether the AC voltage is grounded or not, high voltage to ground will be present at many points.

CONTROL CARD HP SELECT POTENTIOMETER ADJUSTMENT

1. Remove power from the QIII DC Controller.

2. Remove jumpers JP8, JP9, JP10, and JP11 from the Control card.

3. Measure the resistance (with a digital voltmeter [DVM]) between Control card test points "HS" and "ACOM".

4. Adjust potentiometer P3-HP SEL on the control card for the ohm value specified per module catalog number, motor horsepower, shunt catalog number, and DC voltage (see Chart at end of this section).

5. Replace jumpers JP8, JP0, JP10, and JP11 on the Control card (in positions called for per the Programming Software Package) after the appropriate resistance value has been set.

BEFORE APPLYING POWER

NOTE: *Review warning and caution notices at the beginning of this instruction book before proceeding.*

Every QIII DC drive controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are performed:

1. Verify that all incoming wiring is in agreement with the elementary drawings supplied with the drive, and conforms to approved wiring practices (wire size, separation of power from control; etc.).

2. Verify that incoming 3-phase voltage is in accordance with nameplate requirements (+10%, -5%), unless otherwise specified.

3. Confirm that the following jumpers (JP) and switch positions (SW) are in accordance with the jumper selections in Table 1 of this instruction and the Programming Software Package. (Refer to the card layout diagrams, Figures 7-1 thru 7-7, for the location of jumpers.)

   a. Control Card:

Jumper	Description/Position
JP1	Reference voltage range for maximum reference **1-2:** 8 - 15 VDC maximum reference. **2-3:** 5 - 8 VDC maximum reference.
JP8 thru JP11	Horsepower settings (see Table 1).
JP15	Loadmeter Select for percent load output **1-2:** Absoluted (zero left scale) **2-3:** Bipolar (zero center scale)

b. MFC/Power Supply Card:
Version G3, See Figure 7-2

Jumper	Description/Position
JP1, JP2	Set maximum motor field current range (amps).

JP1	JP2	
2-3:	2-3:	0.64 to 1.6 Amps with gain factor (GF) of 0.4
1-2:	2-3:	1.6 to 4.0 Amps with gain factor (GF) of 1.0
1-2:	1-2:	4.0 to 10.0 Amps with gain factor (GF) of 2.5

JP3, JP4 Armature current feedback signal.
- **1-2:** Normal operating mode (must be in this position for proper operation).
- **2-3:** Calibrate mode only (see procedure in Table 1).

Version G2, See Figure 7-2

Jumper	Description/Position
JP1, JP2	Set maximum motor field current range (amps).

JP1	JP2	
2-3:	2-3:	1.5 to 6.0 Amps with gain factor (GF) of 1.5
1-2:	2-3:	6.0 to 12.0 Amps with gain factor (GF) of 3.0
1-2:	1-2:	12.0 to 24.0 Amps with gain factor (GF) of 6.0

NOTE: *Gain Factor required in "Apply AC Power" paragraph of start-up procedure.*

JP3, JP4 Armature current feedback signal.
- **1-2:** Normal operating mode. (must be in this position for proper operation).
- **2-3:** Calibrate mode only (see procedure in Table 1).

c. Power Connection Card (Figure 7-3):
JP1, JP2 DC Armature Voltage range (see Table 1).

d. 3TB Terminal Board Card (Figure 7-4):
XSTP (3TB1) jumpered to REF24 (3TB6 or 3TB56).

CTLN1 (3TB3) jumpered to CTLN2 (3TB4).

e. Encoder Process Interface Card (Figure 7-6) (if present):
SW1 Tach Scale settings (see Table 1).

f. Application Card (Figure 7-5) and 4TB Terminal Board Card (Figure 7-7) (if present):
If an Application Card is present, MODE6 (4TB7) must be jumpered to REF24 (3TB6 or 3TB56).

4. Verify that all connections are tight.

5. Verify that the DC motor has been properly installed and that the shaft is free to rotate.

APPLY AC POWER

WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT HAZARD: Ensure safety of personnel in vicinity of controller and controlled equipment before operation. It is the responsibility of the user to insure that operation does not result in an unsafe condition. Factory provided interlocks and operating limits should not be bypassed or modified.

1. Check for proper direction of rotation of all three-phase blower motors on both the power unit(s) and the DC motor(s).

2. Verify that one Diagnostic LED is shifting from right to left.

NOTE: *If at any time the LEDs remain locked (steady or blinking), indicating a fault message, or are blank, refer to the "Troubleshooting" section for corrective procedures.*

3. Check the field for proper voltage.
   -- Since the drive may be in field economy until a "START" is called for, check the setting in ADR050-FLDEC in the Programming Software Package to determine the proper field voltage.

4. Before continuing, it is recommended that the following be done:
   -- The list of "as shipped" parameter settings provided with the controller be checked against the actual parameters in the drive (use Programming Software Package).
   -- All incoming signals be checked for proper value and correct connection.

NOTE: *The speed reference referred to in the following steps may be preprogrammed in the drive software. Refer to the Programming Software Package and elementaries to determine the reference source.*

5. Start the drive as follows:
   a. Set the speed reference (speed pot, N/C reference, etc.) at their **minimum** settings.
   b. Start the drive (press "START" on an operator station or call for a "RUN" from the control, RUN input at 3TB58).
   c. Verify that the MA or MD contactor picks up and the field current is of the correct value with an amp probe or a field current meter.
6. Ensure that all "STOP" circuits are functional, dropping out MA or MD. ("STOP" from an operator station, 115 VAC START/STOP circuit - 3TB49, 50 & 51.)
7. Restart the drive and slowly increase the speed reference until the motor starts to rotate.
   -- If the motor does not rotate, or rotates in the wrong direction, check the Diagnostic LEDs.
   -- If DC tach feedback is provided, a tach polarity fault may be indicated.
   -- Refer to the fault codes in the Troubleshooting section for corrective action.
8. With the drive **stopped**, increase the speed reference to 100% speed.
   -- Adjust P1-REF SCALE on the Control card, to obtain 4 VDC at test point SRS (**Manual Speed Reference only**; Jumper JP1 on the Control card selects the manual speed reference input voltage range).
   -- Measure with volt-ohmmeter at the appropriate test point and PCOMX.
   -- **If Auto System Reference (ASR) is used** on the optional Application card, adjust P1-ASR SCALE for 4 VDC at test point ASR, also located on the optional Application card.
   -- Set the speed reference to 0% speed.

NOTE: *If no tach or a digital tach is used, SFB will be "0" VDC in the next step.*

9. Start the drive and gradually increase speed reference to **maximum**.
   -- Check for 4 VDC at test points SRS and SFB on the Control card and, if equipped, the optional Encoder Process Interface card.

   NOTE: *SFB reading will be zero if a digital tach is used.*
   -- **If ASR is used,** check test point ASR on the optional Application card. Adjust P1-TACH SCALE (requires optional Encoder Process Interface card) if necessary to reach the correct top speed (only on drives with an AC or DC tach).
   -- Confirm rated armature voltage.

WARNING - ELECTRICAL SHOCK HAZARD: Exercise caution as high voltage may be present.

10. Confirm good motor commutation (absence of sparking between brushes and commutator) under load.
11. Slowly decrease the speed and verify stability through the full speed range.
    -- If the drive is **NOT** stable, refer to the Stability Adjustment procedure in the Troubleshooting section.
12. After the drive is stable throughout the full speed range, confirm that the armature current does not exceed the nameplate rating.
13. The above list of adjustments should result in a properly running drive. Other parameters as sold and shipped with the drive can be accessed and changed as necessary with the Programming Software Package.
    -- A partial listing of "KEY" Parameters, their descriptions, and a Parameter Map are included in this instruction book.

NOTE: *Final responsibility for parameter settings resides with the customer. Always document all changes to parameter settings.*

14. Accidental reprogramming should be prevented by making sure an **even** number is present in ADR003-DGNJP (using the Programming Software Package, subtract "1" if an odd number is present and enter the new value), then place the EEROM write device (JP13 on Control card) in the "PROTECT" (1-2) position.

The drive is now ready for operation.

WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT HAZARD: Ensure safety of personnel in vicinity of controller and controlled equipment before operation. It is the responsibility of the user to insure that operation does not result in an unsafe condition. Factory provided interlocks and operating limits should not be bypassed or modified.

CONTROL CARD P3-HP SEL RESISTANCE SETTINGS

Molule Catalog Number	HP and Shunt Catalog Number at 240 VDC HP	104X149-	500 VDC HP	104X149-	P3-HP SEL Resistance Setting (HS to ACOM Test Points)
7VQD14	1	CC032	2	CC032	175.5
and	1.5	CC043	3	CC043	111.5
7VQD14R	2	CC033	5	CC033	52.5
	3	CC034	7.5	CC034	10.0
7VQD43	1	CC032	2	CC032	175.5
and	1.5	CC043	3	CC043	111.5
7VQD43R	2	CC033	5	CC033	52.5
	3	CC034	7.5	CC034	10.0
	5	CC035	10	CC035	210.0
	7.5	CC036	15	CC036	10.0
	10	CC042	20	CC042	198.5
	--	----	25	CC042	20.7
7VQD127	1	CC032	2	CC032	175.5
and	1.5	CC043	3	CC043	111.5
7VQD127R	2	CC033	5	CC033	52.5
	3	CC034	7.5	CC034	10.0
	5	CC035	10	CC035	210.0
	7.5	CC036	15	CC036	10.0
	10	CC042	20	CC042	198.5
	--	----	25	CC042	20.7
	15	CC037	30	CC037	93.4
	20	CC038	40	CC038	119.7
	25	CC039	50	CC038	60.9
	30	CC039	60	CC039	92.2
	--	----	75	CC040	10.0
7VQD260	40 & Below	CA008	--	----	90.0
and					
7VQD260R	50	CA008	100 & Below	CA008	92.6
	60	CA009	125	CA009	68.3
	75	CA021	150	CA021	10.0

CONTROL CARD P3-HP SEL RESISTANCE SETTINGS (continued)

Molule Catalog Number	HP and Shunt Catalog Number at 240 VDC HP	104X149-	500 VDC HP	104X149-	P3-HP SEL Resistance Setting (HS to ACOM Test Points)
7VQD440 and 7VQD440R	40 & Below	CA008	--	----	10.0
	--	---- Below	100 &	CA008	210.0
	50	CA008	--	----	177.5
	--	----	125	CA009	10.0
	60	CA009	--	----	127.5
	75	CA021	150	CA021	127.5
	--	----	200	CA010	210.0
	100	CA014	--	----	161.0
	--	----	250	CA014	210.0
	125	CA014	--	----	93.0
7VQD525 and 7VQD525R	40 & Below	CA008	--	----	10.0
	--	---- Below	100 &	CA008	210.0
	50	CA008	--	----	177.5
	--	----	125	CA009	10.0
	60	CA009	--	----	127.5
	75	CA021	150	CA021	127.5
	--	----	200	CA010	210.0
	100	CA014	--	----	161.0
	--	----	250	CA014	210.0
	125	CA014	--	----	93.0
	--	----	300	CA011	210.0
	150	CA011	--	----	104.0
7VQD855 and 7VQD855R	200 & Below	CC007	400 & Below	CC007	210.0
	--	----	500	CC001	192.0
	250	CC001	--	----	90.0
7VQD1285 and 7VQD1285R	200 & Below	CC007	400 & Below	CC007	210.0
	--	----	500	CC001	192.0
	250	CC001	--	----	90.0
	--	----	600	CC002	14.0
	300	CC002	--	----	25.0
	--	----	700	CC003	50.0
	--	----	800	CC003	10.0

NOTES:

7. HARDWARE ADJUSTMENTS

This section details adjustments available through hardware jumpers and potentiometers. Table 1 and Figures 7-1 to 7-7 describe the location and function of all jumpers and potentiometers on the QIII DC Controller cards (standard cards and optional cards). (Section 8 describes key software "parameter" adjustments available and includes a "Parameter Map" that relates the adjustments to their function.)

HARDWARE JUMPER AND POTENTIOMETER ADJUSTMENTS

Most of the jumper selections and pototentiometer (pot) adjustments have been factory set. When one of the cards has to be changed, make sure the replacement card switches, jumpers and pots are set to be the same as on the card being replaced. For a full description of all jumper and pot adjustments, see Table 1.

NOTE: *If the Control card is replaced, the EEROM (U12) from the original card should be put into the replacement card. If the same failure symptoms still exist, the new EEROM, shipped with the replacement card, can be reinstalled but must be reprogrammed with the Programming Software Package.*

Hardware jumpers and pots are located on the Control card, Power Supply card, 3TB Terminal Board card, optional Encoder Process Interface card, and optional Application card. Figures 7-1 thru 7-7 show the locations of pots, jumpers, test points and other key components.

Most of the pots and jumpers are used in the interface circuits of the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use. Most adjustments were set in the factory. Adjustments not set in the factory can easily be made by the startup procedure in this book or by use of the Programming Software Package and Table 1.

NOTE: *Only high impedance digital voltmeters should be used to make indicated measurements.*

NOTE: *Before changing any jumper positions, compare actual jumper positions with "as shipped" jumper positions through use of the Programming Software Package.*

TABLE 1. JUMPER AND POT ADJUSTMENTS

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card	Pot	P1	MAN REF	Used to adjust test point SRS to four (4) volts when top speed reference is at MSR, 3TB65 (6PL-5; connector 6PL, pin number 5). Used in conjunction with jumper JP1.
	Pot	P2	IMET	This pot is used to adjust the output of the analog load meter circuit. Adjust the output voltage to be from 1 to 8 volts at rated load. Maximum current load on this signal is 1000 ohms/volt.
	Pot	P3	HP SEL	Adjusted by the manufacturer. If replacing card, set to make it visually the same as on original card.
	Pot	P4	OFFSET	Adjusted by the manufacturer, no need to be adjusted in the field unless directed to do so by fault codes.
	Pot	P5	FCMET	This pot is used to adjust the output of the analog field current voltmeter circuit. Adjust the output from 1 to 5 volts at rated maximum field current. Maximum current load on this signal is 1000 ohms/volt.
	Pot	P6	DA0 SCL	This pot is used to adjust the gain of the 12-bit digital-to-analog converter (DA0) output from the Control card. CCW sets the gain at 0.25, full CW sets the gain at 1.25. The output saturates at 12 VDC.
	Pot	P7	----	Not Used - Set in full CCW position for proper operation.
	Pot	P8	VMET	This pot is used to adjust the output of the analog armature voltage meter circuit. Output is adjustable from 4 to 8 volts at rated voltage. Maximum current load on this signal is 10mA. 1PU Arm Volt = 4.0 VDC @ VMET (CW rotation increases VMET output.)
	Jumper	JP1	MAN REF	This jumper selects the maximum voltage range for the manual speed reference input. **1-2:** 8 to 15 volts **2-3:** 5 to 8 volts
	Jumper	JP2, JP4 JP5, JP6 JP7		JP2 - JP7 should be set as follows: **1-2:** Set in this position for proper operation.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card (continued)	Jumper	JP8 JP9 JP10 JP11		These jumpers are set in accordance with the shunt based HP of the drive per the following Table:

Horsepower

230 VAC	460 VAC	JP8	JP9	JP10	JP11
1	----	1-2	2-3	2-3	2-3
1.5	3	1-2	2-3	2-3	2-3
2	5	1-2	2-3	2-3	2-3
3	7.5	1-2	2-3	2-3	2-3
5	10	1-2	2-3	2-3	1-2
7.5	15	1-2	2-3	2-3	1-2

Horsepower

230 VAC	460 VAC	JP8	JP9	JP10	JP11
10	20	1-2	2-3	1-2	2-3
----	25	1-2	2-3	1-2	2-3
15	30	1-2	2-3	1-2	1-2
20	40	1-2	1-2	2-3	2-3
25	50	1-2	1-2	2-3	1-2
30	60	1-2	1-2	1-2	2-3
----	75	1-2	1-2	1-2	1-2
40	----	1-2	1-2	2-3	2-3
50	100	1-2	1-2	2-3	1-2
60	125	1-2	1-2	1-2	2-3
75	150	1-2	1-2	1-2	1-2
100	200	1-2	1-2	2-3	2-3
125	250	1-2	1-2	2-3	1-2
150	300	1-2	1-2	1-2	2-3
200	400	1-2	1-2	2-3	1-2
250	500	1-2	1-2	1-2	2-3
--	600	1-2	1-2	1-2	1-2
300	--	2-3	1-2	2-3	1-2
--	700	2-3	1-2	1-2	2-3
--	800	2-3	1-2	1-2	1-2

	Jumper	JP12		Allows SCR test **1-2:** Disables SCR test (normal operation). **2-3:** SCR test before operation. An SCR test will be executed the first time a RUN is commanded. Subsequent RUN commands will execute normally.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card (continued)	Jumper	JP13		This jumper provides a hardware write protection for the software jumpers and adjustments. **1-2:** Protected (should be in this position for maximum security and reliability of the drive). **2-3:** Enables reprogramming. Do not perform a "hard" reset or turn power on or off in this position.
	Jumper	JP14		This jumper selects proper line voltage. **1-2:** 440 to 480 VAC **2-3:** Not Used - Do not set in this position. **2-5:** Not Used - Do not set in this position. **3-4:** 220 to 240 VAC
	Jumper	JP15		This jumper selects whether the load meter driver output is to be bipolar or absoluted. **1-2:** Absoluted (always positive) **2-3:** Bipolar (zero center)
	Jumper	JP16 JP17 JP18		JP16 - JP18 should be set as follows: Set by factory - refer to jumper listing in Programming Software Package.
	Jumper	JP19 JP20		JP19 & JP20 set the polarity of RUN, JOG, POL, and XSTOP and should **always** be set as follows: **JP19 JP20 Description** **1-2 3-4:** Inputs are **active** when pulled down between -24 to 0.4 VDC. Inputs are **inactive** when open or tied to +3.5 to +24 VDC.
	Jumper	JP21		Allows interactive card test. **1-2:** Disables card test (normal operation). **2-3:** Initiates card test if drive is not running. To exit card test, move JP21 back to the 1-2 position and hard reset drive (push SW2).
	Jumper	JP24 JP25 JP26 JP27 JP28 JP30 JP31 JP32		JP24-JP28 and JP30-JP32 set as follows: Set by supplier - refer to jumper listing in Programing Software Package.
	Jumper	JP29		This jumper selects whether the DA0 output has a fixed gain or adjustable gain. **1-2:** Fixed gain (see Table 3, RAM Variable Scaling). **2-3:** Gain adjustable by P6-DA0 SCL.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card (continued)	Switch	SW2	Hardware Reset	This switch causes a hardware reset ("hard" reset of the controller) when pressed. Used to clear some drive faults (refer to Table 5, Specific Fault Troubleshooting). The drive will return to the initial power up condition.

WARNING - ELECTRICAL SHOCK HAZARD: High voltage is present on the Power Supply card. Remove power from the drive before handling the plug and jumpers in adjustment procedure.

MFC/Power Supply Card	Pot	P1	FIELD SCALE	P1 is used to calibrate the field current feedback using a known field current. This pot is set and sealed by the manufacturer.
	Pot	P2	CFB	P2 is used to adjust offset of the current isolator circuit so that test point CFB (on Control card) is zero. Factory Set - Do not adjust unless changing cards.
	Pot	P3	CFB ADJUST	P3 is used to adjust the gain of the current isolator circuit so that CFB equals the value indicated on the label inside the right side of the card compartment (in front of the Power Supply card holder). JP3 and JP4 must be in 2-3 position in order to adjust. Factory Set - Do not adjust unless changing cards. Use following procedure if adjustment is necessary. Pots P2 and P3 are interdependent and should be adjusted alternately. P2 should be adjusted prior to moving JP3 and JP4. P3 can be set by disconnecting the current feedback signal (DRPL) and moving jumpers JP3 and JP4 to the 2-3 position (will cause a fault 46; no concern). This applies a calibration voltage (approx. 0.1V) between pin IP and IN of DRPL. Adjust P3 until test point CFB on the Control card measures the value indicated on the drive calibration label. After setting P3, P2 should be rechecked. Return JP3 and JP4 to the 1-2 positions and reconnect DRPL. **CAUTION:** If JP3 and JP4 are left in the 2-3 position, erratic operation and card failures can occur.

Location	Jumper or Pot	Number	Nomenclature	Description
MFC/Power Supply Card (continued)	Jumper	JP1 JP2		JP1 and JP2 select field current feedback signal (FC) gain. Version G3 MFC/Power Supply cards (Figure 18). **JP1 JP2 Gain Factor Field AMP Range** **2-3: 2-3:** 0.4 0.64 to 1.6 AMPS **1-2: 2-3:** 1.0 1.6 to 4.0 AMPS **1-2: 1-2:** 2.5 4.0 to 10.0 AMPS Version G2 MFC/PSC (Figure 19) and External Field Module (Figure 20). **JP1 JP2 Gain Factor Field AMP Range** **2-3: 2-3:** 1.5 1.5 to 6.0 AMPS **1-2: 2-3:** 3.0 6.0 to 12.0 AMPS **1-2: 1-2:** 6.0 12.0 to 24.0 AMPS
	Jumper	JP3 JP4		JP3 and JP4 are used for calibration of the armature current feedback signal. **JP3 JP4 Function** **1-2: 1-2:** Normal Operating Mode **2-3: 2-3:** CFB calibration (See description at pot P3.)
	Jumper	JP5		K2 coil suppression capacitor selection jumper: **1-2:** Capacitor in circuit (normal). **2-3:** Capacitor out of circuit.

WARNING - ELECTRICAL SHOCK HAZARD: High voltage is present on the Power Connection Card. Remove power from the drive before handling the plug and jumpers in these procedures.

Power Connection Card (531X308PCS)	Jumper	JP1 JP2		JP1 and JP2 select DC armature voltage range and must be stabbed on to the appropriate posts, P3 - P10. **NOTE:** *Scaling ranges are dependent on which group is being used.* **Jumper Position JP1 JP2 / G1/G3 Voltage / G2/G4 Voltage** P3-P7 P6-P10 Not Used 550 P4-P7 P5-P10 Not Used 500 P7-P8 P9-P10 500 Not Used P3-P8 P6-P9 240 240
3TB Terminal Board Card	Jumper	JP1		JP1 selects whether optional SLD5 mounted on 3TB is controlled by the Control card or by an external contact closure. **1-2:** Control card control of SLD5, programmed by ADR085, 086 and 087. **2-3:** External control of SLD5 by connecting 3TB26 K5A to 3TB56, REF24.

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional)	Switch	SW1	Tach Range Select	These switches provide coarse scaling of the tach circuit as follows: DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend.

SW1 Position No. "Closed" or "ON"	Tach Volt At Maximum Speed DC Tach	AC Tach
All "open" or "off"	10 - 12.5V	11.1 - 13.9V
1	12.5 - 15.5V	13.9 - 17.2V
2	15.5 - 19V	17.2 - 21.1V
1, 2	19 - 22V	21.1 - 24.4V
3	22 - 25V	24.4 - 27.8V
1, 3	25 - 30V	27.8 - 33.3V
1, 2, 3	30 - 37.5V	33.3 - 41.7V
4	37.5 - 46V	41.7 - 51.1V
1, 2, 4	46 - 56V	51.1 - 62.2V
1, 2, 3, 4	56 - 70V	62.2 - 77.8V
5	70 - 87V	77.8 - 96.7V
2, 3, 5	87 - 105V	96.7 - 117V
1, 2, 4, 5	105 - 120V	117 - 133V
1, 2, 3, 4, 5	120 - 144V	133 - 160V
6	144 - 180V	160 - 200V
1, 2, 4, 6	180 - 205V	200 - 228V
5, 6	205 - 250V	228 - 278V

NOTE: *SW1 postion 7 must be OFF (or open) for DC tachs or ON (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.*

	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.
	Pot	P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA.
	Pot	P3	----	Not Used

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Pot	P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, or 5 to 65mA transducer inputs. To adjust, start with PF gain at the full CCW position and procede as follows: -- Apply the current reference corresponding to minimum speed. -- Adjust PF OFF for 0 volts at test point PFRF. -- Adjust the current reference corresponding to maximum speed. -- Adjust PF GAIN for 10.0 volts at PFRF. -- Repeat as necessary **NOTE:** *To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.*
	Pot	P6	----	Not Used - Set in full CCW position for proper operation.
	Pot	P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. **1-2:** Enables RS422 Reception. **2-3:** Enables RS232C Reception.
	Jumper	JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2 JP3 Value** **1-2: 1-2:** 5-65 mA current loop follower. **1-2: 2-3:** 5-50 mA current loop follower. **2-3: 1-2:** 4-20 mA current loop follower. **2-3: 2-3:** 1-5 mA current loop follower or 10V voltage follower.
	Jumper	JP4		JP4 selects the speed meter (SMET) circuit mode (with analog tachs only). **1-2:** Unipolar speed meter driver mode. **2-3:** Bipolar meter drive mode (DC tach only).

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Jumper	JP7		Not Used - Set in 1-2 position for proper operation.
	Jumper	JP8		JP8 selects analog tach filter mode. **1-2:** Enables 4MS filter in analog tach feedback (required for AC tach operation). **2-3:** Disables 4MS filter - Required for speed regulator performance above 20 radians (DC or Digital Tach only).
	Jumper	JP11		TXD Mode Select **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode **2-3:** 12PL-9 RS232 transmitted data mode
	Jumper	JP12		DTR Mode Select **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode **2-3:** 12PL-8 RS232 DTR handshake mode
	Jumper	JP13		DTR RS232 Logic Polarity **1-2:** 12PL-8 high = drive ready to receive **2-3:** 12PL-8 low = drive ready to receive **NOTE:** *For Series 6 interfaces using the CCM card, set JP1 and JP11 to 1-2 and JP12 and JP13 to 2-3. 12PL-7 should be tied to +5V.*
	Jumper	JP14A JP14B		CTS Handshake Mode Select **JP14A** / **JP14B** / **Description** **1-2:** / **1-2:** / Enable Differential CTS Handshaking **1-2:** / **2-3:** / Enable Single-Ended CTS Handshaking **2-3:** / **2-4:** / Disable CTS Handshaking
	Jumper	JP15		Tach Counter Direction Control **1-2:** Normal control by Tach direction of rotation. **2-3:** Up only
	Jumper	JP16		Tach A Quad B Enable **1-2:** Normal A quad B tach operation. **2-3:** Count on leading edge of track A only.
	Jumper	JP17		Not Used - set in 1-2 position for proper operation.
	Jumper	JP18		Not Used - set in 1-2 position for proper operation.

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Jumper	JP19		Digital Tach Input Signal DTA Scaling **1-2:** 10 Volt Maximum Input **2-3:** 5 Volt Maximum Input **3-4:** 1.5 Volt Maximum Input **1-4:** 15 Volt Maximum Input
	Jumper	JP20		Digital Tach Input Signal DTB Scaling **1-2:** 10 Volt Maximum Input **2-3:** 5 Volt Maximum Input **3-4:** 1.5 Volt Maximum Input **1-4:** 15 Volt Maximum Input
	Jumper	JP21		Not Used - set in 1-2 position for proper operation.
	Jumper	JP22		Voltage Follower Enable **1-2:** Voltage Follower Disabled **2-3:** Voltage Follower Enabled
Application Card (optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
	Pot	P2	----	Not Used - Set in full CCW position for proper operation.
	Pot	P7	----	Not Used - Set in full CCW position for proper operation.
	Pot	P8	----	Not Used - Set in full CCW position for proper operation.
	Jumper	JP1		JP1 selects the voltage range for the auto speed reference input (ASR) at 4TB23 (13PL-23). **1-2:** 8 to 15 volts. **2-3:** 5 to 8 volts.
	Jumper	JP5 JP6		JP5 and JP6 select polarity of the logic inputs, MODE 0-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8) and should be set as follows: **JP5** **JP6** **Description** **1-2:** **3-4:** The control is active low. Inputs are active when pulled below 0.8 volts.
	Jumper	JP7		JP7 selects whether the sixteen-bit parallel input is latched on the high or low level of the strobe. **1-2:** Data latched on low to high transition of PSB. **2-3:** Data latched on low level of PSB.

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card (optional) (continued)	Jumper	JP8		JP8 selects whether the sixteen-bit parallel input and strobe signals are driven by an active high or low driver. **1-2:** Lines internally pulled up to +5 VDC. Must be driven by active pull-down to common dirver. **2-3:** Lines internally pulled down to DCOM. Must be driven by an active pull-up (above 3.5 VDC) drive.
	Jumper	JP14		JP14 set as follows: **1-2:** Should be in this position for proper operation.
	Jumper	JP24, JP25 JP26, JP27 JP28		Not Used - set in 1-2 position for proper operation.
	Jumper	JP29A, B, C JP30A, B, C JP31A, B, C		Not Used - set as follows for proper operation: **Jumper** — **Position** JP29A, JP30A, JP31A — **6-7:** JP29B, JP30B, JP31B — **1-2:** JP29C, JP30C, JP31C — **3-8:**
	Jumper	JP32		Not Used - set in 1-2 position for proper operation.
	Jumper	JP36 JP37		JP36 & JP37 set as follows: **1-2:** Should be in this position for proper operation. **Open:** Only for manufacturer's use.
	Jumper	JP40		Not Used - set in 1-2 position for proper operation.
	Jumper	JP41		Not Used - set in 1-2 position for proper operation.
	Jumper	JP45		Not Used - set in 1-2 position for proper operation.
	Jumper	JP46		Not Used - set in 1-2 position for proper operation.
	Jumper	JP54		Not Used - set in 1-2 position for proper operation.
	Jumper	JP55		Not Used - set in 1-2 position for proper operation.
	Jumper	JP56		Not Used - set in 1-2 position for proper operation.

TABLE 2. TEST POINTS

As a further aid to troubleshooting the QIII DC Controller, a series of test points have been provided to allow for specific signal observation and/or measurements. Test points not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common for analog circuit power supplies.	
CFB	Armature current feedback signal - positive when forward SCRs are conducting. Maximum external loading is 5 mA.	Calibration label voltage level at rated drive current (approx. 1.0 VDC)
DA0	Scaled output from 12-bit digital-to-analog converter (DAC).	
DCOM	Common for digital circuit power supply.	
DFP	Delayed firing power for the SCR gate circuits. Maximum external loading is 25 mA. Enabled only when armature SCR firing is allowed.	+24 VDC when enabled or 0 VDC when disabled
MUP1 (DM1)	Multiuse point 1, connects to 6PL-9. Multiuse points may be used to jumper signals from other test points to the associated connector for custom applications. Refer to elementaries supplied with drive.	
MUP2 (DM2)	Multiuse point 2, connects to 6PL-10.	
MUP3 (DM3)	Multiuse point 3, connects to 6PL-11.	
FC	Field current feedback from Power Supply card. Maximum external loading is 1 mA. FC = Max. rated motor field Amps X 4 VDC *divided by* Max. field current set by JP1and JP2 **NOTE:** *See Table 1, Power Supply card for maximum field current set by JP1 and JP2.*	
JOGR	Jog reference input at 6PL-7.	
-15V	-15 VDC power supply.	-14.25 to -15.75 VDC
+15V	+15 VDC power supply.	+14.25 to +15.75 VDC
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
R/F	High when reverse SCRs being fired. Low when forward SCRs being fired.	High = 2 to 5 VDC Low = 0 to 2 VDC
SRS	Scaled speed reference signal.	4 VDC at top speed reference
VFBB or VFB	Armature voltage feedback signal. Positive when forward motoring. Maximum external loading is .1 mA. Not for external use. **NOTE:** *VFBB is affected by P8-VMET.*	4 VDC at rated armature voltage

POWER SUPPLY CARD TEST POINTS

Test Point Name	Description	Nominal Value
TP1	Field current feedback signal (FC). Maximum loading is 1 mA. FC = I_{field}/Gain Factor (See Table 1, MFC/Power Supply Card JP1 and JP2 for Gain Factor setting)	

OPTIONAL ENCODER PROCESS INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
DVM	Probe connection for digital voltmeter - NOT USED	
SFB	Analog speed feedback signal. No value when digital tach or no tach being used in drive system.	4 VDC at top speed
PFRF	Output of the reference interface circuitry.	10 VDC at top reference
DM5	Dummy point 5, connects to 10PL-16.	
TPN	Digital or AN tach pulse output.	
STPRF	Not Used	
RPU	Not Used	

OPTIONAL APPLICATION CARD TEST POINTS

Test Point Name	Description	Nominal Value
ASR	Auto System Reference signal.	4 VDC at top speed reference
FRA	Not Used	
RPU	Not Used	
OSC	500 KHz TTL oscillator used by the shared bus encoder interface and by the real time clock on MUP.	
CLK	Not Used	
SPT	Not Used	
DM4	Dummy point 4, connects to 3PL-30. May be used to jumper signals from other test points to 3PL for custom applications, or to bring signals from the control card to the general purpose amplifier circuits.	

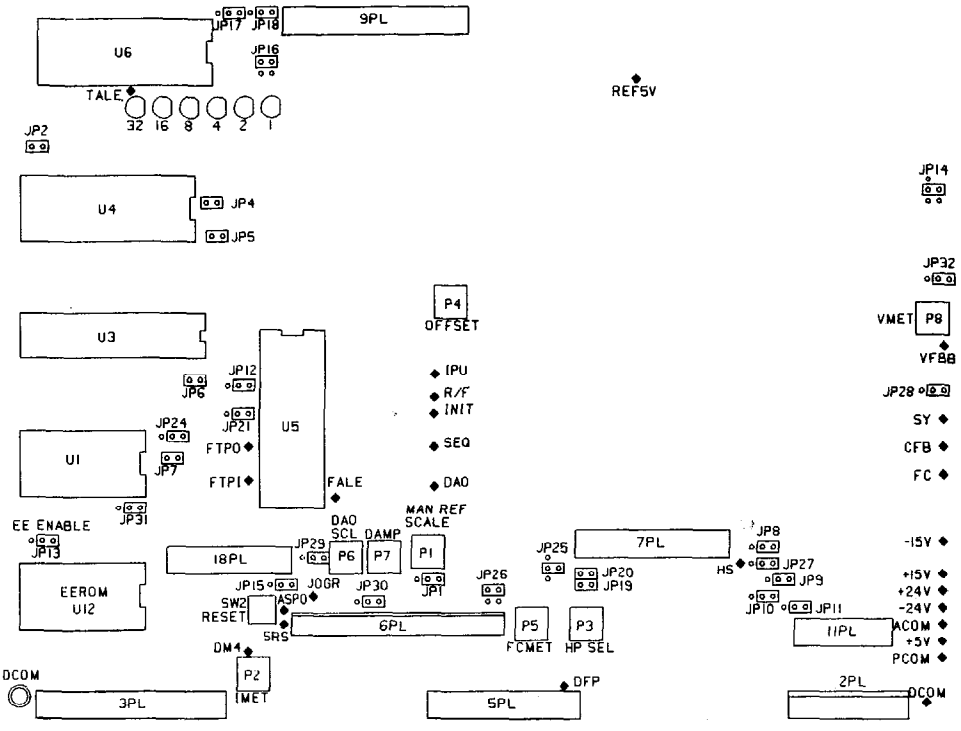


Figure 7-1. CONTROL CARD LAYOUT DIAGRAM

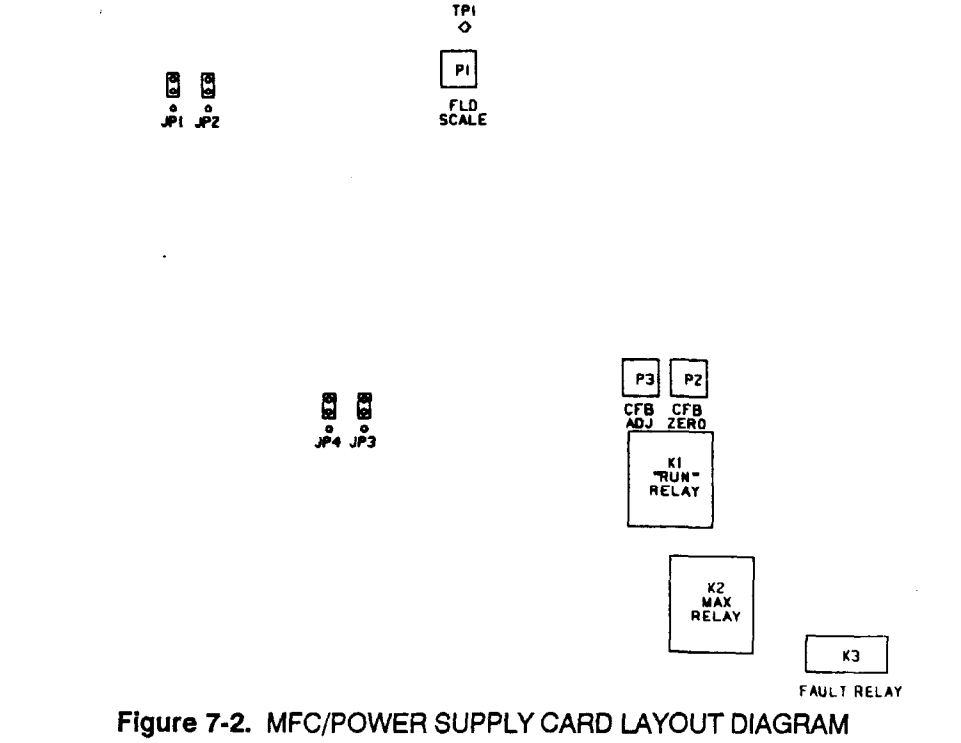


Figure 7-2. MFC/POWER SUPPLY CARD LAYOUT DIAGRAM

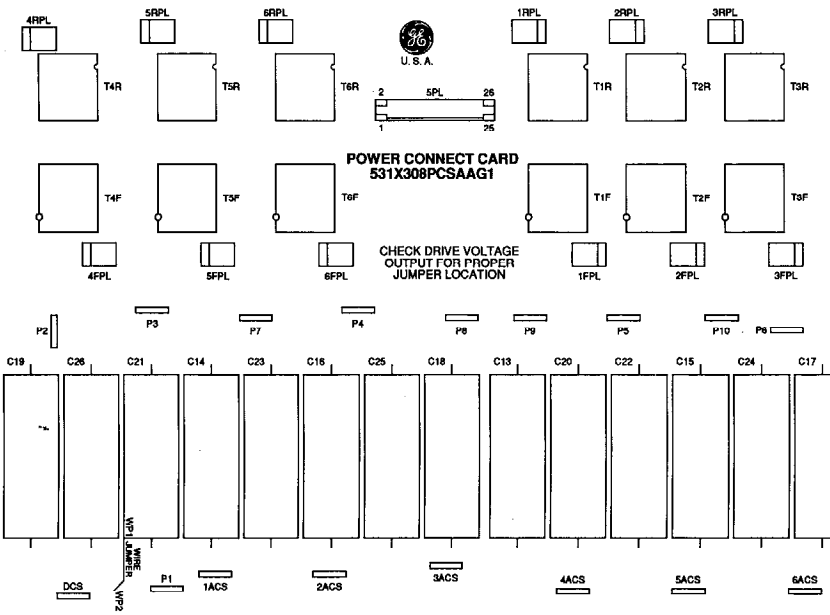


Figure 7-3. POWER CONNECTION CARD LAYOUT DIAGRAM

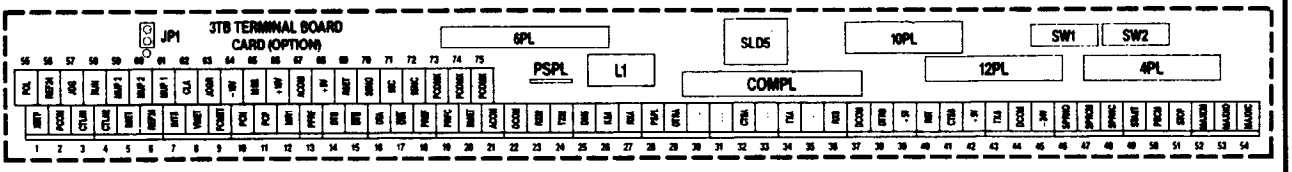


Figure 7-4. 3TB TERMINAL BOARD CARD LAYOUT DIAGRAM

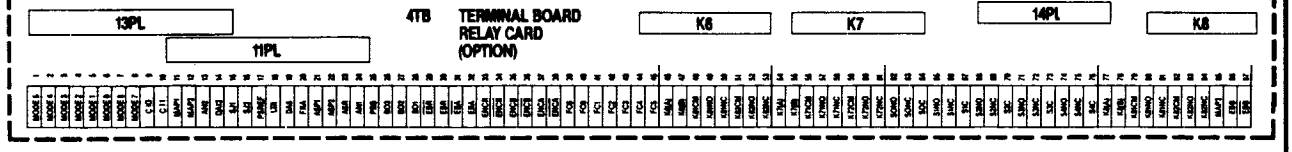


Figure 7-5. OPTIONAL 4TB TERMINAL BOARD CARD LAYOUT DIAGRAM

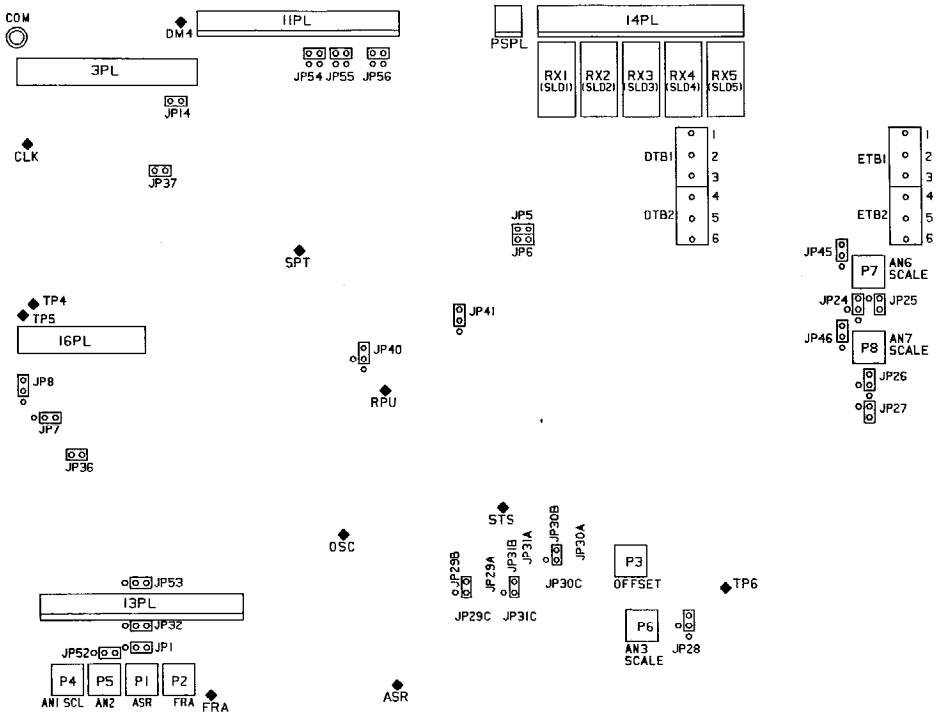


Figure 7-6. OPTIONAL APPLICATION CARD LAYOUT DIAGRAM

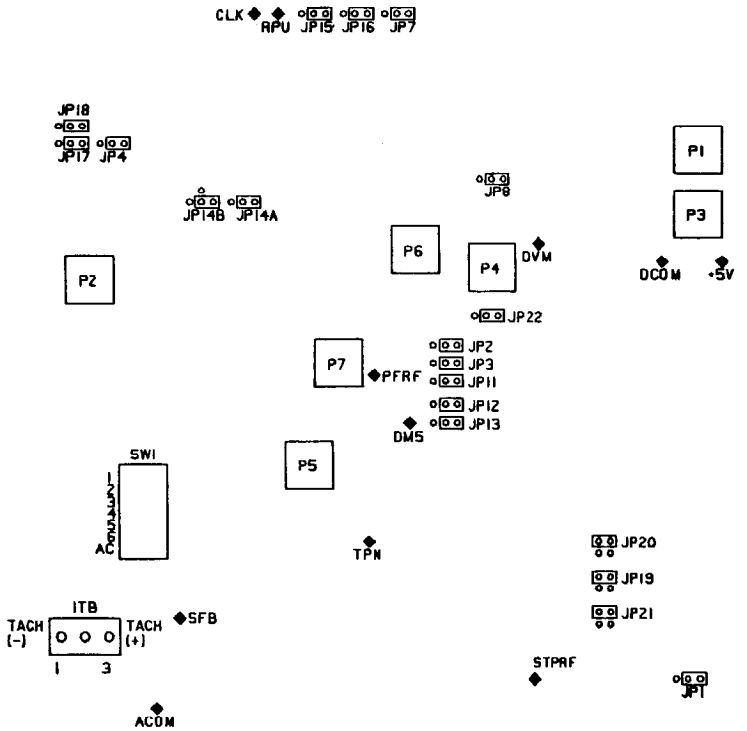


Figure 7-7. OPTIONAL ENCODER PROCESS INTERFACE CARD LAYOUT DIAGRAM

8. SOFTWARE ADJUSTMENTS

The following pages provide a listing of "Key" QIII DC Controller software adjustments (parameters) common in most QIII DC Controller applications. Included is the "Parameter Map" (equivalent analog representation of parameters, Figure 8-1), the "Key Parameter Adjustments Table" (Table 3), and the "RAM Variable Scaling Table" (Table 4), that defines the scaling of RAM addresses available for interrogation of the drive control "circuit". The parameters can be examined and adjusted with the Programming Software Package or via the serial link.

PARAMETER MAP (Figure 8-1)

The Parameter Map is a diagnostic aid designed to reference specific software adjustments and correlate them to their specific regulating functions. For example, ADR015-BSLPG is represented as a potentiometer feedback across the base speed regulating amplifier, giving an effective linear gain adjustment of input to output.

Software adjustments are shown next to their symbol with the mnemonic and the parameter address in a box beside it. Hardware adjustments are represented by a symbol, the adjustment name and the card it is located on (e.g.: FCMET is a potentiometer, P5 on the Control card). RAM addresses that can be used to examine outputs of certain stages of regulation through D-A converters are shown as a three-digit number in a circle, connected to the regulator section it is associated with (e.g.: Speed loop output can be examined by using RAM address 053 in the D-A converters).

KEY PARAMETER ADJUSTMENTS

The following parameter adjustments, listing mnemonic, address number, function name, and description, will prove helpful in setup and troubleshooting of the QIII DC Controller. This is a partial list and should be used in conjunction with the "Parameter Map". For a more complete list of parameters contained in a specific drive, refer to the Programming Software Package.

The following is a description of column headers:

NAME	Mnemonic of the parameter.
ADR	Parameter address.
UNITS	Unit designation of displayed number (example - RPM).
MIN	Minimum allowable value.
MAX	Maximum allowable value.
STEP	Amount by which value can be incremented or decremented.
DEFAULT	Standard value if no special setting is required.

NOTE: *Parameters listed in Table 3 are grouped functionally per the parameter map (Figure 8-1).*

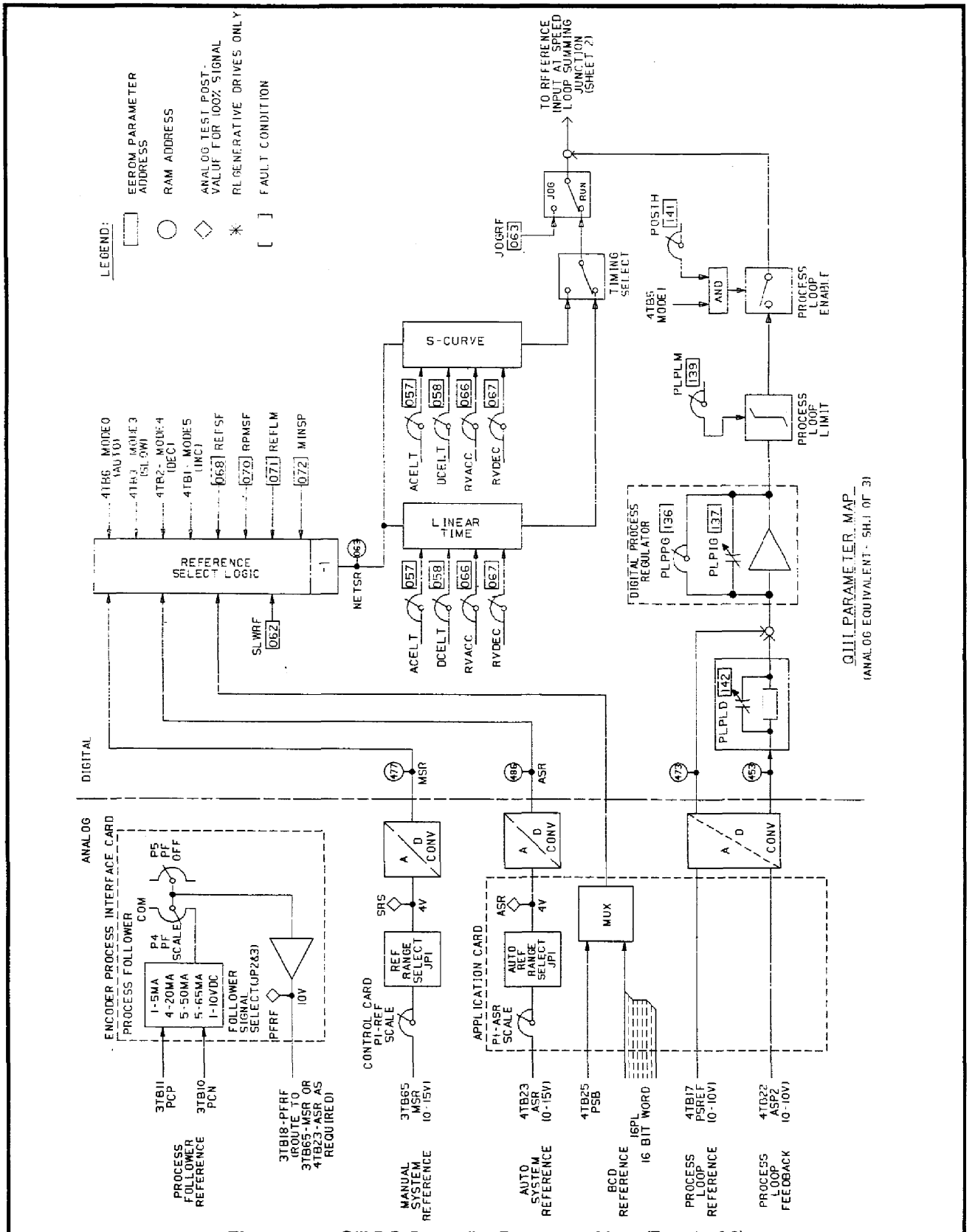


Figure 8-1. QIII DC Controller Parameter Map (Part 1 of 3)

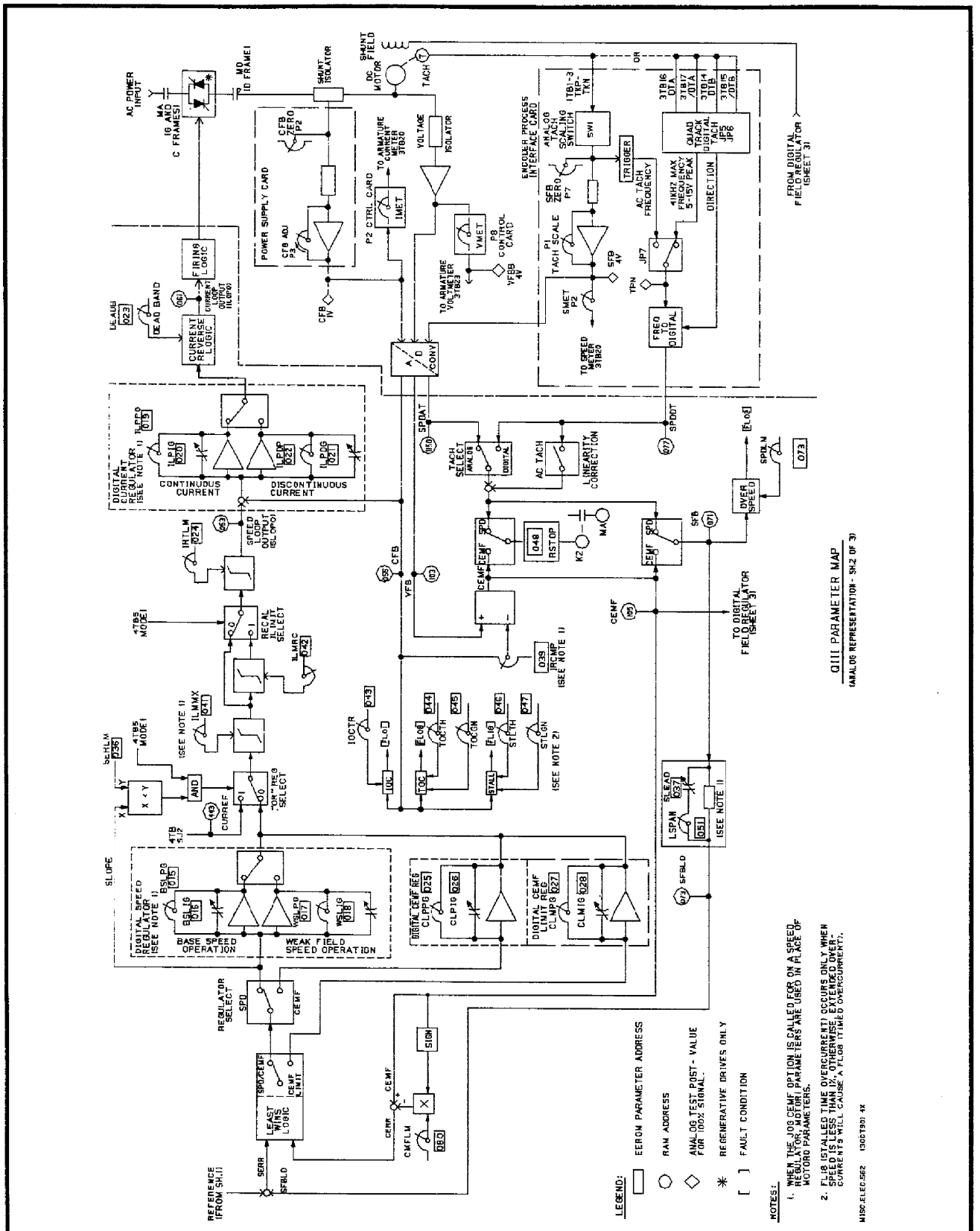


Figure 8-1. QIII DC Controller Parameter Map (Part 2 of 3)

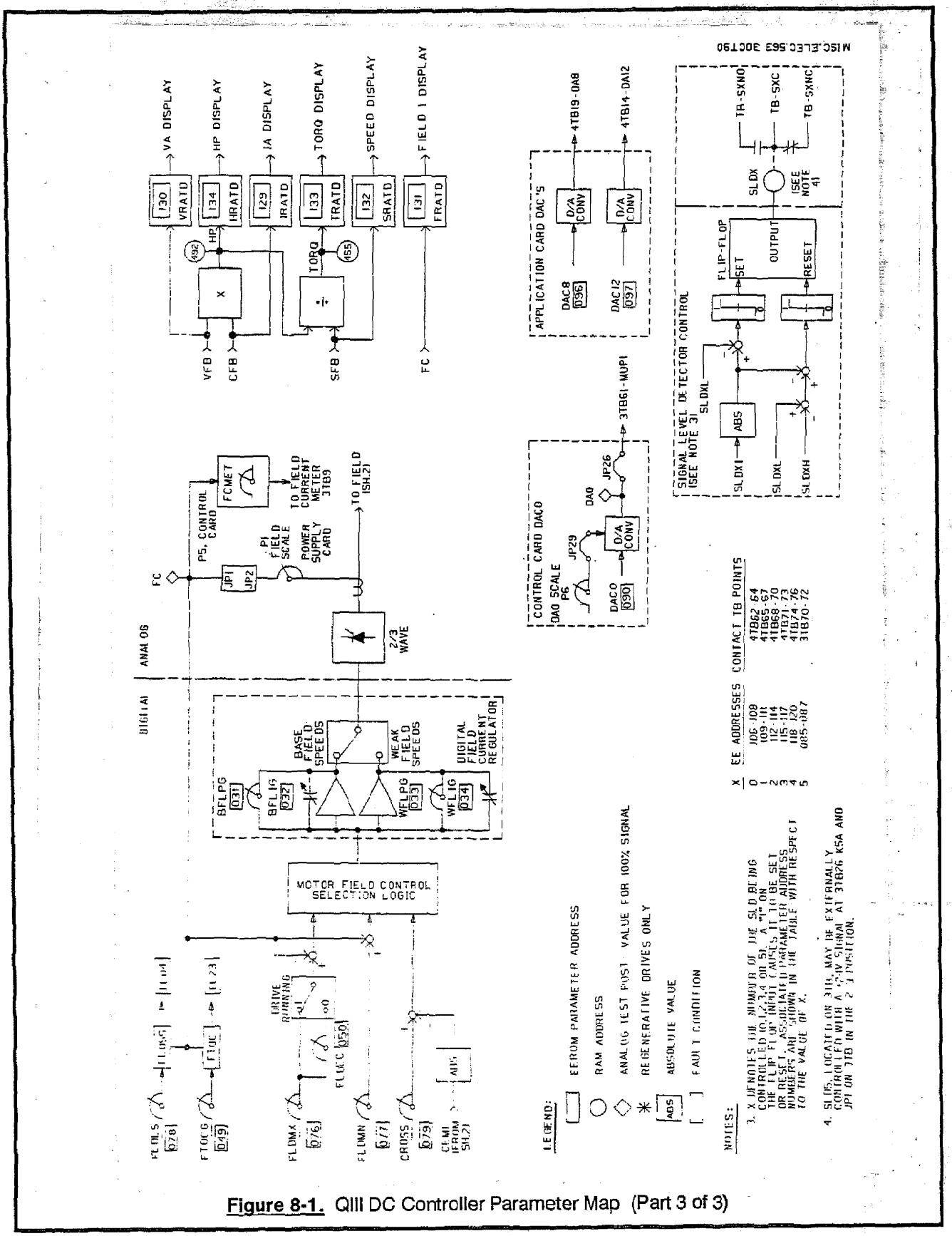


Figure 8-1. QIII DC Controller Parameter Map (Part 3 of 3)

TABLE 3: KEY PARAMETER ADJUSTMENTS

MOTOR FIELD

Name	Adr.	Units	Min.	Max.	Step	Default
FLDMX	**076**	**None**	**0**	**2047**	**1**	**800**

Full Field Current -- sets the motor field current below base speed. Also is the voltage reference in the open loop mode of field operation. In the current regulated mode, 1638 corresponds to 4 volts at test point FC. The field current scaling jumpers on the Power Supply card should be set to give the largest feedback at FC that does not exceed roughly 4.5 volts. In the open loop mode, 0 corresponds to fully retarded firing and 2047 corresponds to fully advanced firing.
FLDMX = 2048 x (If [max] / 5 x Gain Factor)
NOTE: *The gain factor is determined by the field current scaling jumpers on the Power Supply card (refer to Table 1).*

FLDMN	**077**	**None**	**0**	**2047**	**1**	**200**

Minimum Field Current -- sets the minimum field current level the drive will regulate. 1638 equals 4 volts at FC.
FLDMN = 0.6 x 2048 x If (min) *divided by* 5 x Gain Factor.
NOTE: *The gain factor is determined by the field current scaling jumpers on the Power Supply card (refer to Table 1).*

FLDLS	**078**	**None**	**0**	**2047**	**1**	**100**

Field Loss Trip Setpoint -- sets the level of field current below which the drive will trip. 1638 equals 4 volts at FC.
FLDLS = 2048 x If (min) *divided by* 12.5 x Gain Factor.
NOTE: *A setting of 60,000 disables the field loss trip function.*

CROSS	**079**	**None**	**0**	**1884**	**1**	**1850**

Field Crossover -- sets the CEMF level at which the field will begin to weaken. A setting of 1638 corresponds to rated motor voltage (4 volts at VFB). Determine % of Rated Armature volts desired at Crossover. Multiply that % by 1638 to obtain CROSS parameter setting.

FLDEC	**050**	**None**	**0**	**255**	**1**	**0**

Field Economy -- sets level of field current when drive is stopped. Determine % of FLDMX desired for field economy. Then FLDEC will equal that percent x 255.
EXAMPLE: 70% of maximum field current desired for field economy.
0.70 x 255 = 178.5
Setting of FLDEC = 179.
NOTE: *A setting of zero defaults FLDEC to 70% field current also.*

BFLPG	**031**	**None**	**0**	**255**	**1**	**40**

Base Field Loop Proportional Gain -- sets the proportional gain of the field current regulator below base speed.

BFLIG	**032**	**None**	**0**	**255**	**1**	**20**

Base Field Loop Integral Gain -- sets the integral gain of the field current regulator below base speed.

MOTOR FIELD (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
WFLPG	033	None	0	255	1	40
		Weak Field Loop Proportional Gain -- sets the proportional gain of the field current regulator above base speed.				
WFLIG	034	None	0	255	1	20
		Weak Field Loop Integral Gain -- sets the integral gain of the field current regulator above base speed.				
FTOCG	049	None	0	255	1	82
		Field TOC Gain -- sets the time constant of the field timed overcurrent integrator. Set to 0 to inhibit this feature. For field forcing of FF per unit over a setting of FLDMX, the drive will trip in 6990/(FLDMX x FF x FTOCG) minutes. The default allows 0.1 P.U. forcing over a nominal FLDMX of 850 for 1 minute.				

REFERENCE TIMING

ACELT	057	Decisec	0	32000	1	150
		Acceleration Time -- sets the rate at which motor speed will increase in response to speed change command. For linear time, enter time in tenths of a second for accel from 0 to top speed. ACELT = Time (in sec.) x 10. Some drives may have an optional extended linear time range. Refer to the Programming Software Package for information.				
DCELT	058	Decisec	0	32000	1	150
		Deceleration Time -- sets the rate at which motor speed will decrease in response to speed change command. For linear time, enter time in tenths of a second for decel from top speed to 0. DCELT = Time (in sec.) x 10. Some drives may have an optional extended linear time range. Refer to the Programming Software Package for information.				
RVACC	066	Decisec	0	32000	1	150
		Reverse Acceleration Time -- sets the optional independent reverse acceleration. Operation is same as ACELT (057) when enabled and references are negative (check REFJP). For S-curve, sets the acceleration limit, with 255 being the maximum (fastest) value.				
RVDEC	067	Decisec	0	32000	1	150
		Reverse Deceleration Time -- sets the optional independent reverse deceleration. Operation is same as DCELT (058) when enabled and references are negative.				
LINT1	154	Decisec	0	65535	1	0
		Motor #1 Linear Time -- sets the accel and decel (non-independent) times while motor #1 is selected.				

REFERENCES AND SCALING

Name	Adr.	Units	Min.	Max.	Step	Default
REFSF	**068**	**None**	**0**	**6553**	**1**	**6553**

Reference Scale Factor -- sets the factor by which references entered from the serial link will be multiplied and by which speeds sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). Important - affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLKRF, parallel input scaling, and speed reference display. With Tach, REFSF = Top Speed RPM; for CEMF regulator, REFSF = Base Speed RPM at rated armature volts.

SLWRF	**062**	**RPM**	**0**	**6553**	**1**	**0**

"Slow" Key Reference -- sets the preset reference for SLOW commands from the serial link. Normally, SLWRF = Slow Speed (RPM).

JOGRF	**063**	**RPM**	**0**	**6553**	**1**	**10% of REFSF**

"Jog" Key Reference -- sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. JOGRF = Jog Speed (RPM).

RPMSF	**070**	**RPM**	**0**	**6553**	**1**	**6553**

RPM Scale Factor -- sets the speed in RPM which corresponds to 4 volts at SFB.

NOTE: *Change parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective.*

IMPORTANT: *Affects setting of REFLM, MINSP, SPDLM, tapered current limit speed setting and digital tach scaling. Same as setting for REFSF.*

REFLM	**071**	**RPM**	**0**	**6656**	**1**	**6560**

Maximum Speed -- sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1.015 x RPMSF. Typically, REFLM = RUNRF + 100.

MINSP	**072**	**RPM**	**0**	**6553**	**1**	**0**

Minimum Speed -- sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. MINSP = desired Minimum Speed (RPM). Maximum value should not exceed RMPSF.

SPDLM	**073**	**RPM**	**0**	**7936**	**1**	**7208**

Speed Limit Trip Setpoint -- sets the speed at which the drive will trip. SPDLM = RPM at drive overspeed trip. Should not exceed 1.2 x RPMSF.

SPEED REGULATOR

Name	Adr.	Units	Min.	Max.	Step	Default
BSLPG	015	None	0	255	1	7
		Base Speed Loop Proportional Gain -- sets the proportional gain of the outer speed regulator below base speed when tachometer feedback is used.				
BSLP1	143	None	0	255	1	0
		Motor #1 Base Speed Loop Proportional Gain -- sets the proportional gain of the outer speed regulator below base speed when motor #1 is selected in a multimotor environment. **NOTE:** *Changes to motor #1 parameters while the drive is running will not become effective until the drive is stopped.*				
BSLIG	016	None	0	255	1	8
		Base Speed Loop Integral Gain -- sets the integral gain of the outer speed regulator below base speed when tachometer feedback is used.				
BSLI1	144	None	0	255	1	0
		Motor #1 Base Speed Loop Integral Gain -- sets the integral gain of the outer speed regulator below base speed when motor #1 is selected in a multimotor environment.				
TCHLN	052	None	0	31	1	0
		AN Tach Linearization Loop Response -- sets the response of the AN (low PPR) tach digital speed correction loop. Program to 0 to inhibit the loop. Time constant in seconds is 25.6/TCHLN for DTCHR = 100.				
DTCHR	053	Millisec	0	255	1	15
		Digital Tach Speed Loop Update Rate -- sets the frequency at which the digital tach speed calculation is done. Set to 15 milliseconds for most digital tachs, 50 milliseconds for 18PPR AN tachs.				
TCPPR	074	None	0	4096	1	18
		Digital Tach Pulses Per Revolution -- sets the assumed number of pulses at TPN from the Encoder Process Interface card corresponding to one revolution.				
WSLPG	017	None	0	255	1	7
		Weak Speed Loop Proportional Gain -- sets the proportional gain of the outer speed regulator above base speed.				
WSLIG	018	None	0	255	1	8
		Weak Speed Loop Integral Gain -- sets the integral gain of the outer speed regulator above base speed.				
SLEAD	037	None	0	255	1	10
		Speed Feedback Lead -- sets the lead time constant on tachometer feedback.				
SLEA1	150	None	0	255	1	0
		Motor #1 Speed Feedback Lead -- sets the lead time constant on tachometer feedback when motor #1 is selected in a multimotor environment. If LSPA1 (184) is set to 128 or larger, filter becomes a lag only.				

SPEED REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
LSPAN	051	None	0	15	1	0
		Speed Feedback Lead/Lag Span -- sets the span of the tachometer lead/lag network. A setting of 0 is treated as a two-to-one span.				
LSPA1	184	None	0	15	1	0
		Motor #1 Speed Feedback Lead/Lag Span -- sets the span of the tachometer lead/lag network when motor #1 is selected in a multimotor environment.				
RSTOP	048	None	0	255	1	100
		Regenerative Stop Speed -- sets the speed where the contactor opens during a regenerative stop. One Unit = 0.04% speed (for Speed Reg.). One Unit = 0.12% volts (for CEMF Reg.).				

CEMF REGULATOR

CMFLM	080	None	0	1884	1	1800
		CEMF Limit -- sets the CEMF level where the drive will clamp. A setting of 1638 corresponds to rated motor voltage (4 volts at VFB). CMFLM = 1638 x (% motor rated voltage desired for CEMF limit). **CAUTION:** Do not set CMFLM above 1704 on regenerative drives. This may cause fuse and SCR failure to occur on regeneration.				
CLPPG	025	None	0	255	1	8
		(Tachless) CEMF Loop Proportional Gain -- sets the proportional gain of the outer CEMF regulator when CEMF is used as a feedback instead of a tachometer.				
CLPIG	026	None	0	255	1	10
		(Tachless) CEMF Loop Integral Gain -- sets the gain of the outr CEMF regulator in no tach mode.				
CLMPG	027	None	0	255	1	8
		CEMF Limit Proportional Gain -- sets the proportional gain of the spillover CEMF limit regulator.				
CLMIG	028	None	0	255	1	10
		CEMF Limit Integral Gain -- sets the integral gain of the spillover CEMF limit regulator.				
IRCMP	039	None	0	255	1	0
		IR Compensation -- sets positive or negative IR compensation for CEMF calculation.				
IRCM1	151	None	0	255	1	0
		Motor #1 IR Compensation -- sets positive or negative IR compensation for CEMF calculation when motor #1 is selected in a multimotor environment.				

ARMATURE CURRENT REGULATOR

Name	Adr.	Units	Min.	Max.	Step	Default
ILMMX	041	% Rated	10	510	2	150
		Maximum Current Limit -- sets the maximum DC current level allowed by the drive. One unit = 1% current.				
ILMM1	152	%/2	10	255	1	0
		Motor #1 Maximum Current Limit -- sets the maximum DC current level allowed by the drive when motor #1 is selected in a multimotor environment. One unit = 2% current.				
ILMRC	042	% Rated	10	510	2	150
		Recalibrated Current Limit -- sets the current limit in the limited torque mode commanded by RECAL input from the optional Application card #1. One unit = 1% current.				
IOCTR	043	% Rated	10	510	2	400
		Instantaneous Overcurrent Trip Setpoint -- sets the current level at which the drive will trip. One unit = 1% current.				
TOCTH	044	% Rated	10	120	2	120
		Timed Overcurrent Sense Threshold -- sets the current level at which the TOC integrator begins to time-up. One unit = 1% current.				
TOCGN	045	None	0	255	1	24
		Timed Overcurrent Response -- sets the time constant of the TOC integrator. TOCGN = 66,666 / ([Trip TIme] x [X-TOCTH]) **NOTE:** *X = % of armature current corresponding to desired trip time.* **NOTE:** *The default values for TOCTH and TOCGN allow the drive to operate at 150% current for 90 seconds.*				
STLTH	046	% Rated	10	50	2	30
		Stalled Overcurrent Sense Threshhold -- sets the current level at which the stalled motor TOC integrator begins to time up. One unit = 1% current.				
STLGN	047	None	0	65535	1	55
		Stalled Overcurrent Response -- sets the time constant of the stalled motor TOC integrator. Defining equation is same as TOCGN (045). The default allows operation at 150% for 10 seconds.				
ILPPG	019	None	0	255	1	40
		Current Loop Proportional Gain -- sets the proportional gain of the inner current regulator in continuous current.				
ILPP1	147	None	0	255	1	0
		Motor #1 Current Loop Proportional Gain -- sets the proportional gain of the inner current regulator in continuous current when motor #1 is selected in a multimotor environment.				
ILPIG	020	None	0	255	1	20
		Current Loop Integral Gain -- sets the integral gain of the inner current regulator in continuous current.				

ARMATURE CURRENT REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
ILPI1	148	None	0	255	1	0

Motor #1 Current Loop Integral Gain -- sets the integral gain of the inner current regulator in continuous current when motor #1 is selected in a multimotor environment.

ILPDG	021	None	0	255	1	30

Current Loop Integral Discontinuous Gain -- sets the integral gain of the inner current regulator in discontinuous current.

ILPD1	149	None	0	255	1	0

Motor #1 Current Loop Integral Discontinuous Gain -- sets the integral gain of the inner current regulator in discontinuous current when motor #1 is selected in a multimotor environment.

ILPDP	022	None	0	255	1	0

Current Loop Proportional Discontinuous Gain -- sets the proportional gain of the inner current regulator in discontinuous current.

DEADB	023	None	0	255	1	0

Current Regulator Deadband -- controls the sensitivity of the current loop reversal logic. A setting of 0 is maximum sensitivity (either mode).

IRTLM	024	PU/S	1	50	1	25

Current Rate Limit -- controls the maximum rate of change of current in response to commanded changes.
Exact relation is: Current Rate = 0.9 (IRTLM) per unit per second.

OUTPUT SIGNALS

DAC0	090	None	0	65535	1	0

D-A #0 Output Control -- controls what internal variable is sent to the 12-bit D-A converter (DAC0) on the Control card. Enter the desired RAM variable number into ADR090 to program DAC0.

RAM Adr.	Variable Name	RAM Adr.	Variable Name
492	Horsepower	073	SFB - Speed Feedback
055	CFB - Current Feedback	103	VFB - Voltage Feedback
455	Torque	105	CEMF
065	Timed Speed Reference	463	Field Current
501	TOC Trip		

NOTE: *Refer to the "RAM Variable Scaling", Table 3 at the end of this listing for relative value of RAM addresses through various outputs.*

DAC8	096	None	0	65535	1	0

D-A #1 Output Control -- controls what internal variable is sent to the 8-bit D-A converter (DAC8) on the optional Application card. See DAC0 for programming information.

OUTPUT SIGNALS (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
DAC12	097	None	0	65535	1	0

D-A #2 Output Control -- controls what internal variable is sent to the 12-bit D-A converter (DAC12) on the optional Application card. See DAC0 for programming information.

Name	Adr.	Units	Min.	Max.	Step	Default
BIT #1	098	None	0	65535	1	0

Bit#1 Output Control -- controls what internal bit is sent to bit output #1 (B01) on the optional Application card. Possible outputs include:

RAM Adr.	Variable Name	RAM Adr.	Variable Name
061	Direction of Firing	148	Reference Enable
137	Fault	152	Precondition
140	Current Limit Flag	156	Zero Current Flag
141	CEMF Limit Flag		

NOTE: *In addition, the sign of all of the variables given in DAC8 output control may be used as outputs.*

Name	Adr.	Units	Min.	Max.	Step	Default
BIT #2	099	None	0	65535	1	0

Bit#2 Output Control -- controls what internal bit is sent to bit output #2 (B02) on the optional Application card. See Bit #1 for programming information.

Name	Adr.	Units	Min.	Max.	Step	Default
BIT #3	100	None	0	65535	1	0

Bit#3 Output Control -- controls what internal bit is sent to bit output #3 (B03) on the optional Application card. See Bit #1 for programming information.

DIAGNOSTICS

Name	Adr.	Units	Min.	Max.	Step	Default
DIAG1	122	None	0	65535	1	0

Diagnostic Variable #1 -- selects 1 of 2 variables which can be displayed on the serial link. See DAC0 output control for programming information.

Name	Adr.	Units	Min.	Max.	Step	Default
DIAG2	123	None	0	65535	1	0

Diagnostic Variable #2 -- selects second variable which can be displayed on the serial link. See DAC0 output control for programming information.

Name	Adr.	Units	Min.	Max.	Step	Default
IRATD	129	None	0	3276	1	100

Displayed Arm Current Scaling -- controls the factor by which per unit current is multiplied for display on the serial link.

Name	Adr.	Units	Min.	Max.	Step	Default
VRATD	130	None	0	2621	1	100

Displayed Arms Volts Scaling -- controls the factor by which per unit armature voltage is multiplied for display on the serial link.

Name	Adr.	Units	Min.	Max.	Step	Default
FRATD	131	None	0	43690	1	1000

Displayed Field Current Scaling -- controls the factor by which per unit field current is multiplied for display on the serial link.

Name	Adr.	Units	Min.	Max.	Step	Default
SRATD	132	None	0	6200	1	100

Displayed Speed Scaling -- controls the factor by which per unit speed is multiplied for display on the serial link.

DIAGNOSTICS (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
TRATD	133	None	0	3276	1	100
		Displayed Torque Scaling -- controls the factor by which per unit torque is multiplied for display on the serial link.				
HRATD	134	None	0	3276	1	100
		Displayed HP Scaling -- controls the factor by which per unit horsepower is multiplied for display on the serial link.				
BAUDR	126	None	300	4800	1	4800
		External Serial Link Baud Rate -- selects baud rate of serial link. Standard rates are 300, 1200, 2400, and 4800. Changes to this address become effective only on hard resets.				

SIGNAL LEVEL DETECTORS (SLDs)

SLD5I	085	None	0	65535	1	0
		SLD #5 Input -- controls which variable is used as the input to the signal level detector #5 on the 3TB Terminal Board card.				
SLD5L	086	None	0	65535	1	0
		SLD #5 Set Point -- controls the threshhold where SLD #5 picks-up or drops-out.				
SLD5H	087	None	0	255	1	0
		SLD #5 Hysteresis -- controls the hysteresis of SLD #5.				
SLD0I	106	None	0	65535	1	0
		SLD #0 Input -- controls which variable is used as the input to the signal level detector #0 on the optional Application card. BIT variables are 0 when cleared and -32640 when set.				
SLD0L	107	None	0	65535	1	0
		SLD #0 Set Point -- controls the threshhold where SLD #0 picks-up or drops-out.				
SLD0H	108	None	0	255	1	0
		SLD #0 Hysteresis -- controls the hysteresis of SLD #0.				
SLD1I	109	None	0	65535	1	0
		SLD #1 Input -- controls which variable is used as the input to the signal level detector #1 on the optional Application card. BIT variables are 0 when cleared and -32640 when set.				
SLD1L	110	None	0	65535	1	0
		SLD #1 Set Point -- controls the threshhold where SLD #1 picks-up or drops-out.				

SIGNAL LEVEL DETECTORS (SLDs) (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
SLD1H	111	None	0	255	1	0
		SLD #1 Hysteresis -- controls the hysteresis of SLD #1.				
SLD2I	112	None	0	65535	1	0
		SLD #2 Input -- controls which variable is used as the input to the signal level detector #2 on the optional Application card. BIT variables are 0 when cleared and -32640 when set.				
SLD2L	113	None	0	65535	1	0
		SLD #2 Set Point -- controls the threshhold where SLD #2 picks-up or drops-out.				
SLD2H	114	None	0	255	1	0
		SLD #2 Hysteresis -- controls the hysteresis of SLD #2.				
SLD3I	115	None	0	65535	1	0
		SLD #3 Input -- controls which variable is used as the input to the signal level detector #3 on the optional Application card. BIT variables are 0 when cleared and -32640 when set.				
SLD3L	116	None	0	65535	1	0
		SLD #3 Set Point -- controls the threshhold where SLD #3 picks-up or drops-out.				
SLD3H	117	None	0	255	1	0
		SLD #3 Hysteresis -- controls the hysteresis of SLD #3.				
SLD4I	118	None	0	65535	1	0
		SLD #4 Input -- controls which variable is used as the input to the signal level detector #4 on the optional Application card. BIT variables are 0 when cleared and -32640 when set.				
SLD4L	119	None	0	65535	1	0
		SLD #4 Set Point -- controls the threshhold where SLD #4 picks-up or drops-out.				
SLD4H	120	None	0	255	1	0
		SLD #4 Hysteresis -- controls the hysteresis of SLD #4.				

PROCESS LOOP/"OR" REGULATORS

PLPPG	136	None	0	255	1	0
		Outer Position Loop Proportional Gain -- sets the proportional gain of the outer position loop for spindle orient or set-point controller.				
PLPIG	137	None	0	255	1	0
		Outer Position Loop Integral Gain -- sets the integral gain of the outer loop regulator.				

PROCESS LOOP/"OR" REGULATORS (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
PLPLM	139	None	0	6553	1	0

Position Loop Output Limit -- sets the level of the output clamp of the position loop. Limits the speed of the drive when the position loop is active. Top speed corresponds to 6553.

PLPLD	142	None	0	255	1	0

Position Loop Lead -- sets the time constant of the position loop feedback lead network. Transfer function is I + ST, where T = .065 x PLPLD seconds.

IRATI	054	None	0	65535	1	0

Current Reference Ratio -- sets the ratio scale factor of the current follower or "OR" regulator current reference. When set greater than 0, the reference is multiplied by IRATI/4096 before being stored at CUR_REF (00443). The default of 0 gives a scale factor of 1. IRATI is treated as a two's complement number, with settings greater than 32767 having the value IRATI-65535. This allows a total ratio range of -8 to +8.

BCOMP	210	None	0	2047	1	0

Block Compensation -- sets the gain of the block compensation function that adds a current to the armature current reference as a function of acceleration. The default setting of 0 inhibits block compensation. The percent current compensation is given by 1.68 x BCOMP/(seconds to accellerate from 0 to top speed). This feature requires an 8032 processor for RUP.

TABLE 4. RAM VARIABLE SCALING FOR DAC OUTPUTS

RAM Adr.	Variable Name	Value At 100% Signal Dec.	Value At 100% Signal Hex(1)	Volts On DAC(2) At 100% Signal
055	**CFB** - Current Feedback	100	0064	2.0
065	**TR** - Timed Reference (if used)	6553	1999	8.0
071	**SFB** - Speed Feedback	6553	1999	8.0
103	**VFB** - Voltage Feedback	1638	0666	8.0
105	**CEMF** - Counter Electromotive Force	1638	0666	8.0
455	**TORQ** - Torque	800	320	3.9
463	**FC** - Field Current (rated current, not scaled)	2047	07FF	10.0
492	**HP** - Horsepower	10238	27FE	3.1
501	**TOCT** - Timed Overcurrent Timer (trips at 1024)	1024	0400	0.3

NOTES: *(1) Negative numbers are represented in Two's complement.*
(2) D/A converters DAC8 and DAC12 have an op-amp with a gain of two.

NOTES:

9. TROUBLESHOOTING

This section contains information on detecting and correcting problems that may be encountered when working on a QIII DC Controller. The QIII DC Controller contains a built-in fault detection system that indicates when a fault occurs and the nature of the fault. Tables 5 and 6 are intended to aid in the correction of problems both indicated by the drive fault system and those not indicated by the fault display. Following Table 6 is a "Stability" Troubleshooting paragraph and Table 7, "Stability Troubleshooting".

Table 5, "BugBuster™ Troubleshooting Chart" is for aid in troubleshooting problems that have no associated fault or error indication or cannot be remedied through the standard fault system. Table 4 includes a "Quick Reference Guide", an outline of the possible problem descriptions included in the chart.

Table 6, "Specific Fault Troubleshooting" defines drive fault and error codes. Fault codes are displayed on the Diagnostic LEDs and broadcast on the optional Serial Link. Error codes are only related to Serial Link errors, and therefore, only broadcast on the Serial Link.

FAULT CODES

The QIII DC Controller displays faults with a series of Diagnostic LEDs (interpreted as a binary number).

NOTE: *In the normal condition the drive's LEDs will flash from right to left as follows:*

***One** at a time in the standby mode.*
***Two** at a time in the RUN or JOG modes.*

Table 6 lists predefined fault codes in order of the assigned fault number. To determine the fault number from the Diagnostic LEDs, add the number value of the LEDs printed on the card above or below the LEDs (also add to this total the value of 64 if the LEDs are blinking). See EXAMPLE.

TYPES OF FAULTS

The faults listed in Table 6 may be divided into four categories:

L - The fault is **latched**, i.e., once it occurs it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

N - The fault is **non-latched**, i.e., it will clear itself if the cause of the fault no longer exists.

T - The fault **trips the drive**, i.e., it causes the controller to remove power from the motor. The controller must be reset. **All trip faults are latched.**

A - The fault is **annunciated only**, i.e., if the drive is running it will not cause the drive to shut down. However, an annunciated fault will prevent the drive from starting if it is not running. Annunciated faults may or may not be latched.

There are several ways to reset the controller.

-- A **software reset** may be executed by serial command from a device connected to the serial link.

-- A **hardware reset** may be executed by toggling the reset switch on the Control card, by momentarily connecting 6PL-1 to 6PL-4 (3TB69 to 3TB66), or by momentarily interrupting incoming power to the drive.

Most latched faults may be adequately cleared by a software reset, however, some faults should be cleared by a hardware reset. These are indicated in Table 6 by the letter "H". If a software reset does not seem to restore the drive to normal operation, a hardware reset should be attempted.

EXAMPLE:

FAULT CODE IDENTIFICATION

LED Indication	Binary Number						Decimal Number	Fault Number
	32	16	8	4	2	1		
Steady display	0	1	1	1	0	0	16+8+4 = 28	FL28
Blinking display	0	1	0	1	0	0	16+4+64 = 84	FL84

NOTE: *0 = LED OFF 1 = LED ON 1 = BLINKING LED*

BUGBUSTER™ TROUBLESHOOTING CHART QUICK REFERENCE GUIDE

PROBLEM DESCRIPTION

DRIVE FAULTS, LED INDICATION

1. All six diagnostic LEDs off
2. All six diagnostic LEDs on
3. Fault messages displayed (repetitive)
4. Fault messages displayed (non-repetitive, random)

FUSE BLOWING, SCR PROBLEMS, MOVE FAILURES, CARD FAILURES

5. SCR Test Failure
6. Fuse blowing, AC or DC power fuses
7. SCR test blowing AC or DC power fuses
8. Field fuses blowing
9. Power supply fuses blowing
10. MOV failures
11. Repetitive card failures
12. Random SCR misfirings
13. Line voltage problems

OPERATIONAL PROBLEMS

14. 24 Volt logic not operating properly
15. Drive/Motor does not run
16. Field current or voltage wrong value
17. Drive runs in wrong direction
18. Drive does not reverse
19. Drive speed is incorrect
20. Serial link does not function properly
21. Erratic operation
22. Stability problems
23. Signal level detector chatter
24. Too much coast time on deceleration (regen. drives)

MOTOR PROBLEMS

25. Motor sparking
26. Motor runs hot
27. Motor audible noise
28. Motor vibration

NOISE, WIRING, POWER SUPPLIES, ETC.

29. EEROM memory rewrites
30. Noise or random nuisance faults
31. Wiring procedures
32. Checking power supplies

Problem	Check/Adjust/Replace
6. Fuse Blowing (AC or DC Power Fuses)	1. Compare the fuse size with the Renewal Parts List. 2. Check for failed SCRs by running the SCR test. 3. Check the setting of ADR080-CMFLM on regenerative drives; should be 1704 or less. 4. Check for out of tolerance or sagging AC line voltage on regenerative drives - refer to "Line Voltage Problems" and "Noise or Random Nuisance Faults" this section. 5. Check the field power module for a short. 6. Using a megger, check for a ground in motor field, armature circuit or blower circuit. 7. Is the drive forced into Current Limit by an overhauling load (disables CEMF limit while in regeneration)? 8. Adjust the field control parameters to provide a lower armature voltage at top speed - lower ADR079-CROSS on field weakening drives or ADR076-FLDMX on base speed drives. 9. Refer to "Random SCR Misfirings" this section.
7. SCR Test Blowing AC or DC Power Fuses	1. Check for shorted SCRs with an ohmmeter from each MA or MD contactor pole to both sides of the DC bus. 2. Check for ground fault in the motor armature circuit. 3. Check the phase rotation and connection of the AC sync wires at the incoming fuse breaker and Power Supply card. 4. Check the phase rotation and connection of the AC power wires at the incoming fuses, breaker, MA or MD contactor, and SCRs. 5. Check that the phasing of AC voltages at line fuses, power supply, and field exciter input are all exactly in phase with each other.
8. Field Fuses Blowing	1. Check for a ground in the motor field or conduit wiring. 2. Check the field power module for a short or failure. 3. Verify that all wires on the Power Supply card are connected to the correct stab-on. 4. Verify that FSPL is plugged into the proper position with the correct polarity. 5. Check for shorted MOVs on the Power Supply card. 6. Check for shorted Control Power Transformer (CPT) primary.
9. Power Supply Fuses Blowing	1. Verify that the fuses are correct size. 2. Check for any shorted or over-loaded power supplies (Refer to "Checking Power Supplies", this section). 3. If the +15 VDC supplies a digital tach, verify that a cooling fan is installed below the card assembly on the QIII DC Controller and is operating properly. 4. Replace the Power Supply card.
10. MOV Failures	1. Refer to "Line Voltage Problems", this section. 2. Motor fault causes AC fuses to open during regeneration. Check for grounded armature, field, or wiring.

Problem	Check/Adjust/Replace
11. Repetitive Card Failures	1. Check all connections at 2TB, 3TB, and 4TB. 2. Verify that all ribbon cables and plugs are connected properly (i.e., not offset by one row or pin). 3. Check the stab-on connection points on the Power Supply and Power Connection cards. 4. Verify that all voltages (incoming AC and internal DC) are correct. 5. Check for moisture. 6. Check for damage due to airborne chemical contamination.
12. Random SCR Misfirings	1. Check for inoperative SCRs. 2. Check for loose connections on the fuses, breaker, MA or MD contactor, SCRs, or any armature connections. 3. Check for loose connections on SCR gate leads, Power Connection card, or 5PL. 4. Check for notching on the AC line - this could be caused by other drives on the same incoming AC line of a horsepower equal to or greater than **four to six times** the horsepower of the problem drive. 5. Refer to "Line Voltage Problems", this section.
13. Line Voltage Problems	1. Verify that the line voltage is within drive specifications, -5% to +10%. 2. Verify that the AC line voltage is balanced line-to-line and line-to-ground. 3. Check for AC line notches or spikes. 4. Verify that the line frequency is within $\pm$1 Hz of rated. 5. When AC line is supplied by a diesel driven alternator, check for changes in line frequency during drive acceleration or regeneration. 6. Check for more than one and one-half cycles of ringing following commutation notches in the AC line caused by the drive. Amplitude should be less than or equal to 200 volts. 7. Check for an ungrounded secondary on the power system transformer directly in front of the drive (if used). (The transformers are generally a delta/wye type with the neutral of the "wye" grounded through a high resistance or directly to ground.)
14. 24 Volt Logic Not Operating Properly	1. Check the $\pm$24 volt power supplies. 2. Check logic polarity control Jumpers, JP19 and 20 on the Control card, JP5 and JP6 on the optional Application card. 3. Check control logic connections at 3TB and 4TB. 4. 24 volt logic being switched too quickly to optional 4TB or optional Application card - there must be at least a 70 millisecond delay between two logic inputs changing state. 5. Check for proper polarity and common circuit if 24 volt supply is from a source other than drive. 6. Check that the drive is in the proper motor mode (0, 1, 2 or 3) for the particular motor or function being operated.

Problem	Check/Adjust/Replace
15. Drive/Motor Does Not Run	1. No connections to REF 24, ±15V, or common should be made at 4TB. Move any such connections to 3TB. 2. FU1, 2, 3, or 4 open (or optional circuit breaker open). 3. CFU1, 2, 3, 4, or 5 open and/or MOVs shorted on Power Supply card. 4. Check all hardware jumpers, particularly ADR012-OPTJP, JP19 and JP20 on the Control card and JP5 and JP6 on the optional Application card. 5. Check for the speed reference at 3TB or 4TB (MSR, JOGR or ASR). 6. Check control logic voltage and state at 3TB and 4TB (RUN, JOG, SLOW, MODE 0 thru MODE 7, etc.) as well as power supply voltage levels. 7. Check that the drive is in the proper operate mode (AUTO, MANUAL or ORIENT). 8. Drive may be stalled in current limit.
16. Field Current or Voltage Wrong Value	1. Is the drive in field economy? 2. Check for proper motor field connections. 3. Check JP1 and JP2 on the Power Supply card for proper maximum field settings. 4. Check the P1-FIELD SCALE pot seal - factory adjusted for field current. (Do not adjust.) 5. Check ADR076-FLDMX and ADR077-FLDMN per the Programming Software Package. 6. Check optional 4TB-20 for connection to FRA if external field forcing is used. 7. Check for any external parallel resistors or diodes in the field circuit. 8. Check the field power module for shorted elements. 9. Check for saturated fields when not able to produce required torque (symptom is also high current, high speed versus reference). Field current is reduced, but not enough to cause (FL04) Field Loss trip. (Field Undercurrent fault is disabled on QII controllers). (May be due to an open in one of the components feeding the MFC package, a failed fuse, wire connection, or diode in the MFC package, or too high of a required field voltage.) 10. Check the field current (Test point FC). Replace the Power Supply card if failed. 11. Replace the Control card.
17. Drive Runs in the Wrong Direction	1. Check for logic signal on 3TB15 (POL). 2. Armature connections reversed (correct armature and tach leads). 3. Field connections reversed (correct field and tach leads). 4. Check all software jumpers (ADR000 to ADR014) per the Programming Software Package. 5. Check the reference polarity. 6. Check logic polarity control jumpers, JP19 and JP20 on the Control card, JP5 and JP6 on the optional Application card. 7. Check the Programming Software Package to verify that the reverse function is enabled when operating from the serial link.

Problem	Check/Adjust/Replace
18. Drive Does Not Reverse	1. Verify that the drive is regenerative. 2. Check the settings of the soft jumpers (ADR000 to ADR014) per the Programming Software Package. 3. Check the reference polarity. 4. Check control logic to POL (3TB55) if used for reversing. 5. Check logic polarity control jumpers, JP19 and JP20 on the Control card, JP5 and JP6 on the optional Application card. 6. Check the Programming Software Package to verify that the function is enabled when operating from the serial link.
19. Drive Speed is Incorrect	1. Verify that the motor design is suitable for the application. 2. Is the proper reference (MSR, ASR, RUNRF, R1RF, R2RF, SLWRF, JOGRF) enabled? Refer to Programming Software Package. 3. Check JP1 on the Control card for proper scaling of MSR input. 4. Adjust P1-REF SCALE on the Control card for MSR input. 5. ADR068-REFSF set incorrectly? 6. Check JP1 on the optional Application card for proper ASR scaling. 7. Adjust P1-ASR SCALE on optional Application card. 8. Adjust P7-SFB ZERO on optional Encoder Process Interface card if analog AC or DC tach is used. 9. Confirm the setting of SW1-TACH SCALE, on the optional Encoder Process Interface card if an analog AC or DC tach is used. 10. Check output voltage at the drive if a pulse tach is used. May be low due to excessive cable length. 11. RPMSF or REFSF may be misadjusted, check per the Programming Software Package. 12. Verify the proper settings in ADR053-DTCHR and ADR074-TCPPR per the Programming Software Package. 13. Check for noisy tach signal. Refer to "Wiring Procedures", this section. 14. Set ADR001-FDBJP per the Programming Software Package. 15. Confirm that JP15, JP16 and JP19, JP20 on the optional Encoder Process Interface card are in accordance with application requirements. 16. Check the setting of ADR041-ILMMX if the drive is suspected of being in current limit. 17. Check the setting of ADR080-CMFLM if the drive is suspected of being in CEMF limit. 18. Check ADR079-CROSS per the Programming Software Package. 19. Check ADR076-FLDMX and ADR077-FLDMN per the Programming Software Package and by calculations in the key parameter adjustment list. 20. Check ADR059-RATIO per the Programming Software Package. (Should be set at "10000" for no draw.)

Problem	Check/Adjust/Replace
20. Serial Link Does Not Function Properly	1. Verify that JP11, JP12, JP13 and JP14 on the optional Encoder Process Interface card are set correctly. 2. Check the baud rate of the drive (ADR126-BAUDR) compared to serial link device (hard reset must be performed after BAUDR is changed). 3. Check serial link connections at 3TB22, 23, and 24 (DCOM, R232, and 3TB22). 4. If RS422 is used, verify that the handshaking lines are properly connected. Verify that sending device is responding to the drive's DTR signal (low level) when the device is allowed to transmit to the drive. 5. Replace the optional Encoder Process Interface card.
21. Erratic Operation	1. Check power supplies for magnitude and ripple (refer to "Checking Power Supplies"). 2. Verify that 1CPL on the MFC/Power Supply card is inserted correctly (not backwards or offset). 3. Check connections on all ribbon cables. 4. Tighten all connections on 2TB, 3TB, optional 4TB, and all power connections. 5. Confirm correct hardware jumper setting per the Programming Software Package. 6. Check all parameters per the Programming Software Package. 7. Check for noisy ground connection - refer to "Wiring Procedures". 8. Examine CFB and VFB with an oscilloscope for stability - refer to "Stability Problems", this section. 9. If a digital tach is used, check the integrity of the signal.
22. Stability Problems	1. Examine CFB with an oscilloscope to determine if any SCRs have failed. 2. Check settings of all parameters per the Programming Software Package. 3. Check the setting of SW1-TACH SCALE switch on the optional Encoder Process Interface card if an analog AC or DC tach is used. 4. Verify that all contacts on the MA or MD contactor are closing. 5. Check and adjust for any offset on CFB, P2-CFB ZERO, P3-CFB ADJUST (on the Power Supply card), JP3 and JP4 on the Power Supply card or the P4-OFFSET (on the Control card). 6. Refer to the "Troubleshooting Stability Problems" section in this manual.
23. Signal Level Detector Chatter	1. Is SLD looking at a "noisy" signal (e.g., CFB, tach feedback, etc.)? 2. Is enough hysteresis programmed into the appropriate parameter address?
24. Too Much Coast Time on Deceleration (Regenerative Drives)	1. Lower the value in ADR048-RSTOP. 2. Check settings per Programming Software Package and confirm regeneration is enabled.

Problem	Check/Adjust/Replace
25. Motor Sparking	1. Check motor commutator for contamination. 2. Check the setting in ADR024-IRTLM, may be too high. 3. Check the field current - refer to "Field Current or Voltage Wrong Value". 4. Check the field current stability - refer to "Stability Troubleshooting" section. 5. Check the commutator condition. 6. Confirm that all brushes are seated properly with proper brush tension. 7. Check the brush grade. 8. Check the brush neutral adjustment. 9. Check for sagging AC line voltage - refer to "Line Voltage Problems", this section.
26. Motor Runs Hot	1. Motor may be overloaded, check for mechanical problems. 2. Check the motor winding temperature. If the motor has Class F insulation and the winding temperature is less than 130 C, this is normal. 3. Check the rotation of the motor blowers and that filters are not dirty or clogged. 4. Check the field current with respect to the motor nameplate rating. Adjust ADR076-FLDMX if necessary. 5. Check the field current when motor is not running. If the field current is full nameplate rating when the motor or blower is not running, check ADR050-FLDEC. 6. If motor runs hot at very low speeds, check motor application guide for full-torque speed range. 7. Confirm that all SCRs are firing properly by monitoring current feedback CFB.
27. Motor Audible Noise	1. Loose pole bolts - tighten per manufacturer's torque specifications. 2. SCRs misfiring - check current feedback. 3. Cross-firing SCRs - increase deadband. 4. Unstable regulator - check "Troubleshooting Stability Problems" section.
28. Motor Vibration	1. Coast Stop Drive - if the vibration is electrically induced the vibration will disappear. If it does not disappear, check for mechanical causes. 2. Refer to "Stability Problems", this section. 3. Motor not mounted level. 4. Motor base vibrating. 5. Motor not mounted as designed - i.e., horizontal design motor mounted with shaft vertical.
29. EEROM Memory Rewrites	1. JP13 on the Control card must be in the 1-2 position during "Power UP" or hard reset.

Problem	Check/Adjust/Replace
30. Noise or Random Nuisance Faults	1. Verify that the 5-volt DC supply is not used externally. 2. Check if ACOM, DCOM, or PCOM are used externally - PCOMX is the only point that should be used for common connections. 3. Verify that PCOM passes through a ferrite core (not required if PCOMX is on 3TB73-75). 4. Check for suppression on all relays, contactors, and starters. 5. Place JP16 on the Control card in the 2-3 position on all drives not using a frequency reference. 6. Check all power supplies - refer to "Checking Power Supplies", this section. 7. When an encoder is used, verify that all of the encoder wires except the shield have been passed through a ferrite core on optional balun assembly. 8. Refer to "Wiring Procedures", this section.
31. Wiring Procedures	1. Check the wire sizes on the armature, field, and AC power connections. 2. Verify that the control wires (24-volt DC, 115-volt AC) do not run in the same conduit or run parallel closer than four inches to the power wires, causing noise pick-up. 3. Verify that all of the signal wires (i.e., tach, encoder, reference, load cell, or RS-232C/422) are twisted pairs, shielded, and shields grounded at the drive. 4. Verify that the signal wires are not grounded at more than one end or other than the drive end only. 5. Verify that the signal wires do not run in the same conduit or parallel closer than four inches to the power wires. 6. Verify that there is only one system ground. 7. Verify that the ground bus is large enough to prevent voltage drops or noise pick up.
32. Checking Power Supplies	1. Check the +15V test point on the Control card for +15VDC, $\pm$5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 2. Check the -15V test point on the Control card for -15VDC, $\pm$5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 3. Check the +5V test point on the Control card for +5VDC, $\pm$1%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 4. When the 5V power supply is used to power an encoder or digital tach, check for power supply heating if no fan is supplied.

TABLE 6: SPECIFIC FAULT TROUBLESHOOTING

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
000001	FL01	IOC	**Instantaneous DC overcurrent (L,T)** -- Tighten armature connections. -- Armature grounded. -- Examine motor commutator for damage or evidence of flashover. -- Verify proper brush seating and tension (refer to motor manual). -- Mechanical Problems - binding shaft, unreleased brake, vibration, gearbox, or load problems. -- IOC parameter set too low. Set at least 128 units above current limit settings. -- Current feedback isolator failure - check 1PL & DRPL. -- SCR misfiring - Check VFB or CFB waveforms. -- Check input at SJ2 (4TB16 or 13PL-16). -- ADR011-SJAJP set incorrectly. -- SCR failure: Try SCR test. (Refer to Control card JP12 description, Table 1.) -- Current regulator mistuned - adjust ILPPG, ILPIG, ILPDG, ILPDP (ADR019, 020, 021, 022).
000010	FL02	OVRSP	**Overspeed (L,T)** -- ADR073-SPDLM set too close to ADR068-REFSF and ADR071-REFLM - set a minimum of 5% greater than REFSF and REFLM. -- Speed overshoot occurs - speed regulator response too high, reduce gain. (Refer to Table 7, "Troubleshooting Stability Problems".) -- Noise on digital tach input. -- Loss of field current.
000011	FL03	----	**SPARE**

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
000100	FL04	FLOSS	**Loss of field current (L,T)** -- CFU1, CFU2, or CFU3 on Power Supply card open. -- 1PL disconnected or damaged. -- Field grounded. -- Field lead connections made improperly. -- Open field circuit components - resistors, contactors, or DC fuses. -- ADR078-FLDLS set too high - set at least 25% below ADR077-FLDMN and 50% below ADR076-FLDMX. -- FC gain jumpers set wrong - JP1 and JP2 on Power Supply card. -- External field supply - set ADR078-FLDLS at "60000" to inhibit field loss. -- Field regulator gain too low - adjust BFLPG, BFLIG, WFLPG, WFLIG (ADR031, 032, 033, 034).
000101	FL05	TPOL	**Tach polarity incorrect (L,T)** **CAUTION:** Remove power from digital tachs before moving wiring. -- Reversed tach connections. -- Switch SW1-7 on the optional Encoder Process Interface card set improperly - open for DC tach, closed for AC tach. -- Armature or field reversing contactors - drive must have AN Tach feedback.
000110	FL06	JMPER	**Software jumper illegally changed while running (L,T)** -- Drive must be stopped to change parameters in ADR000 to ADR014.
000111	FL07	RESET	**Reset while running (L,T)** -- Drive powered up with RUN or JOG closed - check 6PL-12 and 6PL-13 (3TB58 and 57). -- Momentary loss of incoming power while running. -- Incorrect logic polarity, check JP19 and JP20 on the Control card and JP5 and JP6 on the optional Application card.
001000	FL08	TOC	**Timed overcurrent (L,T)** -- Motor mechanically overloaded - check for change in process load, worn bearings, gears, or brakes. -- CFU2 on Power Supply card open. -- Field current is less than 85% of nameplate rated current - readjust field current parameters. -- ADR044-TOCTH and ADR045-TOCGN set too low - check against Programming Software Package. -- Current feedback isolator failure or 1PL disconnected or damaged. -- P3-CFB ADJ on the Power Supply card misadjusted - readjust per procedure in Table 1.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
001000 (continued)	FL08	TOC	**Timed overcurrent (L,T)** -- Armature winding failed. -- Dynamic braking failure (if used) - i.e., failure to close DB circuit.
001001	FL09	LNKR	**Serial Link trip (drive told to trip via serial link) (L,T)** -- Drive received external trip signal - check for external fault. -- 12PL or 8PL incorrectly connected or damaged.
001010	FL10	LNKFL	**Serial Link fault (communications failure) (L,T)** -- Baud rate set incorrectly - check ADR126-BAUDR. -- 12PL or 8PL incorrectly connected or damaged. -- Intermittent connection between 3TB32 (CTSA) and 12PL-7 or 3TB41 (CTSB) and 12PL-14. -- FL36 appears momentarily previous to FL10 latching - refer to FL36.
001011	FL11	DFPON	**Delayed firing power on when it should be off (L,T)** -- Q2 failed on - check test point DFP. -- Replace Control card.
001100	FL12	DFPOF	**Delayed firing power off when it should be on (L,T)** -- Q2, Q3, R33, or U14 failed on Control card - replace Control card.
001101	FL13	EEXXX	**Checksum error or parameter out of range where XXX equals a 3-digit number, indicating an out-of-range parameter address or altered checksum. (L, T, H)** -- Spontaneous parameter alteration - check all parameters and verify EEROM is not write "ENABLED" during power up or power down, or "hard" reset. -- Value other than zero in ADR255-UNDFD - replace EEROM.
001110	FL14	ILOOP	**Current loop regulator saturated (L,T)** -- Main contactor not picking up - check pilot relay operation. -- DC fuse open (regenerative drives only). -- Loss or incorrect scaling of voltage feedback - check jumpers on the Power Connection card and Control card TP, VFB, or VFBB. -- Loss of current feedback - check 1PL-20 and DRPL. -- Armature not connected. -- Regeneration enabled by soft jumper on single quadrant drive. -- Brushes missing or unseated. -- Controllers using the inner voltage loop (optional) - hardware failure in voltage regulator; replace Control card. -- Controllers using the inner current loop, ADR019-ILLPG, ADR020-ILPIG, and ADR021-ILPDG set too high - check against the Programming Software Package.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
001110 (continued)	FL14	ILOOP	**Current loop regulator saturated (L,T)** -- Current feedback miscalibrated - see Table 1, Power Supply card. -- Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card per Table 1. -- Power Supply card failed - replace. -- Control card failed - replace.
001111	FL15	VFBLS	**Loss of armature voltage feedback (L,T)** -- Voltage feedback wires disconnected or damaged - check P1 and P2 between SCR conversion module and Power Connection card. -- DC fuse (FU4) open - regenerative drives only. -- 5PL disconnected or damaged. -- SFB circuit misadjusted - adjust P7-SFB ZERO on the optional Encoder Process Interface card per Table 1. -- CFB jumpers incorrect - check JP3 and JP4 on the Power Supply card per Table 1. -- VFB circuit failure - check test point VFB. -- Run SCR Test - refer to Control card description of JP12.
010000	FL16	NOTRP	**Trip from N.O. on Optional Application card (L,T)** -- External input at 4TB8 (13PL-8). -- Optional Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card is present. -- If preceded by FL07 and soft reset, check logic polarity jumpers JP19 and JP20 on the Control card and JP5 and JP6 on the optional Application card. -- Replace optional Application card.
010001	FL17	XSTOP	**Coast stop trip (L,T)** -- Check for input at XSTOP on 3TB1 (6PL-16).
010010	FL18	STALL	**Stalled overcurrent (L,T)** -- Mechanical problems - failed brake, frozen bearings, application problem, etc. -- ADR046-STLTH and ADR047-STLGN set wrong - check against Programming Software Package. -- Drive running less than 1% of top speed and armature current greater than ADR046-STLTH. -- CFU2 on Power Supply card open. -- Field current is less than 85% of nameplate rated current while the drive is supplying armature current. -- Current feedback isolator failure or 1PL disconnected or damaged. -- P3-CFB ADJ on the Power Supply card misadjusted - readjust per procedure in Table 1.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
010010 (continued)	**FL18**	**STALL**	**Stalled overcurrent (L,T)** -- Armature winding failed. -- Dynamic braking failure (if used) - i.e., failure to open DB circuit.
010011	**FL19**	**OVERV**	**Excessive armature voltage (L,T)** -- ADR080-CMFLM set too high - compare to Programming Software Package -- Drive unstable - see "Stability Troubleshooting", Table 7. -- VFB circuit failure - check test point VFB.
010100	**FL20**	**TLOSS**	**Loss of tach feedback (L,T)** -- Tach leads disconnected. -- Tach feedback circuit failure - 7PL disconnected or damaged. -- Optional Encoder Process Interface card jumpers or switches incorrect. Refer to Table 1. -- P7-SFB ZERO on optional Encoder Process Interface card misadjusted - set test point SFB for zero volts at zero speed. -- AC or digital tach connected but CEMF regulator being used - remove JP7 from optional Encoder Process Interface card. -- ADR053-DTCHR set too high when AC Tach being used - lower value to 10. -- ADR002-REGJP set incorrectly for optional Encoder Process Interface card (531X134EPRA_G1), refer to Programming Software Package. -- JP17 on Control card in 1-2 position - move to 2-3 position. -- Optional Encoder Process Interface card failed - replace.
010101	**FL21**	**NCTRP**	**Trip from N.C. interlock on Optional Application card (L,T)** -- Loss of external input on 4TB7 (13PL-7). -- Optional Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present, check per Programming Software Package. -- If preceded by FL07 and soft reset, check logic polarity jumpers JP19 and JP20 on the Control card and JP5 and JP6 on the optional Application card.
010110	**FL22**	**PHLOS**	**AC phase loss (L,T)** -- JP14 on Control card in wrong position - refer to Table 1. -- Fuse open in AC line - check FU1-3. -- 1PL disconnected or damaged. -- Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. -- Loose AC power connections. -- CFU1 open.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
010110 (continued)	**FL22**	**PHLOS**	**AC phase loss (L,T)** -- Verify that ADR187-FUSLS contains a zero - change if necessary. -- ±24-volt or ±15-volt power supplies failed - check CFU6 and CFU7 on Power Supply card. -- ±15-volt power supplies outside limits of ±5% - replace Power Supply card. (Check possible external overload to power supplies.)
010111	**FL23**	**OVERF**	**Field overcurrent (L,T)** -- Field regulator unstable - refer to "Stability Problems", this section. -- Field forcing at FRA (4TB20, 13PL-20) - check input scaling, see Table 1. -- ADR076-FLDMX or ADR049-FTOCG incorrect - check against Programming Software Package. -- Field SCR shorted - replace Motor Field Module. -- Voltage regulated field - check for miswired or grounded field.
011000	**FL24**	**U3 FL**	**Faulty RAM on U3 8155 (L,T)** -- Replace Control card.
011001	**FL25**	**COMTR**	**SCR commutation failure (L,T)** -- Low AC line voltage (less than 90% rated). -- ADR080-CMFLM set too high - should not exceed 1.13 x AC RMS in-line voltage. -- 1CPL disconnected. -- 1CPL reversed - phase 1 CT should be wired to bottom of 1CPL at Power Supply card. -- P3-HP SEL and JP8-JP11 gains set too high on Control card. -- 1PL disconnected or damaged. -- Gate plug(s) incorrectly connected. -- SCR(s) failed - try SCR self test.
011010	**FL26**	**ACIOC**	**Circulating AC line fault (regenerative controllers only) (L,T)** -- Low or noisy AC line. -- Incorrect setting of P3-HP SEL and JP8-JP11 on Control card. -- Noise on SCR gate leads. -- 1PL disconnected or damaged. -- SCR(s) failed - try SCR self test.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
011011	FL27	LYSNC	**Loss of AC line signal (L,T)** -- Fuse open in AC line - check FU1-3. -- 1PL disconnected or damaged. -- Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. -- Loose AC power connections. -- CFU1 open. -- 15-volt power supplies outside limits of ±5% - replace Power Supply card. -- AC line frequency below 55Hz or above 63Hz. -- Check 60Hz square wave at test point SY on Control card - replace Control card if no signal.
011100	FL28	RMODE	**Illegal selection of RUN mode (N,A)** -- Do not command DIAGNOSTIC RUN or STATIC while running or vice versa.
011101	FL29	CNLON	**Control on input opened (L,T)** -- Open circuit between 3TB3 and 3TB4. -- 6PL incorrectly connected or damaged. -- Noise between 3TB3 and 3TB4.
011110	FL30	NOAPP	**Optional Application card failure (L,T)** -- Optional Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present. -- 3PL incorrectly connected or damaged. -- Replace Optional Application card.
011111	FL31	NOSEQ	**Phase sequence indeterminate (L,T)** -- CFU1 on Power Supply card open. -- Excessively noisy or notched AC line. -- Line unbalance phase to ground by more than 15%. -- 15-volt power supplies outside limits of ±5% - replace Power Supply card. -- 1PL incorrectly installed or damaged.
100000	FL32	TOCAN	**Timed overcurrent annunciation (N,A)** -- Motor mechanically overloaded - check for change in process load, worn bearings, gears, or brakes. -- CFU2 on Power Supply card open. -- Field current is less than 85% of nameplate rated current. -- ADR044-TOCTH and ADR045-TOCGN set too low - check against the Programming Software Package. -- Current feedback isolator failure or 1PL disconnected or damaged.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
100000 (continued)	**FL32**	**TOCAN**	**Timed overcurrent annunciation (N,A)** -- P3-CFB ADJ on the Power Supply card misadjusted - readjust per procedure in Table 1. -- Armature winding failed. -- Dynamic braking failure (if used) - i.e., failure to close DB circuit.
100001	**FL33**	**PHSEQ**	**Phase sequence reversed (N,A)** -- Reverse any two AC input lines, recheck rotation of any 3-phase motors. -- Excessively noisy or notched AC line. -- Line unbalance phase to ground by more than 15%. -- 15-volt power supplies outside limits of $\pm$5%, noisy, or excessive ripple - replace Power Supply card. -- 1PL incorrectly installed or damaged.
100010	**FL34**	**PLLSY**	**AC line PLL out of sync (N,A)** -- Excessively noisy or notched AC line. -- CFU1 on Power Supply card open. -- Line unbalance phase to ground by more than 15%. -- 15-volt power supplies outside limits of $\pm$5% - replace Power Supply card. -- On variable frequency AC lines - ADR029-PLLGN too low.
100011	**FL35**	**O SPD**	**Motor rotating when RUN commanded (N,A)** -- RUN command initiated during motor rotation. -- ADR048-RSTOP set too low. -- Test point SFB contains an offset at zero speed - adjust P7-SFB ZERO on the optional Encoder Process Interface card.
100100	**FL36**	**LNKER**	**Uncipherable messages detected on serial link by drive or other device (N,A)** -- Excessive noise on 12PL-7 thru 12PL-14. -- 12PL incorrectly connected or damaged. -- 5 Volt DC power supply noisy. -- Baud rate incorrectly set - ADR126-BAUDR normally set at 4800. -- Check optional Encoder Process Interface card jumpers JP1, JP11, JP12, JP13, and JP14. -- Serial link message protocol incorrect, message too long, checksum byte incorrect or handshaking incorrect - refer to "Serial Link Users Guide" . -- Replace optional Encoder Process Interface card. -- Replace Control card.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
100101	FL37	IHUNG	**Armature current not zero when fully retarded (N,A)** -- JP3 and JP4 on Power Supply card in 2-3 position when running - should be in 1-2 position. -- Offset voltage at test point CFB when zero armature current flowing - adjust P2-CFB ZERO on the Power Supply card. -- DRPL disconnected or damaged. -- 1PL disconnected or damaged. -- ADR080-CMFLM set too high (only if fault occurs during regeneration). -- Current isolator failed - replace Power Supply card.
100110	FL38	ELOCK	**Attempt to change software adjustments when soft locked (N,A)** -- ADR003-DGNJP must be odd to program EEROM - check per Programming Software Package.
100111	FL39	EFAIL	**Software adjustment not stored in EEROM (N,A)** -- JP13 (or SW1) on Control card in hard lock position. -- EEROM failed - replace.
101000	FL40	MXSPD (N,A)	**Speed reference greater than REFLM speed adjustment** -- Manual Reference jumper JP1 on Control card improperly set - see Table 1. -- P1-REF SCALE on the Control card misadjusted - see Table 1. -- ADR071-REFLM set below ADR070-RPMSF. -- P1-ASR SCALE, on the optional Application card misadjusted (if used) - see Table 1. -- P4-PF GAIN or P5-PF OFF, misadjusted on optional Encoder Process Interface card - see Table 1. -- Top speed reference for parallel input (if used) is set higher than ADR068-REFSF and ADR070-RPMSF. -- Preset references (RUNRF, R1RF, R2RF, SLWRF) set too high (if used).
101001	FL41	----	**NOT USED**
101010	FL42	RLOGC	**Run logic error (N,A)** -- JOG, RUN, or DIAGNOSTIC RUN called for with XSTOP open. -- Incorrect logic polarity - check jumpers JP19, and JP20 on the Control card, JP5 and JP6 on the optional Application card.
101011	FL43	U11FL	**A-D converter failure (L,T)** -- Check for noise on ACOM, move wiring to PCOMX. -- Control card failed - replace.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
101100	FL44	U4LAT	**U4 to U5 communication error (L,T)** -- Check PROM revision compatibility. -- Check for noise on power supplies or analog inputs. -- Replace Control card.
101101	FL45	SLOOP	**Speed tracking fault (L,T)** -- Drive entered current limit or ADVLM limit when tracking enabled. -- ADR036-SERLM set too low - should be between 1 and 7. -- If using inner voltage loop, check software revision - if earlier than 33, replace software. -- Speed regulator unstable - see "Stability Troubleshooting", Table 7.
101110	FL46	IOFST	**Current feedback offset fault (L,T)** -- Current feedback signal at testpoint CFB offset by 35%. -- P2-CFB ZERO on MFC/Power Supply card misadjusted.
101111	FL47	OADJ>	**A-D converter offset too high (N,A)** -- P4-OFFSET on the Control card misadjusted - turn CW. -- Check for noise on ACOM.
110000	FL48	OADJ<	**A-D converter offset too high (N,A)** -- P4-OFFSET on the Control card misadjusted - turn CCW. -- Check for noise on ACOM.
110001	FL49	NO U4	**U4 to U6 communications error (L,T)** -- Check for PROM revision compatibility. -- Check for noise on power supplies or analog inputs. -- Replace Control card.
110010	FL50	U4BAD	**U4 RAM, timer or ALU is faulty (L,T)** -- Verify settings of ADR011, 014, 056, 089, 187, 209, and 210 per Programming Software Package. -- Incorrect software installed for application - check revision level. -- U4 failed - replace Control card.
110011	FL51	U1ROM	**U1 ROM is faulty (L,T)** -- Replace U1 on the Control card. -- Replace Control card.
110100	FL52	NCLSD	**Annunciate N.C. interlock on Application card (N,A)** -- Check external input on 4TB7 (13PL-7). -- Optional Application card absent but selected by soft jumper - ADR010-APPJP should be zero if no Application card is present.

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
110101	FL53	SCLER	**SCALE ERROR (L,T)** -- Occurs if position Feedback is not enabled from the optional Encoder Process Interface card and the product of ADR070-RPMSF and ADR074-TCPPR is greater than 2,490,367 - check setting in ADR070 and 074.
110110	FL54	NO U5	**U5 to U4 communication error (L,T)** -- Check for PROM revision compatibility. -- Replace Control card or U5.
110111	FL55	U5BAD	**U5 RAM, timer or ALU is faulty (L,T)** -- U5 failed - replace. -- Replace Control card.
111000	FL56	U5ROM	**U5 ROM is failed (L,T)** -- U5 failed - replace.
111001	FL57	NOPEN	**Annunciate N.O. interlocks on optional Application card (N,A)** -- External input on 4TB8 (13PL-8). -- Optional Application card absent but selected by soft jumper - ADR010-APPJP should be zero if no Application card is present; check per Programming Software Package.
111010	FL58	U5COM	**U4 not communicating correctly with U5 (L,T)** -- Check for PROM revision compatibility. -- Digital tach pulse rate exceeds 41,000 Hertz. -- Noise on tach input. -- Replace Control card.
111011	FL59	NO U6	**U6 does not communicate properly with U4 (L,T)** -- Check for PROM revision compatibility. -- Replace Control card.
111100	FL60	U6BAD	**U6 RAM, timer or ALU is faulty (L,T)** -- U6 failed - replace. -- Control card failed - replace.
111101	FL61	U6ROM	**U6 ROM is faulty (L,T)** -- U6 failed - replace. -- Control card failed - replace.
111110	FL62	----	**SPARE**

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
111111	FL63	U6COM	**U4 not communicating correctly with U6 (L,T)** -- Check for PROM revision compatibility. -- Digital tach pulse rate exceeds 410,000 when using optional Encoder Process Interface card (531X134EPRA_G1). -- ADR002-REGJP not set for optional Encoder Process Interface card (531X134EPRA_G1); check per Programming Software Package. -- Check for noise on tach input signal. -- U1 extended memory - move JP24 and JP31 to the 2-3 position. -- U5 and U6 (TUP) in wrong sockets. -- Replace Control card. -- Motor fields are open.
000000	FL64	----	**SPARE**
000001	FL65	A1FOP	**SCR A1F is not operating (L,T)** -- Gate circuit 1F disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
000010	FL66	A2FOP	**SCR A2F is not operating (L,T)** -- Gate circuit 2F disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
000011	FL67	A3FOP	**SCR A3F is not operating (L,T)** -- Gate circuit 3F disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
000100	FL68	A4FOP	**SCR A4F is not operating (L,T)** -- Gate circuit 4F disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.

**(see door label for module location)*

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
000101	FL69	A5FOP	**SCR A5F is not operating (L,T)** -- Gate circuit 5F disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
000110	FL70	A6FOP	**SCR A6F is not operating (L,T)** -- Gate circuit 6F disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
000111	FL71	FWDOP	**All forward SCRs are not operating (L,T)** -- 5PL disconnected or damaged. -- Replace Power Connection card. -- Replace Control card.
001000	FL72	ARMOP	**Armature circuit open (L,T)** -- DC fuse open (regenerative drives only). -- Armature circuit open. -- Brushes missing or disconnected. -- 5PL disconnected or damaged. -- Main contactor not picking up - check pilot relay operation. -- Loss of current feedback - check 1PL-20 and DRPL. -- Armature not connected. -- Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card. -- Power Supply card failed - replace. -- Control card failed - replace.
001001	FL73	A1ROP	**SCR A1R is not operating (L,T)** -- Gate circuit 1R disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
001010	FL74	A2ROP	**SCR A2R is not operating (L,T)** -- Gate circuit 2R disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Control card.

**(see door label for module location)*

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
001011	FL75	A3ROP	**SCR A3R is not operating (L,T)** -- Gate circuit 3R disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
001100	FL76	A4ROP	**SCR A4R is not operating (L,T)** -- Gate circuit 4R disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
001101	FL77	A5ROP	**SCR A5R is not operating (L,T)** -- Gate circuit 5R disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
001110	FL78	A6ROP	**SCR A6R is not operating (L,T)** -- Gate circuit 6R disconnected or open. -- 5PL disconnected or damaged. -- Failed SCR - check and replace if necessary.* -- Replace Power Connection card. -- Replace Control card.
001111	FL79	REVOP	**All reverse SCRs are not operating (L,T)** -- 5PL disconnected or damaged. -- ADR002-REGJP not set correctly for non-regenerative drives. -- Replace Power Connection card. -- Replace Control card.
010000	FL80	SCTFL	**SCR short circuit test failed (L,T)** -- Check for shorted SCRs - replace.*
010001	FL81	A1FSH	**SCR A1F or A4R is shorted (L,T)** -- Check for shorted SCRs - replace.*
010010	FL82	A2FSH	**SCR A2F or A5R is shorted (L,T)** -- Check for shorted SCRs - replace.*
010011	FL83	A3FSH	**SCR A3F or A6R is shorted (L,T)** -- Check for shorted SCRs - replace.*

**(see door label for module location)*

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
010100	FL84	A4FSH	**SCR A4F or A1R is shorted (L,T)** -- Check for shorted SCRs - replace.*
010101	FL85	A5FSH	**SCR A5F or A2R is shorted (L,T)** -- Check for shorted SCRs - replace.*
010110	FL86	A6FSH	**SCR A6F or A3R is shorted (L,T)** -- Check for shorted SCRs - replace.*
010111	FL87	----	**NOT USED**
011000	FL88	----	**NOT USED**
011001	FL89	----	**NOT USED**
011010	FL90	----	**NOT USED**
011011	FL91	----	**SPARE**
011100	FL92	----	**SPARE**
011101	FL93	----	**SPARE**
011110	FL94	----	**SPARE**
011111	FL95	WATCH	**FUP watchdog detected RUP has misoperated (L,T)** -- Same as FL58.
100000	FL96	PASSED	**Annunciates when SCR test has passed.**
100001	FL97	OCTFL	**SCR open circuit test failed (L,T)** -- DC fuse open (regenerative drive only). -- Armature circuit open. -- Brushes missing or disconnected. -- 5PL disconnected or damaged. -- Main contactor not picking up - check pilot relay operation. -- Loss of current feedback - check 1PL-20 and DRPL. -- Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card. -- Power Connection card failed - replace. -- Power Supply card failed - replace. -- Control card failed - replace.
100010	FL98	OCTMF	**Multiple open SCRs (L,T)** -- Check all gate plugs. -- 5PL disconnected or damaged. -- Check SCRs for failure - replace. -- Replace Power Connection card. -- Replace Control card.

**(see door label for module location)*

Diagnostic Readout	Fault Number	Name	Description/Probable Causes/Remedies
100011	FL99	ALLOP	**All SCRs tested open (L,T)** -- DC fuse open (regenerative drive only). -- Armature circuit open. -- Brushes missing or disconnected. -- 5PL disconnected or damaged. -- Main contactor not picking up - check pilot relay operation. -- Loss of current feedback - check 1PL-20 and DRPL. -- Armature not connected. -- Offset voltage at CFB with zero armature current - adjust P2-CFB ZERO on the Power Supply card. -- Power Supply card failed - replace. -- Control card failed - replace.

TROUBLESHOOTING STABILITY PROBLEMS

There are three broad categories of drive symptoms broadly classified as "Stability" problems. Technically, two of these are not actually stability problems but, due to their behavior, are included in this troubleshooting procedure. These categories are as follows:

"Stability" problems: "Hunting", oscillations in speed or current, etc. In general all problems in this category are cyclical in nature, with well-defined sinusoidal periods of oscillation. Usually these problems are electrically induced and can be tuned out with proportional and integral gain adjustments.

There are at least **eight** regulating loops in most QIII DC Controllers, any one which can be mistuned, causing stability problems under the right conditions. They are as follows:

(1) Current regulator - continuous current
(2) Current regulator - discontinuous current
(3) Speed regulator - base speed
(4) Speed regulator - weak field speed
(5) CEMF regulator
(6) CEMF limit regulator
(7) Field current regulator - base speed
(8) Field current regulator - weak field speed

NOTE: *Various forms of position regulators are also available, which, due to their variety are not addressed here.*

Mechanical Oscillation Problems: These are a direct result of mechanical deficiencies but under certain conditions can be aggravated or excited by electrical frequencies. Examples of this are vibration, audible motor noise, gear chatter, etc.

Erratic Operation Problems: Intermittent speed or current fluctuations, mechanical "bumping", or other disturbances non-sinusoidal or non-periodic in nature. These may be induced by misfiring SCRs, intermittent grounds, "noise", or mechanical problems.

NOTE: *The Programming Software Package or Serial Link connection is required to make parameter checks called for in Part I and adjustments as required in Part II.*

I. Before troubleshooting the drive, first verify the following:

- -- All software jumpers, ADR000 through ADR014 are in accordance with the Programming Software Package.
- -- All hardware jumpers are in accordance with the Programming Software Package.
- -- All wiring is correct and in agreement with the drive elementaries.
- -- All "gain" parameters are in agreement with the Programming Software Package. Specifically, the following parameters* should be checked:

Speed Loop:	ADR015-BSLPG
	ADR016-BSLIG
	ADR017-WSLPG
	ADR018-WSLIG
	ADR037-SLEAD
	ADR051-LSPAN
Current Loop:	ADR019-ILPPG
	ADR020-ILPIG
	ADR021-ILPDG
	ADR022-ILPDP
CEMF Loop:	ADR025-CLPPG
	ADR026-CLPIG
CEMF Limit Loop:	ADR027-CLMPG
	ADR028-CLMIG
Field Current Loop:	ADR031-BFLPG
	ADR032-BFLIG
	ADR033-WFLPG
	ADR034-WFLIG

**These parameters assume Motor 0 operation. If Motor 1, Motor 2, or Motor 3 stability problems exist, other parameters will affect the speed and current loop (e.g.: Motor 1 base speed loop proportional gain is designated as ADR143-BSLP1).*

II. Use Table 7 to decide which type of problem actually exists, keeping the following pointers in mind:

- -- QIII DC Controller current loops are almost always stable.
- -- Most stability problems, but not all, are corrected by adjusting speed loop parameters.
- -- It is always worthwhile to verify field current stability first.
- -- In general, when troubleshooting cascading or series regulating loops, the correct procedure is to stabilize the inner regulator first and then move outward; i.e., current regulator, then speed regulator, then position regulator (if furnished).
- -- If at any time the drive becomes stable as a result of a step, record any parameter changes, and do not continue with the remaining procedures.

TABLE 7: STABILITY TROUBLESHOOTING

Symptom	Procedures
A. Unstable when unloaded but stable when loaded, drive is stable but exhibits gearbox or motor noise **or** runs erratically.	1. Check for failed SCRs - monitor test point CFB on the Control card with an oscilloscope. 2. Current reversal too sensitive - increase ADR023-DEADB. 3. "Light Load" instability - increase ADR021-ILPDG. If frequency of instability inceases, raise ADR021 as high as possible without encountering erratic operation. (If ADR021 is changed, set ADR020-ILPIG to the same value.) 4. Reduce ADR019-ILPPG as low as possible, even zero if necessary. 5. If instability still exists, return adjusted parameters to their original setting and skip to Symptom E.
B. Unstable above and /or below base speed and over a wide speed range (Stabilizing the Field Current Regulator).	1. Confirm Field Current Regulator stability - monitor test point FC on the Control card with an oscilloscope. If stable, skip to the next appropriate symptom. 2. Field current is unstable - reduce ADR031-BFLPG until FC is stable. Do not reduce below the setting in ADR032-BFLIG **NOTE:** *ADR031 may already be lower than ADR032 in some applications, maintain this relationship. Lower ADR032 if necessary.* 3. Set ADR033-WFLPG equal to the new value in ADR031-BFLPG. 4. Set ADR034-WFLIG equal to the new value in ADR032-BFIPG. 5. Document any change and proceed to Symptom C or D as necessary.
C. Speed instability **below** base speed.	1. Confirm that the values in ADR015-BSLPG, ADR016-BSLIG, and ADR037-SLEAD agree with the Programming Software Package and satisfy the following equation: BSLIG/BSLPG = .127 (SLEAD) **NOTE:** *If ADR037-SLEAD = 0, skip to step 5.* 2. If ADR037-SLEAD contains a value (typically "10" or less) readjust BSLPG to satisy the above equation. **Do not** change SLEAD unless different from the Programming Software Package value. 3. Raise or lower ADR016-BSLIG and ADR015-BSLPG maintaining the above ratio. Raising the parameters makes the drive more responsive; lowering them makes the drive less responsive. 4. If SLEAD is 0 and the drive is slow and unresponsive, increase ADR015-BSLPG and adjust ADR016-BSLIG upward or downward as necessary.

Symptom	Procedures
C. (Continued)	5. If no improvement, return the adjusted parameters to their original settings. 6. If the drive is a CEMF regulator - lower ADR039-IRCMP to zero. 7. If there is some improvement but instability still exists - document changes and skip to Symptom E.
D. Speed stability problems **above** base speed.	1. Confirm that the values in ADR017-WSLPG, ADR018-WSLIG, and ADR037-SLEAD agree with the Programming Software Package and satisfy the following equation: WSLIG/WSLPG = .127 (SLEAD) **NOTE:** *If ADR037-SLEAD = 0, skip to step 5.* 2. If ADR037-SLEAD contains a value (typically "10" or less) readjust BSLPG to satisy the above equation. **Do not** change SLEAD unless different from the Programming Software Package value. 3. Raise or lower ADR018-WSLIG and ADR017-WSLPG maintaining the above ratio. Raising the parameters makes the drive more responsive; lowering them makes the drive less responsive. 4. If SLEAD is 0 and the drive is slow and unresponsive, increase ADR017-WSLPG and adjust ADR018-WSLIG upward or downward as necessary. 5. If no improvement, return the adjusted parameters to their original settings. 6. Drive is in CEMF limit - to confirm, lower ADR079-CROSS (should be approximately 150 to 200 counts lower than ADR080-CMFLM). 7. If there is some improvement but instability still exists - document changes and skip to Symptom E.
E. Drive is unstable: over a small part of the speed range, **or** when loaded but stable when unloaded, **or** such that the oscillating frequency is proportional to speed.	1. Confirm that the problem is not tach related - Check for tach feedback problems (misaligned or loose coupling, belt slippage, wiring, or excessive ripple). 2. Verify that the power supplies are clean with an oscilloscope (refer to "Checking Power Supplies" in the Troubleshooting section). 3. Run drive to top speed and push the "RESET" button on the Control card. The drive will **coast** stop. Mechanically induced problems will still be present as the motor decelerates. If problem disappears, continue to next step.

Symptom	Procedures
E. (Continued)	4. Lower settings in ADR015, ADR016, ADR017, and ADR018 as low as possible while maintaining the following ratios: a. $\frac{\text{FINAL ADR016}}{\text{FINAL ADR015}} = \frac{\text{ORIGINAL ADR016}}{\text{ORIGINAL ADR015}}$ b. $\frac{\text{FINAL ADR018}}{\text{FINAL ADR017}} = \frac{\text{ORIGINAL ADR018}}{\text{ORIGINAL ADR017}}$ 5. Reduce ADR037-SLEAD and double ADR015 and ADR016. 6. Determine resonant frequency "Fr" in Hz with an oscilloscope, if possible. Convert to radians per second as follows: "Wr" = 6.28 x "Fr". 7. Adjust ADR037-SLEAD and ADR051-LSPAN as follows: Set ADR037-SLEAD = .615 x "Wr" Set ADR051-LSPAN = 5 and readjust as necessary 8. If no improvement, return the parameters to their original settings and skip to Symptom G.
F. Undirectional Stability Problems.	1. Try running the motor in the opposite direction to confirm that stubbing brushes or brush position is not the problem. 2. Inspect the load to make sure that nothing is wrong mechanically. 3. If a regenerative drive, make certain that all SCRs are firing in both directions by monitoring test point CFB on the Control card with an oscilloscope.
G. Additional Procedures.	If the above procedures do not result in stable operation, additional steps may be required including: 1. Checkout of the current loop. 2. Running the field regulator open loop to verify stability. **If such procedures are required, contact the supplier of the QIII DC Controller for instructions.**

10. TYPICAL WAVEFORMS

The following are examples of typical waveforms that can be seen on an oscilloscope. Each figure is accompanied by the name of the signal, the test point which it is derived from (in parenthesis), and the mode of operation if applicable.

Also attached is the proper time setting for the waveform. (For information on using an oscilloscope, consult the owner's manual.)

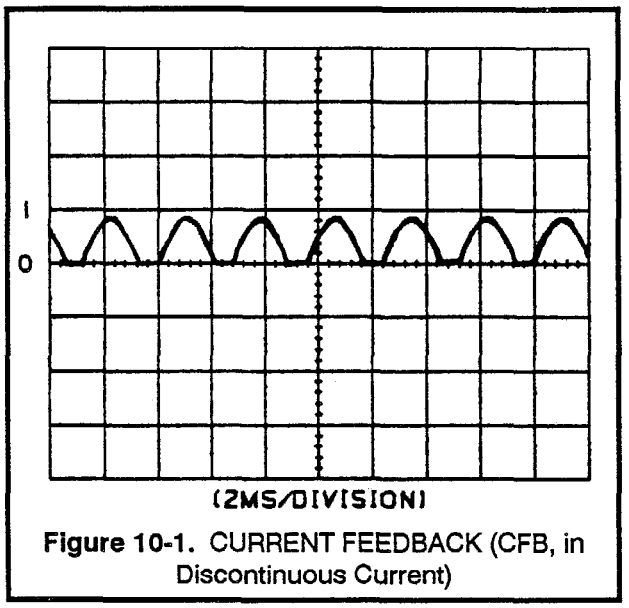


Figure 10-1. CURRENT FEEDBACK (CFB, in Discontinuous Current)

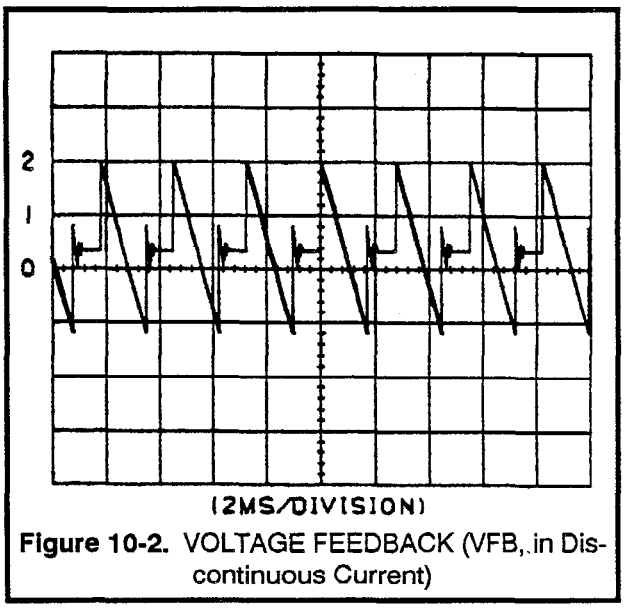


Figure 10-2. VOLTAGE FEEDBACK (VFB, in Discontinuous Current)

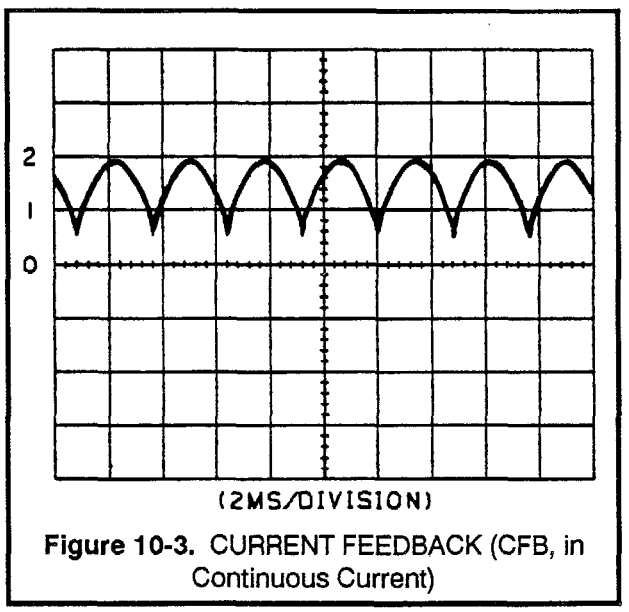


Figure 10-3. CURRENT FEEDBACK (CFB, in Continuous Current)

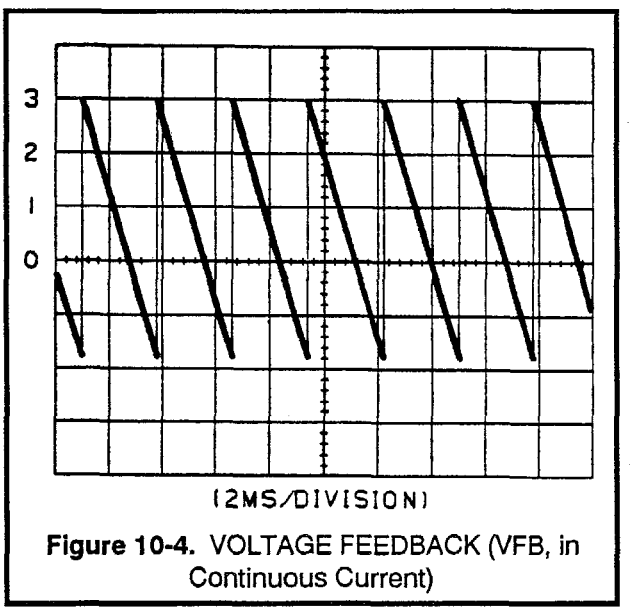


Figure 10-4. VOLTAGE FEEDBACK (VFB, in Continuous Current)

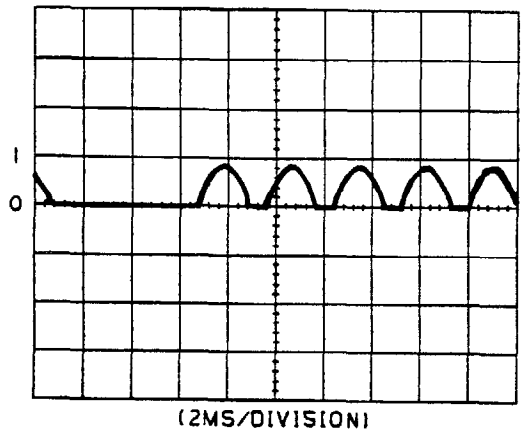


Figure 10-5. CURRENT FEEDBACK (CFB, with One SCR Open or Disconnected)

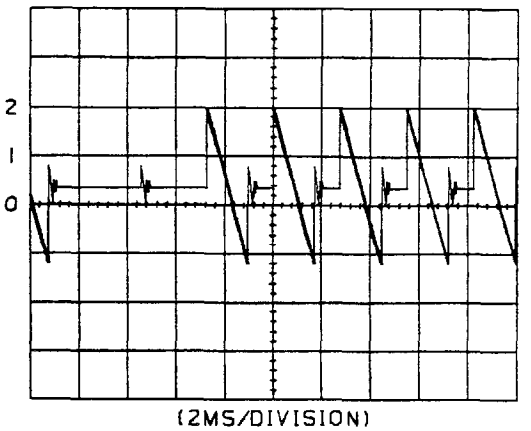


Figure 10-6. VOLTAGE FEEDBACK (VFB, with One SCR Open or Disconnected)

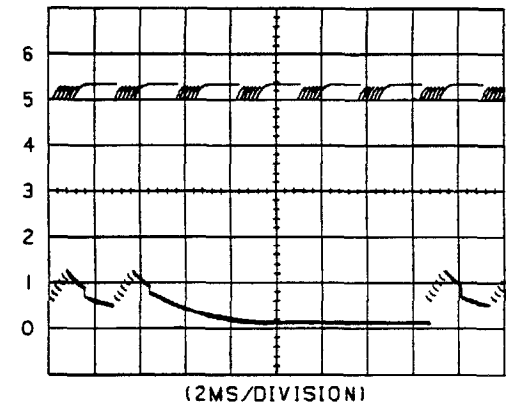


Figure 10-7. TOP - DELAYED FIRING POWER
BOTTOM - GATE PULSES FOR A1 SCR
(Discontinuous Current)

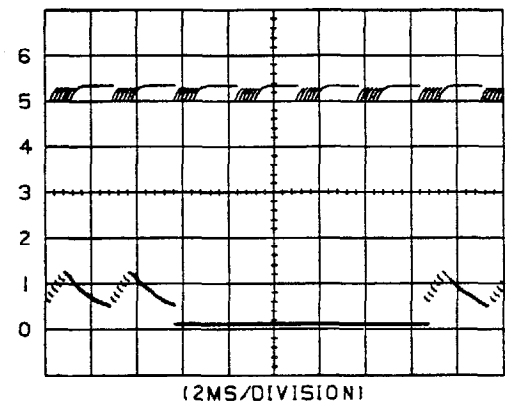


Figure 10-8. TOP - DELAYED FIRING POWER
BOTTOM - GATE PULSES FOR A1 SCR
(Continuous Current)

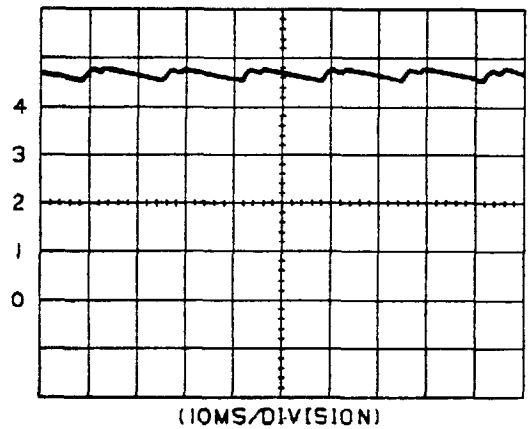


Figure 10-9. FIELD CURRENT (FC) AT FULL FIELD

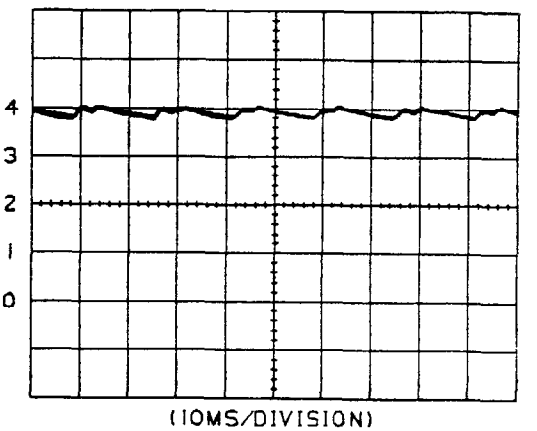


Figure 10-10. FIELD CURRENT (FC) AT WEAK FIELD

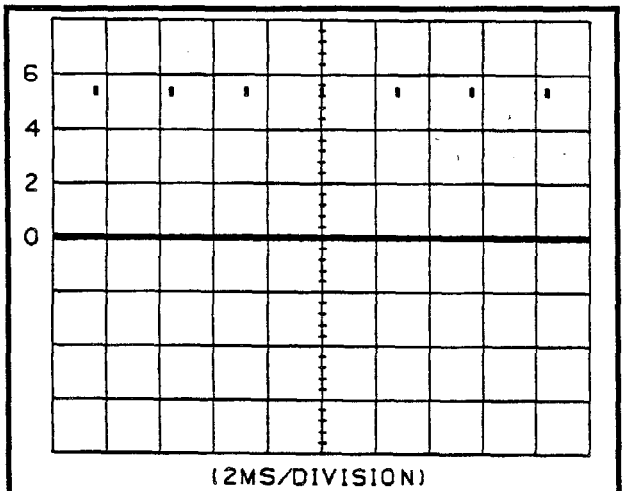


Figure 10-11. INITIAL PULSE UNIT (IPU)
(Indicated Initial Pulse of Each Armature
SCR Firing Pulse Burst)

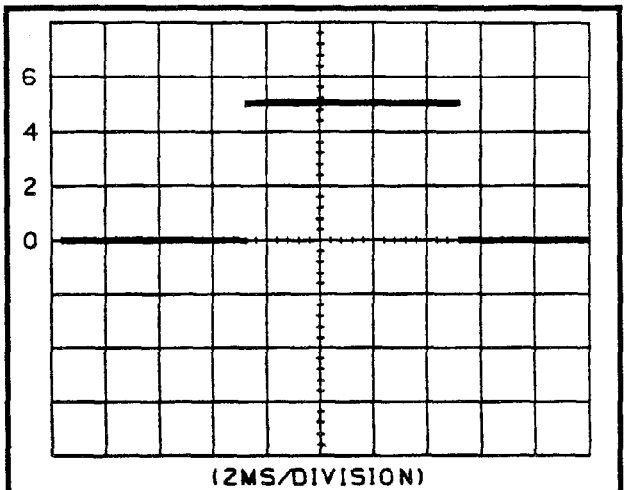


Figure 10-12. SYNC SIGNAL (SY)
(Square Wave Synchronized to
AC Line Frequency)

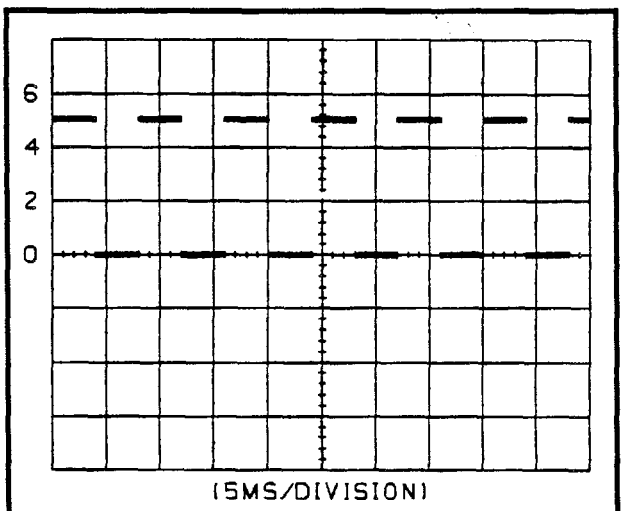


Figure 10-13. AC TACH SIGNAL (TPN)
AT 450 RPM

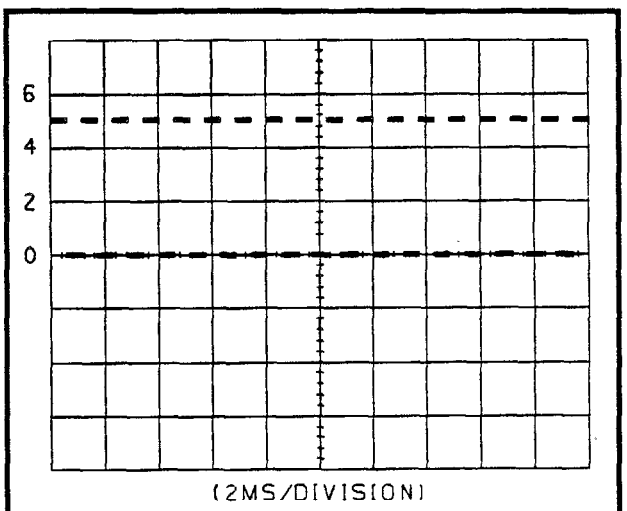


Figure 10-14. AC TACH SIGNAL (TPN)
AT 3000 RPM

NOTES:

11. PARTS REPLACEMENT

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when handling cards and store cards in the anti-static bags they are shipped in.

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the controller before performing any work on controller components. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled machinery.

CARD REPLACEMENT PROCEDURE
(All Frames, Figures 11-1 thru 11-4)

1. Verify power to the controller that the card is to be installed is OFF.
2. Access card to be changed and carefully disconnect all cables (see Figure 11-1 and Figure 11-2).
3. Release all hold-down tabs and remove card (see Figure 11-3).
4. Set all jumpers on the replacement card exactly as the jumpers on the failed card (refer to Table 1 for jumper descriptions and GEK Instruction that is shipped with replacement card).
5. Install the new card verifying that all tabs snap tightly into position.
6. Reconnect all cables verifying that they are properly seated at both ends.
7. Set all pots in the same physical position as the pots on the failed card and fine tune as required per instructions in Table 1 of the Adjustments section.
8. If the Control card is changed, remove the EEROM from the failed card and install in the new card. If the EEROM from the failed card is suspected to be damaged, it will be necessary to reprogram a new EEROM using the Programming Software Package. See Figure 11-4 for typical EEROM removal tool and grounding techniques.

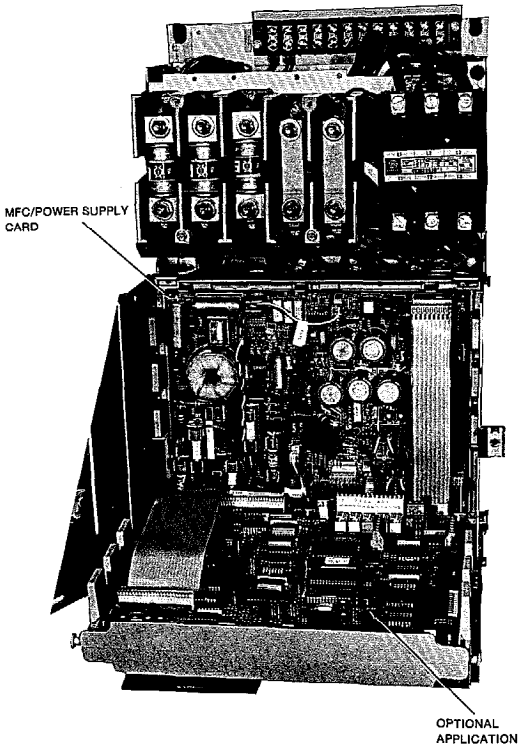


Figure 11-1. CONTROLLER WITH FRONT CARD RACK LOWERED

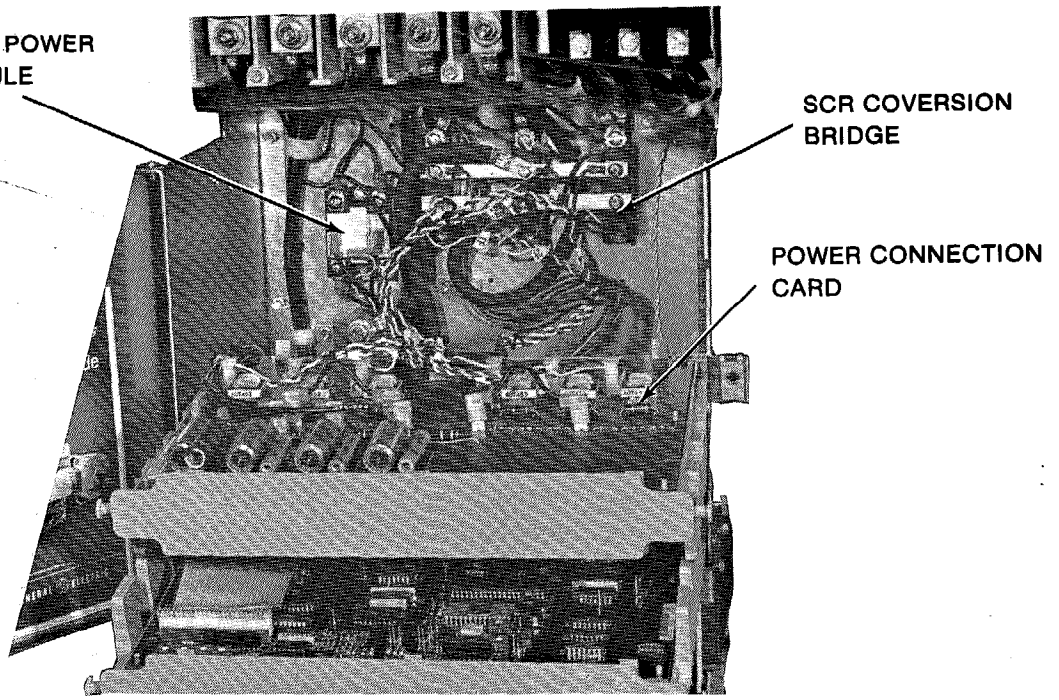


Figure 11-2. CONTROLLER WITH REAR CARD RACK LOWERED

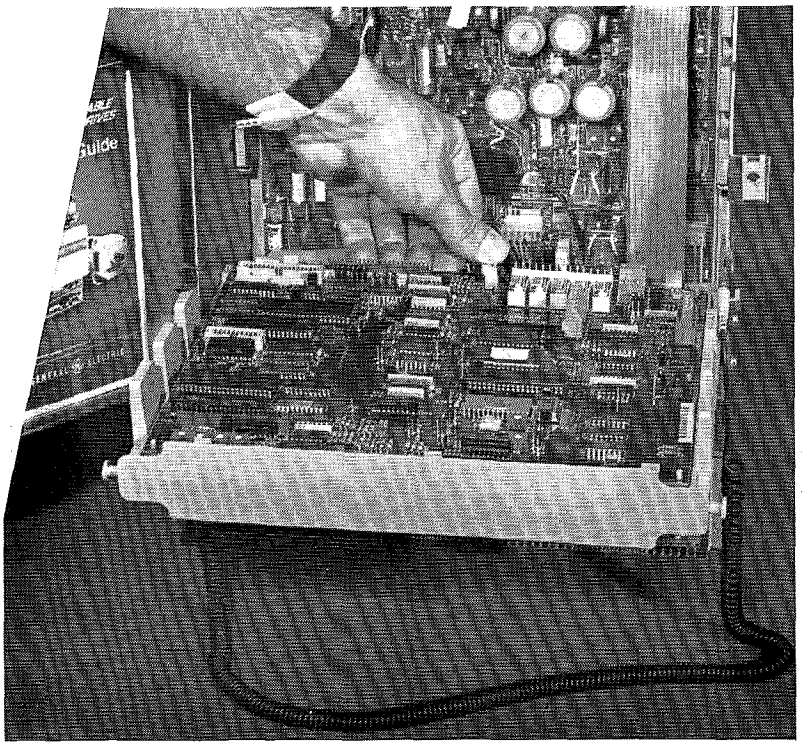


Figure 11-3. CONTROLLER CARD WITH SIX CARD RELEASE TABS

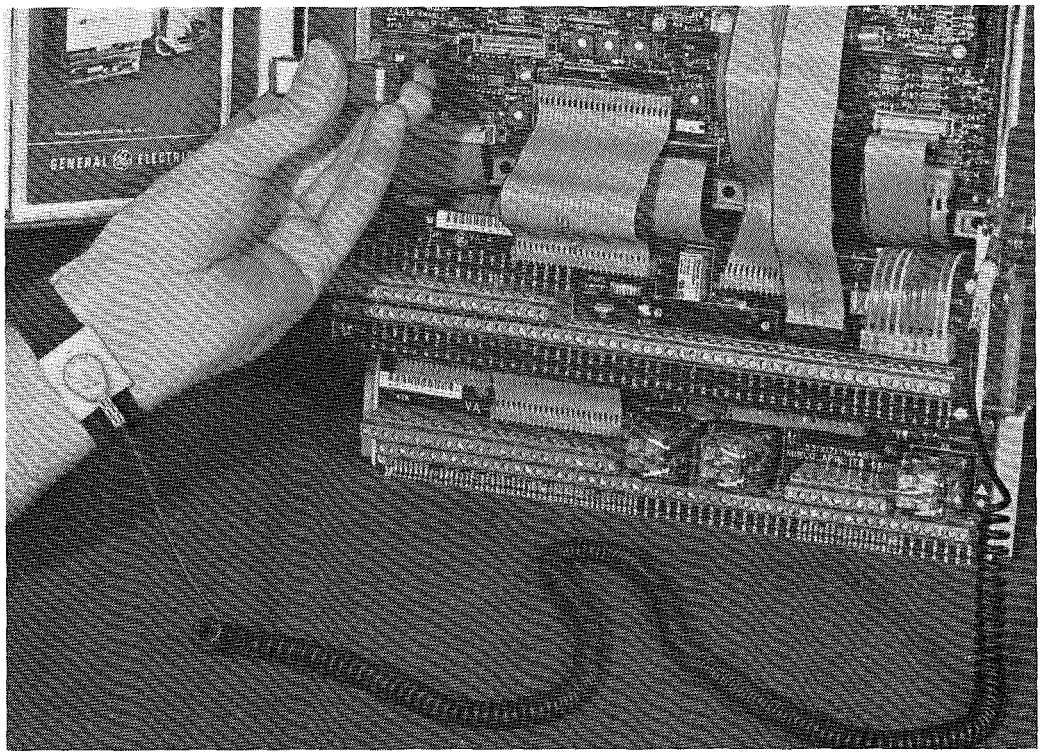


Figure 11-4. TYPICAL EEROM REMOVAL AND/OR REPLACEMENT EQUIPMENT

REPLACEMENT OF SCR MODULES

("G" and "C" Frames, Figures 11-1, 11-2, and 11-5)

1. Verify that power to the controller that the SCR Module is to be installed is OFF.

2. Lower the card carriers to expose the rear of the unit.

3. Disconnect the wires from the damaged module and remove the bus bar and mounting screws.

4. Clean the mounting surface of old lubricant and dirt.

5. Apply a thin film of General Electric GL322 Versilube® Plus lubricant (Part No. 104X214AB059) to both contacting surfaces and install the module. Tighten screws to the torque specifications below.

SCR MODULE PART NUMBER	HEATSINK MOUNTING SCREWS	*BUS BAR MOUNTING SCREWS
104X125DC044	Max. 40 inch pounds	Max. 106 inch pounds
104X125DC045	Max. 53 inch pounds	Max. 133 inch pounds
104X125DC053	Max. 53 inch pounds	Max. 133 inch pounds
104X125DC054	Min. 40 inch pounds	Min. 22 inch pounds

**Support the bus bar while tightening screws to avoid excess torque on connected modules.*

REMOVAL OF SCR STACK ASSEMBLIES

("D" Frames, Figures 11-6 thru 11-9)

Right Hand SCR Removal

1. Verify that power to the controller that the SCR is to be installed is OFF.

2. Remove the SCR firing circuit plugs from the snubber card for the SCRs located in the right hand module only. (It is not necessary to remove the SCR firing circuit plugs from the left hand module.)

3. Remove the AC wires (1AC_ _, 2AC_ _, 3AC_ _) from the snubber card.

4. Remove the six mounting nuts that hold the snubber card in place being careful not to let the bolts drop into the bottom of the controller.

5. Remove the harness retainers from their captive positions.

6. Roll the snubber card off the stack and over onto the left hand stack carefully.

7. Remove the three AC line input bolts, again being careful not to lose any of the hardware.

8. Remove the electrical connection (bus) at the bottom of the stack.

9. Remove the four stack mounting bolts at the top and bottom of the stack.
   - -- The stack assembly can now be lifted off the panel and placed on a workbench for further disassembly (see "SCR Replacement Procedure" that follows).

NOTE: *For **Left Hand Stack Removal**, follow the same procedure as for right hand stack removal **except** the following:*

- -- *It is not necessary to remove the snubber card.*
- -- *Remove AC wires (4AC_ _, 5AC_ _, 6AC_ _) from the snubber card.*

SCR REPLACEMENT PROCEDURE

("D" Frames, Figures 11-6 thru 11-9)

1. Identify the location of the defective SCR (from fault codes and verify with an ohmeter) and SCR location.

2. Place the stack assembly on its side and disassemble the section containing the defective SCR.

3. Alternately loosen each nut on the stack assembly and place them on a clean surface in the same orientation and order that they were removed in.

4. Remove the defective cell, then clean and lubricate the heatsinks as follows before reassembling:
   - -- Clean the heatsinks and lubricate them by placing a few drops of GE, SF1153 silicone oil in the center of both heatsink mounting surfaces.
   - -- Polish the interface area until clean, using a combination of back and forth, sideways, and circular motions with a lint-free cloth or tissue.
   - -- Place a few drops of GE, SF1153 silicone oil on the mating surfaces of the heatsink and use a lint-free cloth or tissue to spread the oil evenly.
   - -- Wipe off any excess oil.

5. Clean and lubricate the SCR cell by performing all substeps of step 4 (above).

6. Coat the threads, nuts, and washers used in the clamp mechanism with a small amount of graphite spray.

7. Replace the SCR cell carefully, ensuring that its orientation and polarity are correct.
   - -- Use care when placing the cell on the dowel pin to prevent scratching the cell post surface.
   - -- Rotate the cell back and forth a few times to ensure proper seating.

8. Reassemble the SCR cell assembly, taking care not to damage the cell with the heatsink.
   - -- Tighten the clamp nuts only finger tight.
   - -- Ensure that the number of exposed threads on each stud are approximately equal.

9. Tighten each nut alternately as shown below:

SCR Part #	Number of Turns*
104X125DA154	1-3/8
104X125DA152	3/4
104X125DA151	1/2

**In 1/4 turn increments from finger tight starting position.*

10. Replace the repaired stack assembly by following, **in reverse**, the procedure outlined for removal.

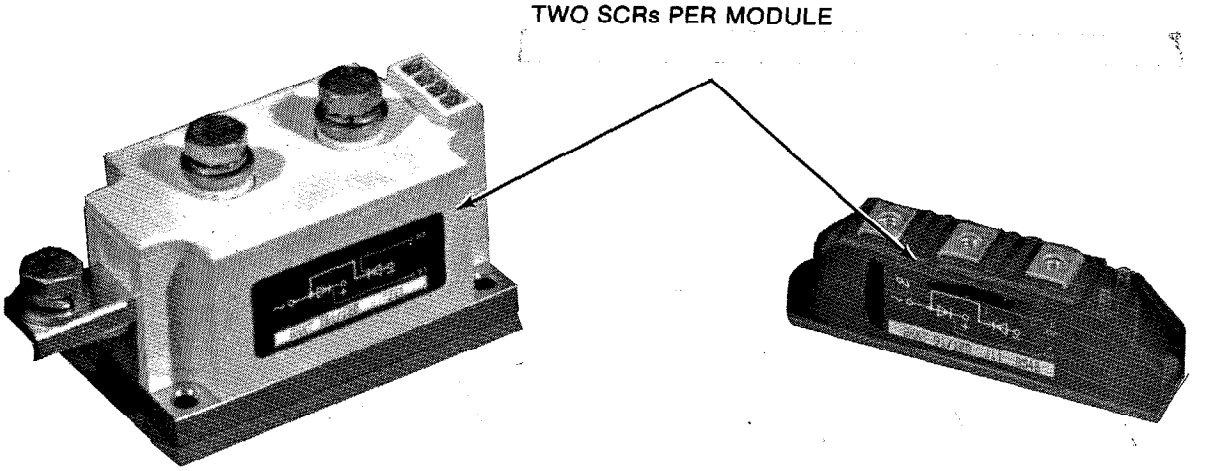


Figure 11-5. TYPICAL ARMATURE SCR MODULES ("G" and "C" Frames)

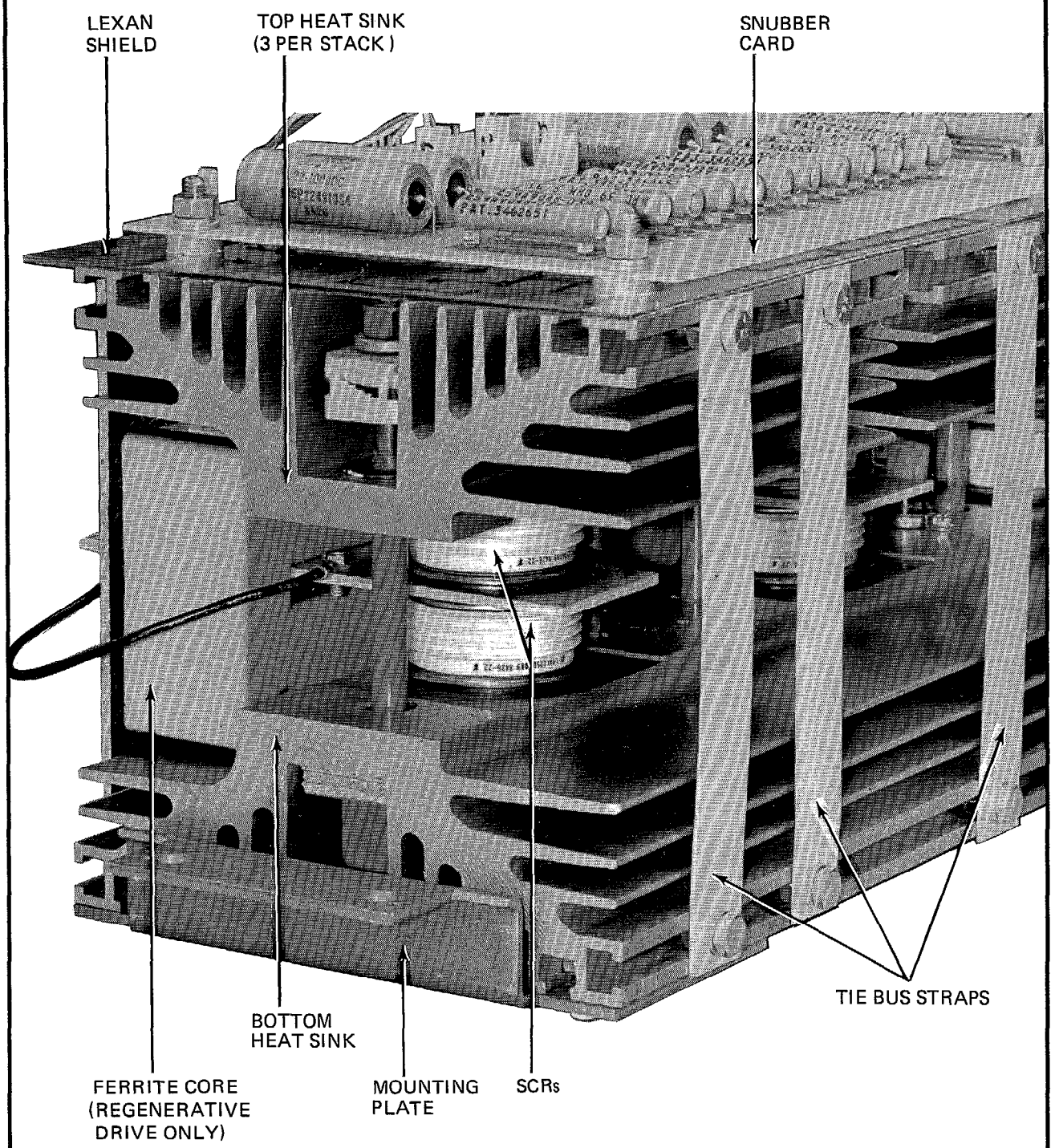


Figure 11-6. SCR STACK ASSEMBLY ("D" Frames)

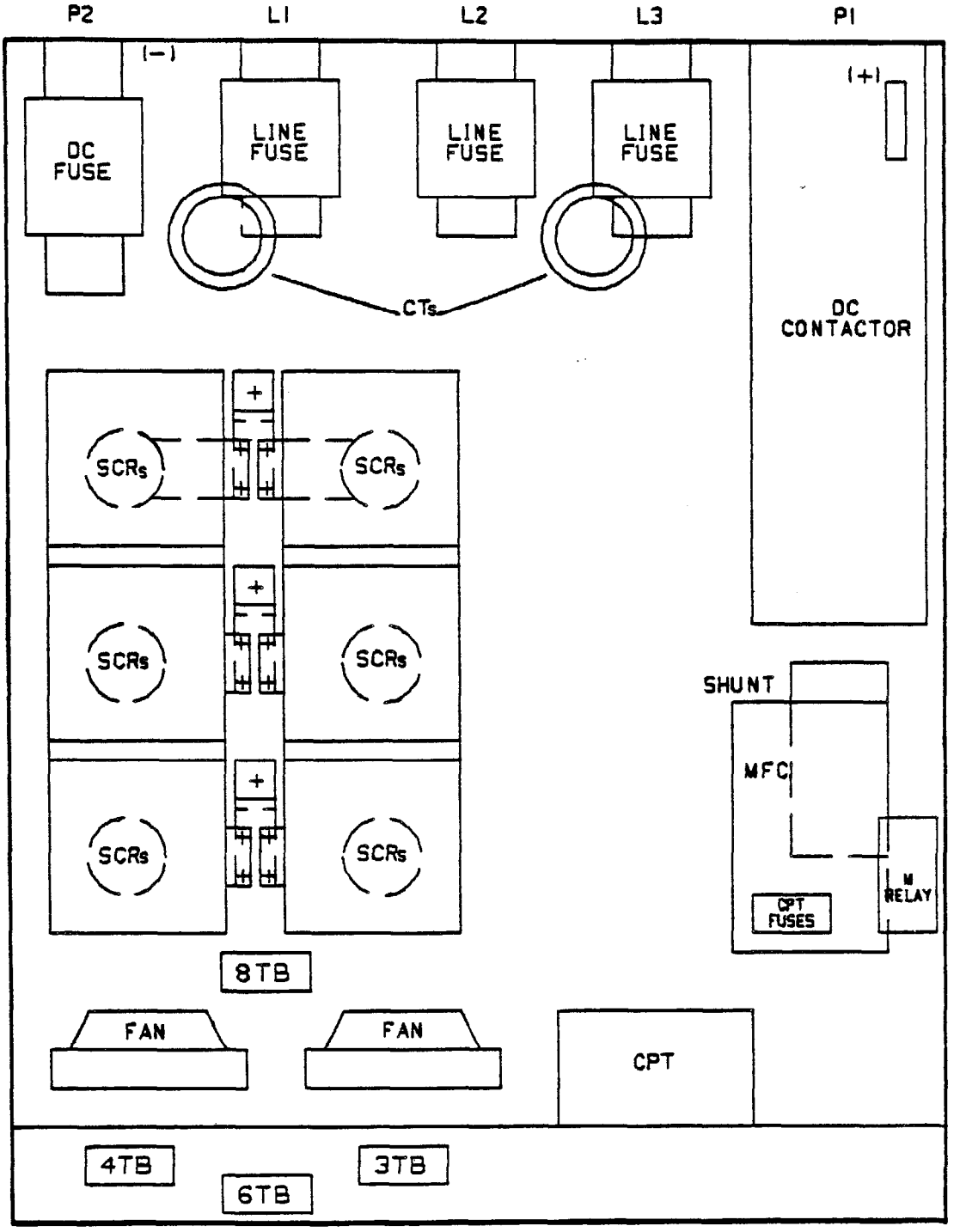
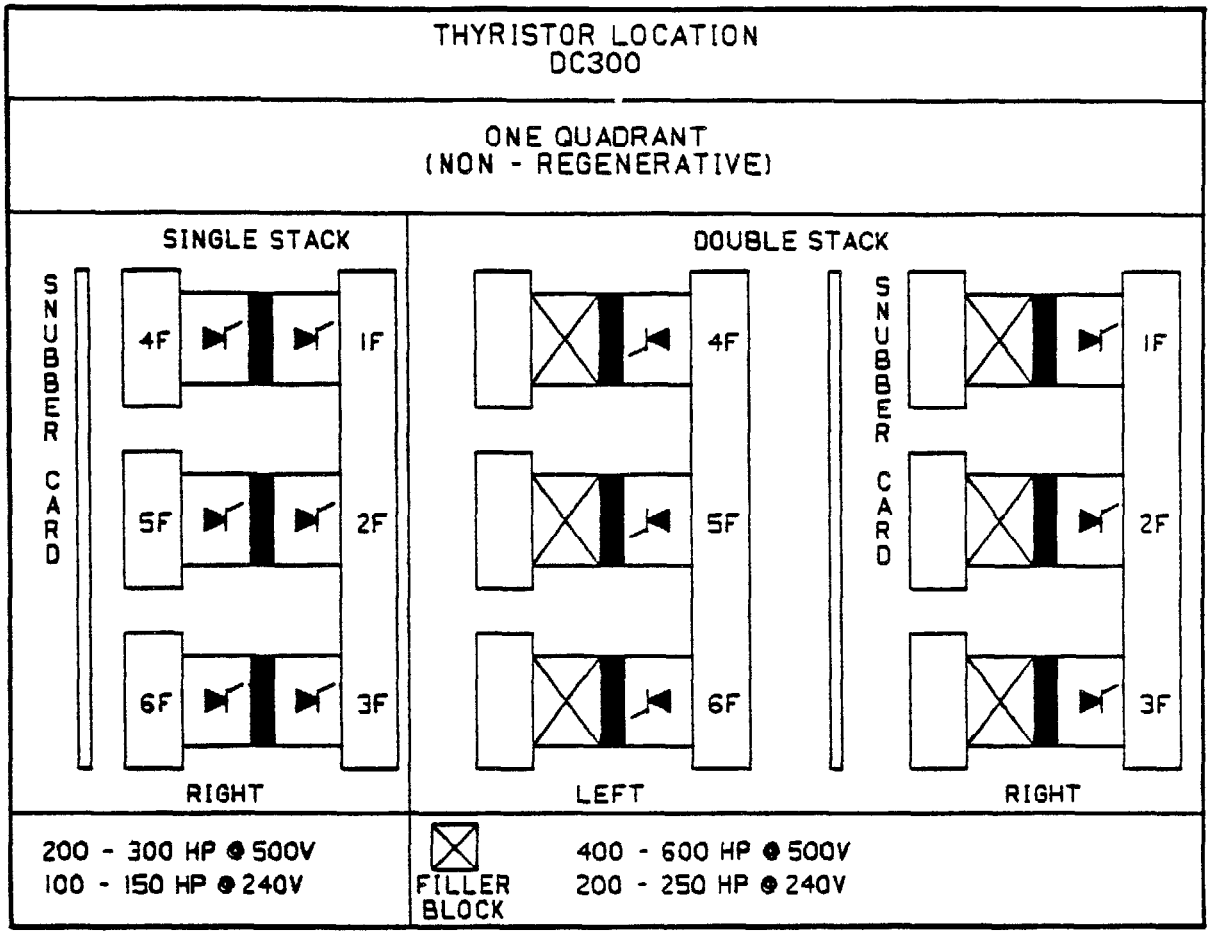


Figure 11-7. SCR LOCATION - GENERAL ("D" Frames)

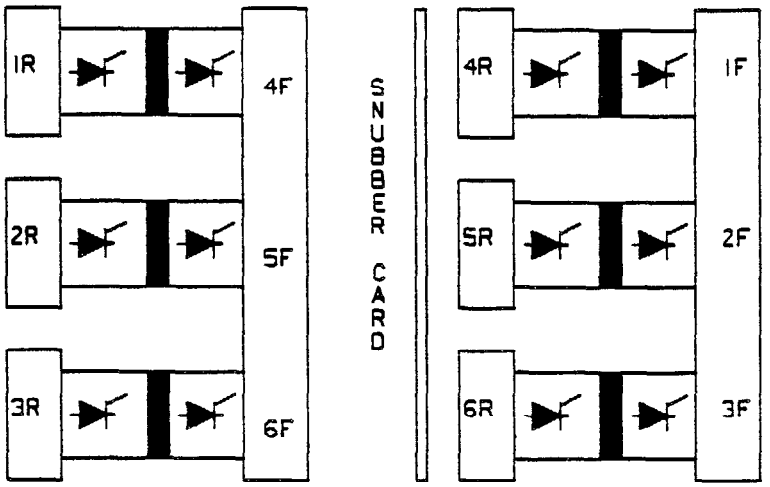


FUSE REPLACEMENT 230VAC					FUSE REPLACEMENT 460VAC				
HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*	HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*
100 TO 150	FU1-3	500	500	A50P	200 TO 300	FU1-3	500	500	A50P
200 TO 250	FU1-3	800	500	A50P	400	FU1-3	800	500	A50P
					500 TO 600	FU1-3	1000	500	A50P
* GOULD - SHAWMUT OR EQUIVALENT									

Figure 11-8. SCR LOCATION - NON-REGENERATIVE ("D" Frames)

THYRISTOR LOCATION
DC300

FOUR QUADRANT
(REGENERATIVE)



200 - 600HP AT 500V
100 - 250HP AT 240V

FUSE REPLACEMENT 230VAC					FUSE REPLACEMENT 460VAC				
HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*	HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*
100 TO 150	FU1-3	500	500	A50P	200 TO 300	FU1-3	500	500	A50P
200 250	FU1-3 FU1-3	800 800	500 500	A50P A50P	400 500	FU1-3 FU1-3	800 1000	500 500	A50P A50P
100	FU4	400	500	A50P	600	FU1-3	1000	500	A50P
125 150	FU4 FU4	500 600	500 500	A50P	200	FU4	400	700	A70P
200 250	FU4 FU4	800 1000	500 500	A50P A50P	250	FU4	500	700	A70P
					300 TO 400	FU4	800	700	A70P
					500 TO 600	FU4	** 700	700	A70P

* GOULD - SHAWMUT OR EQUIVALENT

** *Quantity of two fuses in parallel on DC output.*
Replace both fuses, even though only one may be blown.

Figure 11-9. SCR LOCATION - REGENERATIVE ("D" Frames)

NOTES:

12. SPARE AND RENEWAL PARTS

It is recommended that the following parts lists be reviewed and a set of spare parts be maintained to minimize down time should repairs be required. Renewal parts may be ordered by contacting your local **GE Sales Office** or an **Authorized GE Parts Distributor in your area.**

PART NUMBER IDENTIFICATION

The parts listed are divided into four categories, each with its own part number structure:

Order-Specific Assemblies
Common Assemblies
Component Parts
Circuit Cards

Order-Specific Assemblies are generally major assemblies consisting of a group of common assemblies that make up the specific controller provided. Other items which are obtained specifically for the order may also be provided with a similar part number structure.

Example:

7VXYZ999CD01

7V	--	assembly provided by GE DRIVE PRODUCTS OPERATION
XYZ999	--	identification number (original order)
CD	--	assembly type (CD = Core Drive)
01	--	unit identification number

Common Assemblies are subassemblies, consisting of a group of component parts, which are in common use in GE DRIVE PRODUCTS OPERATION products and are not specific to a particular order. The part number consists of the number "36" followed by an alpha character and may be up to 14 characters in total length.

Example:

36C77345AAG01 is the part number for a High Current Motor Field Control Module for a "D" Frame QIII DC Controller.

Component Parts are the basic components used in the assemblies and generally represent the lowest discrete level. Component part numbers consist of a combination of alphanumeric characters, defining the class of component and the specific item.

Example:

104X109AD038

104X	--	component part
109	--	component type (fuse)
AD038	--	specific item (500 V, 3.2A FNQ)

Printed Circuit Cards are identified by a catalog number giving the product class, card function and revision levels. It should be noted that some cards contain software which is designated by the suffix "Mn" where "n" is the software revision number. These cards are also available less software with suffix "Gn" designation. The factory may substitute later versions of cards based on availability and design enhancements.

Example:

531X300CCHABM5

531X	--	card for GE DRIVE PRODUCTS OPERATION microprocessor based controller
300CCH	--	Control card for QIII DC Controller
ABM5	--	revision level, including software

As an aid in expediting part identification, a partial listing of the parts, functions and associated root part numbers is shown in the following charts.

STANDARD PRINTED CIRCUIT CARDS	CATALOG NUMBER (Root)
Control card (with microprocessors)	531X300CCH_M_
Power Connection card	531X308PSCA_G1/G2/G3/G4
3TB Terminal Board card	531X170TBS_G_
Power Supply card	531X111PSH_G2/G3 (with external field control)
Snubber Card ("D" Frame only)	531X310SNDA_G1

OPTIONAL PRINTED CIRCUIT CARDS	CATALOG NUMBER (Root)
Application card	531X139APMA_G2
Encoder Process Interface card	531X134EPRA_G1
4TB Terminal Board card	531X171TMAA_G2

SUBASSEMBLIES AND COMPONENTS

Subassembly/Component	Catalog Number
Motor Field Module	104X125DC012
High Current Motor Field Control Module	36C774345AAG01 (mounted in "D" frame)
K6, K7, K8, Relays*	104X166AA081
Thermoswitch	104X152AC067
Control Power Transformer (460 VAC)	36B605538AAG02 ("G" Frames)
Control Power Transformer (460 VAC)	36B605538ACG02 ("C" Frames)
Control Power Transformer (460 VAC)	36B605538AEG02 ("D" Frames)
Suppressor	36A358207AAG01
Fan (small, 4-1/2")	104X215CA003
Fan (large, 7")	104X215CA004
Blower Motor	104X215BA008 ("C" Frames)
Blower Motor (460 VAC)	104X215BA020 ("D" Frames)
Current Transformer Assembly	36B05537AAG01 (75 HP, 460 VAC)
Current Transformer Assembly	36B05537ABG01 (150 HP, 460 VAC)
Current Transformer Assembly	36B05633AAG01 (200/300 HP, 460 VAC)
Fuses (Control Power)	
3 Amps (CFU6)	104X109AD004
1.5 Amps (CFU7)	104X109AD088
0.5 Amps (CFU8)	104X109AD009
3.2 Amps (CFU4 & 5)	104X109AD038
15 Amps (CFU1, 2 & 3)	104X109AD076
6 Amps	104X109AD039
Fuses, MFC	
30 Amps	104X109AD084

**Once relay contacts are used on high current or voltage signals, they cannot be used for low level signals.*

INTERCONNECTION CABLES

Connector	Catalog
1PL	36A358208AAG02
2PL	104X154CB111
3PL	36A358208AAG04
4PL-3	36A358208AAG07
5PL	36A358208AAG01
6PL	36A358208AAG08
7PL	36A358208AAG05
7PLC* (5 ft.)	36A358208AAG35
9PL	36A358208CXG01
10PL (G,C Frames)	36A358208AAG09
10PL (D Frames)	36A358218XAG03
11PL (G,C Frames)	36A358208AAG11
11PL (D Frames)	36A358218XAG04
12PL (G,C Frames)	36A358208AAG10
12PL (D Frames)	36A358218XAG05
13PL (G,C Frames)	36A358208AAG12
13PL (D Frames)	36A358218XAG06
14PL (G,C Frames)	36A358208EYG01
14PL (D Frames)	36A358208EYG02
16PL**	36A358028ARG01

**7PLC is used with Serial Link Interface Card (531X120STRA_G1).*

***16PL is used with BCD Assembly Terminal Block (104X154AB038).*

NOTE: *Connectors used may vary with options provided.*

OPTION KITS

Description	Kit Number
Encoder Process Interface Card	7V11CPE
Application Card	7V11CAC
4TB Terminal Board Card	7V11CTB
Serial Link Interface Card	7V11CSL
Balun Assembly	7V11CBA
BCD Assembly	7V11CBD

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "G" AND "C" FRAMES (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	109AG-	007	007	007	007	011	011	011	021
DC Power Fuse*	109AG-	005	005	007	007	009	015	015	021
Arm SCR Modules	125DC-	054	054	054	054	054	054	054	044
MA Contactor	105CB-	003	003	003	003	005	006	006	013
MA Contactor	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	105CB-	007	007	007	007	007	008	008	011
MA Aux. Interlock	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	118BA-	----	----	----	----	----	----	----	011

Part Description	Cat. # 104X-	Horsepower 50	60	75	100	125
AC Power Fuse	109AG-	021	021	021	027	027
DC Power Fuse*	109AG-	021	021	021	025	025
Arm SCR Modules	125DC-	044	044	044	045	053
MA Contactor	105CB-	013	013	013	----	016
MA Contactor	105CA-	----	----	----	069	----
MA Aux. Interlock	105CB-	011	011	011	----	----
MA Aux. Interlock	105CA-	----	----	----	025	----
MA Aux. Interlock	118BA-	011	011	011	----	----

**Regenerative drives only.*

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "G" AND "C" FRAMES (HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	109AG-	007	007	007	007	007	007	007	007
DC Power Fuse*	109AJ-	003	003	003	007	007	009	009	013
Arm SCR Modules	125DC-	054	054	054	054	054	054	054	054
MA Contactor	105CB-	003	003	003	003	003	003	003	005
MA Contactor	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	105CB-	007	007	007	007	007	007	007	007
MA Aux. Interlock	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	118BA-	----	----	----	----	----	----	----	----

Part Description	Cat. # 104X-	Horsepower 50	60	75	100	125	150	200	250	300
AC Power Fuse	109AG-	011	011	015	021	021	021	027	027	027
DC Power Fuse*	109AJ-	013	014	015	015	017	017	019	021	025
Arm SCR Modules	125DC-	054	054	054	044	044	044	045	045	053
MA Contactor	105CB-	005	006	012	013	013	013	----	----	016
MA Contactor	105CA-	----	----	----	----	----	----	069	069	----
MA Aux. Interlock	105CB-	007	008	008	----	----	----	----	----	----
MA Aux. Interlock	105CA-	----	----	----	----	----	----	025	025	----
MA Aux. Interlock	118BA-	----	----	----	011	011	011	----	----	----

**Regenerative drives only.*

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "D" FRAMES (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 100	125	150	200	250	300
AC Power Fuse	109AG-	027	027	027	030	030	032
DC Power Fuse	109AJ-	025	027	028	030	032	032*
Armature SCRs	125DA-	151	151	151	152	152	154
DC Contactor	106AE-	008	008	008	008	009	--
DC Contactor	106MA-	--	--	--	--	--	G01
Armature SCRs With Leads	36A353476AX-	G01	G01	G01	G02	G02	G03
Current Transformer Asm.	36A358218YA-	G01	G01	G01	G02	G02	G02

** Two per drive required.*

NOTE: *On Regenerative Controllers with paralleled DC fuses, replace both fuses even if only one is blown.*

QIII DC CONTROLLER SPARE AND RENEWAL PARTS - "D" FRAME (HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 200	250	300	400	500	600	700	800
AC Power Fuse	109AG-	027	027	027	030	032	032	032	034
DC Power Fuse	019AJ-	019	021	025	026	025*	025*	025*	026*
Armature SCRs	125DA-	151	151	151	152	152	154	154	154
DC Contactor	106AE-	007	008	008	008	009	009	--	--
DC Contactor	106MA-	--	--	--	--	--	--	G01	G01
Armature SCRs With Leads	36A353476AX-	G01	G01	G01	G02	G02	G03	G03	G03
Current Transformer Assembly	36A358218YA-	G01	G01	G01	G02	G02	G02	G02	G02

** Two per drive required.*

NOTE: *On Regenerative Controllers with paralleled DC fuses, replace both fuses even if only one is blown.*

13. WARRANTY PARTS ORDERING

1. Consult the **Warranty Terms and Conditions** for details of the product warranty, including warranty period and parts coverage.

2. Warranty replacement parts may be obtained from the supplier of the QIII DC Controller during regular working hours:

_____

QIII Supplier

_____

Address

_____

Phone Number

_____

Working Hours

3. To expedite your order, always have the following information ready whenever contacting the supplier of the QIII DC Controller:

CONTROLLER SERIAL NUMBER & MODEL NUMBER
PART NUMBER & DESCRIPTION

NOTES:

14. RECOMMENDED POWER STUD WIRING, TERMINALS & CRIMPING TOOLS

RECOMMENDED POWER STUD WIRING AND CRIMPING TOOLS

ITEM	AMP CRIMPING TOOL CAT. NO.	TYPE	FOR WIRE SIZE	TERMINAL TYPE
1	49592	Hand	14--12 AWG	Solistrand*
2	49935	Hand	12--10 AWG	Solistrand
3	69062	Hand-hydraulic with self-contained dies	8--2 AWG	Solistrand

ITEM	AMP ELECTRO-HYDRAULIC POWER UNIT 115VAC (CAT. NO.) Crimp Tool	Head	Die	FOR WIRE SIZE	TERMINAL TYPE
4	69120-1	69065	46323-2	1--10 AWG	Solistrand
5	69120-1	69065	46323-2	2/0 AWG	Solistrand
6	69120-1	69065	46324-2	3/0 AWG	Solistrand
7	69120-1	69065	46325-2	4/0 AWG	Solistrand
8	69120-1	69065	46326-2	250 MCM	AMPower
9	69120-1	69065	46327-2	300 MCM	AMPower

**Trademark of AMP Incorporated.*

RECOMMENDED POWER STUD WIRING AND TERMINALS

Drive	230 VAC, 240 VDC Stud Dia. (c)		Power Wire (a, b)			Terminal - AMP Solistrand GE Cat.#	AMP	Hole	Crimping Tool
HP	AC	DC	Qty.	AWG	MCM	104X161AA	Cat.#	Dia.	See tool list
1-3	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
5	.25		1	10	---	009	33458	.25	Item 2
		.25	1	10	---	009	33458	.25	Item 2
7.5	.25		1	8	---	012	33461	.25	Item 3
		.25	1	8	---	012	33461	.25	Item 3
10	.25		1	6	---	015	33465	.25	Item 3
		.25	1	6	---	015	33465	.25	Item 3
15	.25		1	4	---	059	31811	.25	Item 3
		.25	1	4	---	059	31811	.25	Item 3
20	.25		1	2	---	057	320383	.25	Item 3
		.25	1	2	---	057	320383	.25	Item 3
25	.25		1	1/0	---	067	36915	.25	Item 4
		.25	1	1/0	---	067	36915	.25	Item 4
30	.25		1	1/0	---	067	36915	.25	Item 3
		.25	1	1/0	---	067	36915	.25	Item 3
40	.38		1	2/0	---	023	36923	.38	Item 5
		.38	1	3/0	---	025	36927	.38	Item 6
50	.38		1	3/0	---	025	36927	.38	Item 6
		.38	1	4/0	---	068	321878	.38	Item 7
60	.38		1	4/0	---	068	321878	.38	Item 7
		.38	2	4/0	---	021	36917	.38	Item 4
75	.38		2	1/0	---	021	36917	.38	Item 4
		.38	2	2/0	---	023	36923	.38	Item 5
100	.38		2	3/0	---	025	36927	.38	Item 6
		.38	2	4/0	---	068	321878	.38	Item 7

NOTES:

(a) *Wire size from NEC Table 310-16. Copper wire rated 60 degrees C for 100 amps and less, 75 degrees C for over 100 amps, in 30 degrees C ambient and 1.25 times drive rated amps. These are minimum wire sizes. Consult and conform to local and national codes.*

(b) *Quantity of wires and terminals in parallel per stud.*

(c) *AC stud is not used for wiring on drives that have circuit breakers.*

RECOMMENDED POWER STUD WIRING AND TERMINALS

Drive	460 VAC, 500 VDC Stud Dia. (c)		Power Wire (a, b)			Terminal - AMP Solistrand GE Cat.#	AMP	Hole	Crimping Tool
HP	AC	DC	Qty.	AWG	MCM	104X161AA	Cat.#	Dia.	See tool list
1-3	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
5	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
7.5	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
10	.25		1	10	---	009	33458	.25	Item 2
		.25	1	10	---	009	33458	.25	Item 2
15	.25		1	8	---	012	33461	.25	Item 3
		.25	1	8	---	012	33461	.25	Item 3
20	.25		1	6	---	015	33465	.25	Item 3
		.25	1	6	---	015	33465	.25	Item 3
25	.25		1	6	---	015	33465	.25	Item 3
		.25	1	6	---	015	33465	.25	Item 3
30	.25		1	4	---	059	31811	.25	Item 3
		.25	1	4	---	059	31811	.25	Item 3
40	.25		1	2	---	057	320383	.25	Item 3
		.25	1	2	---	057	320383	.25	Item 3
50	.25		1	1	---	067	36915	.25	Item 4
		.25	1	1	---	067	36915	.25	Item 4
60	.25		1	1	---	067	36915	.25	Item 4
		.25	1	1	---	067	36915	.25	Item 4
75	.25		1	1/0	---	067	36915	.25	Item 4
		.25	1	2/0	---	075	36921	.25	Item 5
100	.38		1	3/0	---	025	36927	.38	Item 6
		.38	1	4/0	---	068	321878	.38	Item 7

NOTES:

(a) *Wire size from NEC Table 310-16. Copper wire rated 60 degrees C for 100 amps and less, 75 degrees C for over 100 amps, in 30 degrees C mabient and 1.25 times drive rated amps. These are minimum wire sizes. Consult and conform to local and national codes.*

(b) *Quantity of wires and terminals in parallel per stud.*

(c) *AC stud is not used for wiring on drives that have circuit breakers.*

RECOMMENDED POWER STUD WIRING AND TERMINALS (cont.)

	460 VAC, 500 VDC					Terminal - AMP Solistrand			Crimping Tool
Drive	Stud Dia. (c)		Power Wire (a, b)			GE Cat.#	AMP	Hole	
HP	AC	DC	Qty.	AWG	MCM	104X161AA	Cat.#	Dia.	See tool list
125	.38		1	4/0	---	068	321878	.38	Item 7
		.38	2	1/0	---	021	36917	.38	Item 4
150	.38		2	1/0	---	021	36917	.38	Item 4
		.38	2	2/0	---	023	36923	.38	Item 5
200	.38		2	3/0	---	025	36927	.38	Item 6
		.38	2	4/0	---	068	321878	.38	Item 7
250	.38		2	4/0	---	068	321878	.38	Item 7
		.38	2	---	300		325803	.38	Item 9

NOTES:

(a) *Wire size from NEC Table 310-16. Copper wire rated 60 degrees C for 100 amps and less, 75 degrees C for over 100 amps, in 30 degrees C mabient and 1.25 times drive rated amps. These are minimum wire sizes. Consult and conform to local and national codes.*

(b) *Quantity of wires and terminals in parallel per stud.*

(c) *AC stud is not used for wiring on drives that have circuit breakers.*

15. GLOSSARY OF TERMS

Base Speed	--	Maximum speed of a motor at rated armature volts and field amps.
DFP	--	Delayed Firing Power to armature SCRs.
EEROM	--	Electrically Erasable Read-Only Memory, also called EE or E^2 (E squared). Contains the drive Parameters.
EE Enable Device	--	Device used to "Enable" or "Protect" EEROM programming. A hardware jumper (JP13) located on the Control card.
Field Weakening	--	Lowering of field current at rated armature volts to allow higher speeds. Actual armature voltage where field weakening begins is called the crossover point (adjusted by Paramater ADR079-CROSS)
FUP	--	Firing Microprocessor, U5 on the Control card.
"Hard" Reset	--	Hardware Reset, resets the drive cards. Used to clear some faults and allow certain parameter changes. Activated by depressing SW2 on the Control card or connecting 3TB69 to 3TB66.
K1	--	"RUN" Relay located on Power Supply card.
K2	--	MA Auxiliary Relay, pilot relay for MA contactor. K2 is located on the Power Supply card.
K3	--	Fault Relay located on the Power Supply card.
LED	--	Light Emitting Diode.
MA	--	Main AC Contactor (see General Description).
MFC	--	Motor Field Control (see General Description).
MOV	--	Metal Oxide Varistor.
Parameters	--	Adjustable software settings used to program and tune the drive regulators.
PI	--	Process Interface card.
PL	--	Plug (may be plug or receptacle).
PSC	--	Power Supply card.
RAM	--	Random Access Memory
Regenerative	--	Ability to return power from the armature to the AC line. Creates braking effect on motor. (Regeneration, regen., etc.)
RS232C	--	Serial Link Communication Standard.
RS422	--	Serial Link Communication Standard.
RUP	--	Regulator Microprocessor, U4 (and U1 PROM) on Control card.
SCR	--	Silicon Controlled Rectifier (Thyristor).
SLD	--	Signal Level Detector, SLD0 thru SLD5, (RX1 or K1 thru RX5 or K5) on optional Application card and SLD5 on 3TB.
TUP	--	Tachometer Microprocessor, U6 on the Control card.

NOTES:



GE Drive Systems

GEK-45167 (12/90)

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